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NON-CONVENTIONAL COUPLING AND TOPOLOGICAL TRANSITIONS IN
MULTIORBITAL PHOTONIC STRUCTURES

THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR
THE DEGREE OF MASTER OF SCIENCE, MENTION IN PHYSICS

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In this thesis, non-conventional coupling phenomena in multi-orbital photonic lattices, fabricated using the femtosecond laser writing technique, are studied. Mechanisms that transcend coupling between fundamental modes are investigated, incorporating orbital degrees of freedom, structural asymmetries, and topological effects. Light propagation is analyzed in configurations where transverse SS and PP modes interact in a controlled manner, revealing new forms of optical manipulation based on symmetry, geometry, and modal structure.

First, the phenomenon of **dipolar invisibility** of P dipole modes is investigated, demonstrating experimentally that, under certain geometric conditions, these modes can remain optically decoupled. This effect is observed in a graphene-type photonic lattice and is modeled theoretically through an extension of coupled-mode theory that incorporates both long-range couplings and non-orthogonality between modes.

Subsequently, the phenomenon of **non-symmetric evanescent coupling** in dimers with different refractive index contrasts is studied. A generalized formulation of coupled-mode theory that accounts for this asymmetry in the dynamics, even in the absence of loss or gain, is presented. The breaking of reciprocity is experimentally validated through measurements of intensity imbalance dependent on the input channel.

Finally, **SP interorbital coupling** is explored. This mechanism allows the hybridization of modes of distinct transverse symmetry through the precise tuning of their propagation constants. Using this technique, a lattice implementing the SP-SSH model—a multi-orbital extension of the Su-Schrieffer-Heeger model—is realized. This system exhibits two distinct transitions between topological phases, determined by geometric dimerization and orbital detuning.

The experimental and numerical results presented in this thesis reveal new ways to control light propagation by utilizing the orbital degrees of freedom of guided modes, paving the way for the design of advanced photonic networks with topological and directional functionality.

*Mas si buscáis descubrimientos
Tierras irrealizables más allá de los cielos
Vegetante obsesión de musical congoja
Volvamos al silencio
Trampas de luz y cascadas lujosas
Trampas de perla y de lámpara acuática
Anda como los ciegos con sus ojos de piedra
Presintiendo el abismo a todo paso
Mas no temas de mí que mi lenguaje es otro
No trato de hacer feliz ni desgraciado a nadie
Ni descolgar banderas de los pechos
Ni dar anillos de planetas
Ni hacer satélites de mármol en torno a un talismán ajeno
Quiero darte una música de espíritu
Música mía de esta cítara plantada en mi cuerpo
Música que hace pensar en el crecimiento de los árboles
Y estalla en luminarias dentro del sueño*

Extracts from *Altazor*, Vicente Huidobro

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1. Introduction

Over the past decade, several Nobel Prizes in physics have been closely linked to optics [1]: for the generation of ultrashort light pulses—femtoseconds [2] and then attoseconds [3–5]—, for pioneering experiments with entangled photons [6–8], for the development of optical tweezers [9], and for the invention of high-efficiency LEDs [10–12]. These fundamental advances have driven applications in industry, biomedicine, telecommunications, and even defense. A prominent everyday application is the optical fiber, which acts as a waveguide for light and is now the primary medium for global data transmission [13, 14].

Many of these developments have been enhanced by the *femtosecond laser writing* technique, which allows for the fabrication of three-dimensional photonic lattices of waveguides with micrometric precision. This technology has enabled the experimental realization of lattices with complex geometries and artificial degrees of freedom, and has been used to study phenomena such as Bloch oscillations [15], Anderson localization [16], flat band dynamics [17–20], topological phases [21–24], formation of nonlinear solitons [25], and even the propagation of quantum light [26–28].

Within this framework, the present work focuses on the study of *multi-orbital* photonic lattices, in which the internal structure of the transverse modes of light can be used as a new control resource. Particularly, the fundamental waveguide modes (denoted here as S) and their first antisymmetric transverse excitations (denoted as P) are studied.

The use of this notation follows a structural analogy with atomic orbitals: the S mode has axial symmetry and lacks transverse nodes, while the P mode presents a lobular structure with a node on its transverse axis. This nomenclature is adopted as a visual and conceptual resource but does not imply the existence of central potentials or angular momentum quantization as in the atomic case. Its purpose is to establish an intuitive language that allows communicating the spatial structure and symmetry of the optical modes and facilitates their analysis in complex configurations.

This analogy extends to the use of terms like *photonic molecules*, which describe a system of two strongly coupled optical waveguides, whose eigenfunctions combine to generate symmetric and antisymmetric hybrid modes. This idea takes inspiration from quantum chemistry, where atomic orbitals combine linearly to form bonding (σ) and antibonding (σ^*) molecular orbitals. In this thesis, we will call a *photonic molecule* a pair of coupled waveguides whose modal hybridization produces collective states analogous to these orbitals.

It is important to emphasize that this is a formal and geometric analogy. It does not imply the existence of shared electrons, bonding forces, quantum levels bound to electronic configurations, or quantum correlations typical of the molecular world. Effects such as binding energy, selection rules, or quantization of electronic states are not observed. The term is used as a visual and structural metaphor that facilitates the understanding of modal coupling and its role as a building block in more complex photonic lattices. This conceptual framework has been used in few works under the name *photonic molecules* [29], and is adopted here for communicative purposes, but with a clear delineation of its physical scope.

Along the same lines, the terminology of σ and π bonds is used to refer to the types of modal coupling between waveguides: σ -type coupling occurs through frontal (axial) superposition of modes, while π -type arises from lateral coupling, with nodes on the central axis. This characterization is based on the geometry of the involved optical fields and will be discussed in detail in the corresponding chapters.

One of the experimental phenomena that motivates this study is the *invisibility angle* [30], described in Chapter 5. In certain geometric configurations, the P modes completely decouple from each other, remaining optically *invisible*. This effect reveals that the transverse symmetry of the modes can be actively exploited as a control tool, allowing for new forms of light localization and routing.

On the other hand, the breaking of symmetry in the evanescent couplings between waveguides with different refractive indices is also analyzed, leading to non-reciprocal dynamics. Although it is traditionally assumed that the coupling $C_{i \rightarrow j}$ between two waveguides is equal to $C_{j \rightarrow i}$, this symmetry can be broken even in passive systems, generating asymmetric couplings that are modeled using non-Hermitian matrices with real spectra. This phenomenon is addressed in Chapter 6.

Finally, the concept of SP inter-orbital coupling is introduced, which allows hybridizing modes with different symmetry through the precise tuning of their propagation constants. This mechanism is experimentally implemented in Chapter 7, where SP-SSH type lattices are designed and characterized: a multi-orbital extension of the Su-Schrieffer-Heeger model [31], with two independent degrees of freedom that generate a double topological transition. The concept of topological transition is analyzed from the perspective of band theory, the existence of edge modes, and bulk polarization.

This approach complements and reinforces the general thrust of this thesis: to study how the internal structure of modes—be it their transverse or orbital symmetry, as well as the difference between them—can be used as a physical resource to control light propagation in photonic lattices.

This thesis is organized as follows:

Chapter 2 introduces the theoretical formalism used throughout the thesis, including the principles of modal propagation, coupled-mode theory in discrete lattices, and the relevant topological concepts in the study of photonic bands.

Chapter 3 presents the numerical tools employed. The Eigenmode Expansion (EME) method, which allows modeling the stationary evolution of light, is described, along with other complementary methodologies such as the Beam Propagation Method (BPM) and Coupled-Mode Theory (CMT).

Chapter 4 details the experimental technique of femtosecond pulsed laser writing, with emphasis on the key parameters for the design and fabrication of three-dimensional multi-orbital photonic lattices.

Chapter 5 studies the phenomenon of P mode invisibility, experimentally characterizing the critical geometry that leads to the optical decoupling of these modes. This property is tested in a graphene-type photonic lattice. To achieve quantitative agreement between theory and experiment, an extended model is introduced that incorporates both modal *non-orthogonality* and *long-range couplings* between non-adjacent waveguides.

Chapter 6 examines the phenomenon of asymmetric evanescent coupling between waveguides of different refractive indices. A correction to conventional coupled-mode theory is presented that allows modeling these effects using non-Hermitian coupling matrices but with real spectra, and the resulting intensity imbalance in asymmetric dimers is experimentally validated.

Chapter 7 analyzes the formation of hybrid modes through photonic molecules for their use as building blocks in the formation of multi-orbital lattices. On this basis, an SP-SSH type lattice, a multi-orbital extension of the Su-Schrieffer-Heeger model, is implemented and characterized, including the study of its topological phases through bulk polarization and edge spectra.

Finally, Chapter 8 presents the conclusions of the work and discusses possible extensions to other photonic platforms.

2. Theoretical Framework

In this chapter, the analytical-theoretical tools that allow for the description of the photonic systems studied in this thesis will be developed. These systems consist of the propagation of low-power laser light¹ propagated in coupled systems of waveguides, which are written inside a borosilicate wafer. These experimental conditions allow the behavior of light to be described using Maxwell's equations applied to a linear, isotropic, non-magnetic medium without free charge or current sources.

2.1 Light Propagation in Dielectric Waveguides from Maxwell's Equations

The Maxwell's equations in this regime, in the International System of Units (SI), are:

$$\nabla \cdot [\varepsilon(\mathbf{r})\mathbf{E}] = 0 , \quad (2.1)$$

$$\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t} , \quad (2.2)$$

$$\nabla \cdot \mathbf{H} = 0 , \quad (2.3)$$

$$\nabla \times \mathbf{H} = \varepsilon(\mathbf{r}) \frac{\partial \mathbf{E}}{\partial t} , \quad (2.4)$$

where $\mathbf{E}$ is the electric field and $\mathbf{H}$ is the magnetic field. In general, the waveguides considered are systems that are invariant in the direction of propagation $\hat{\mathbf{z}}$. Therefore, the refractive index profile can be expressed as $n(x, y) = n_0 + \Delta n(x, y)$, where $n_0 = 1.48$ is the refractive index of the borosilicate substrate and $\Delta n(x, y)$ represents the transverse modulation of the index, with maximum values in the range of $10^{-5} - 10^{-3}$. This spatial variation of the index, $\Delta n(x, y)$, is responsible for the optical confinement in the transverse plane (x, y) , allowing for the guided propagation of electromagnetic modes along the $\hat{\mathbf{z}}$ axis.

After assuming a harmonic time dependence proportional to $e^{-i\omega t}$ Faraday-Lenz's law (2.2) can be substituted into Ampère-Maxwell's law (2.4), yielding:

$$\nabla \times \left(\frac{\nabla \times \mathbf{E}}{i\omega\mu_0} \right) = -i\omega\varepsilon_0 n^2 \mathbf{E} \implies \nabla \times \nabla \times \mathbf{E} = k_0^2 n^2 \mathbf{E} , \quad (2.5)$$

where $k_0 \equiv \omega/c$ is the vacuum wavenumber. By vector calculus identity, it holds that: $\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$. Gauss' law implies that (2.1) implica que $\nabla \cdot \mathbf{E} = -\frac{\nabla n^2}{n^2} \cdot \mathbf{E}$.

¹1 mW output power, within the safety regime of Class 3b lasers.

Substituting into equation (2.5), we obtain the following:

$$(\nabla^2 + k_0^2 n^2) \mathbf{E} = -\nabla \left(\frac{\nabla n^2}{n^2} \cdot \mathbf{E} \right) . \quad (2.6)$$

Similarly, it is possible to combine Ampère-Maxwell's (2.4) and Faraday-Lenz's (2.2) equations together with the null divergence of the magnetic field $\mathbf{H}$ (2.3):

$$\begin{aligned} \nabla \times \left(\frac{\nabla \times \mathbf{H}}{-i\omega\epsilon_0 n^2} \right) &= i\omega\mu_0 \mathbf{H} \implies \nabla \times \left(\frac{1}{n^2} \nabla \times \mathbf{H} \right) = k_0^2 \mathbf{H} , \\ \nabla \times \nabla \times \mathbf{H} + \nabla n^2 \times (\nabla \times \mathbf{H}) &= k_0^2 n^2 \mathbf{H} . \end{aligned}$$

The analogous equation to (2.6) for $\mathbf{H}$ is therefore:

$$(\nabla^2 + k_0^2 n^2) \mathbf{H} = i\omega\epsilon_0 \nabla n^2 \times \mathbf{E} . \quad (2.7)$$

In this thesis, we will work with systems in which light travels through a waveguide structure that is invariant in the $\hat{\mathbf{z}}$ direction and periodic in the transverse direction. It is therefore useful to separate the longitudinal and transverse components of the fields, assuming a plane-wave dependence of the form $e^{ik_z z}$ in the spatial variable z :

$$\nabla_{\perp} \times \mathbf{E} + ik_z \hat{\mathbf{z}} \times \mathbf{E} = i\omega\mu_0 \mathbf{H} , \quad (2.8)$$

$$\nabla_{\perp} \times \mathbf{H} + ik_z \hat{\mathbf{z}} \times \mathbf{H} = -i\omega\epsilon_0 n^2 \mathbf{E} . \quad (2.9)$$

If we consider the decompositions $\mathbf{E} = \mathbf{E}_{\perp} + \hat{\mathbf{z}}E_z$ and $\mathbf{H} = \mathbf{H}_{\perp} + \hat{\mathbf{z}}H_z$, $\nabla_{\perp} \equiv -\hat{\mathbf{z}} \times (\hat{\mathbf{z}} \times \nabla)$, the Maxwell's equations involving curls can be written as:

$$\nabla_{\perp} \times \mathbf{E}_{\perp} + ik_z \hat{\mathbf{z}} \times \mathbf{E}_{\perp} + \nabla_{\perp} \times (\hat{\mathbf{z}}E_z) = i\omega\mu_0 (\mathbf{H}_{\perp} + \hat{\mathbf{z}}H_z) , \quad (2.10)$$

$$\nabla_{\perp} \times \mathbf{H}_{\perp} + ik_z \hat{\mathbf{z}} \times \mathbf{H}_{\perp} + \nabla_{\perp} \times (\hat{\mathbf{z}}H_z) = -i\omega\epsilon_0 n^2 (\mathbf{E}_{\perp} + \hat{\mathbf{z}}E_z) . \quad (2.11)$$

After applying the cross product in the $\hat{\mathbf{z}}$ direction to equations (2.10) and (2.11), E_z y H_z can be expressed in terms of $\mathbf{E}_{\perp}$ and $\mathbf{H}_{\perp}$:

$$\begin{aligned} -i\nabla_{\perp} E_z &= k_z \mathbf{E}_{\perp} + \omega\mu_0 \hat{\mathbf{z}} \times \mathbf{H}_{\perp} , & -i\nabla_{\perp} H_z &= k_z \mathbf{H}_{\perp} - \omega\epsilon_0 n^2 \hat{\mathbf{z}} \times \mathbf{E}_{\perp} , \\ i\hat{\mathbf{z}} \times \nabla_{\perp} H_z &= k_z \hat{\mathbf{z}} \times \mathbf{H}_{\perp} + \omega\epsilon_0 n^2 \mathbf{E}_{\perp} , & i\hat{\mathbf{z}} \times \nabla_{\perp} E_z &= \omega\mu_0 \mathbf{H}_{\perp} + k_z \hat{\mathbf{z}} \times \mathbf{E}_{\perp} . \end{aligned}$$

Finally, the perpendicular components of the fields, $\mathbf{H}_{\perp}$ y $\mathbf{E}_{\perp}$, can be expressed in terms of

the longitudinal components, H_z y E_z :

$$\begin{aligned}\hat{\mathbf{z}} \times \mathbf{H}_\perp &= i \frac{(\omega \varepsilon_0 n^2 \nabla_\perp E_z - \hat{\mathbf{z}} \times \nabla_\perp H_z k_z)}{k_0^2 n^2 - k_z^2}, & \hat{\mathbf{z}} \times \mathbf{E}_\perp &= -i \frac{(k_z \hat{\mathbf{z}} \times \nabla_\perp E_z + \omega \mu_0 \nabla_\perp H_z)}{k_0^2 n^2 - k_z^2}, \\ \mathbf{H}_\perp &= \frac{i}{k_0^2 n^2 - k_z^2} (k_z \nabla_\perp H_z + \omega \varepsilon_0 n^2 \hat{\mathbf{z}} \times \nabla_\perp E_z), & \mathbf{E}_\perp &= \frac{i}{k_0^2 n^2 - k_z^2} (k_z \nabla_\perp E_z - \omega \mu_0 \hat{\mathbf{z}} \times \nabla_\perp H_z).\end{aligned}\tag{2.12}$$

Equations (2.6), (2.7), and (2.12) will serve as the analytical tools for the two case studies in the following sections.

2.2 Analytical solutions for one-dimensional slab-type waveguides

The simplest system that can be studied is a slab waveguide, whose analytical form for the refractive index $n(x)$ is as follows:

$$n(x) = \begin{cases} n_1, & |x| \leq a \\ n_0, & |x| > a \end{cases},$$

with $n_1 > n_0$.

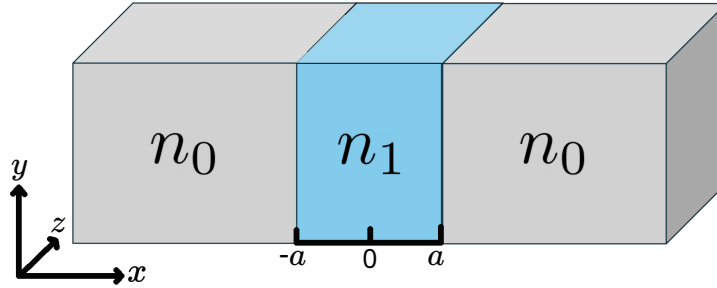


Figure 2.1: Shape of a slab-type waveguide. The structure is invariant in the $\hat{\mathbf{y}}$ (vertical) and $\hat{\mathbf{z}}$ (into the page) directions.

Since $\nabla n^2 = 0$ para $|x| \neq a$, the right-hand sides of equations (2.6) and (2.7) are of the Helmholtz type. Defining $\Psi = \{E_z, H_z\}$:

$$(\nabla_\perp^2 + k_0^2 n^2 - k_z^2) \Psi = \frac{d^2 \Psi}{dx^2} + (k_0^2 n^2 - k_z^2) \Psi = 0.$$

Since $n(x) = n(-x)$, the solutions Ψ must be even or odd. Indeed, if $\Psi(x)$ is a solution, the change $x \rightarrow x' = -x$ implies that $\Psi(-x) = \pm \Psi(x)$, since $\Psi(x)$ is a real field. To find solutions whose energy is localized within the waveguide and decays outside it, we impose

$k_0^2 n_0^2 \leq k_z^2 \leq k_0^2 n_1^2$. It becomes natural to define $\alpha^2 \equiv k_0^2 n_1^2 - k_z^2$ y $\beta^2 \equiv k_z^2 - k_0^2 n_0^2$. With all this,

$$\Psi_s = \begin{cases} \Psi_{s1} \cos(\alpha x) , & |x| \leq a \\ \Psi_{s0} e^{-\beta|x|} , & |x| > a \end{cases} \implies \nabla_{\perp} \Psi_s = \begin{cases} -\hat{x} \alpha \Psi_{s1} \sin(\alpha x) , & |x| \leq a \\ -\hat{x} \frac{|x|}{x} \beta \Psi_{s0} e^{-\beta|x|} , & |x| > a \end{cases} .$$

Thus, the even vertical components E_y and H_y are expressed, due to equation (2.12) as:

$$E_y = \frac{i\omega\mu_0}{k_0^2 n^2 - k_z^2} \begin{cases} \alpha H_{s1} \sin(\alpha x) , & |x| \leq a \\ \frac{|x|}{x} \beta H_{s0} e^{-\beta|x|} , & |x| > a \end{cases} , \quad H_y = \frac{-i\omega\varepsilon_0 n^2}{k_0^2 n^2 - k_z^2} \begin{cases} \alpha E_{s1} \sin(\alpha x) , & |x| \leq a \\ \frac{|x|}{x} \beta E_{s0} e^{-\beta|x|} , & |x| > a \end{cases} .$$

On the other hand, the odd solutions have the form:

$$\Psi_a = \begin{cases} \Psi_{a1} \sin(\alpha x) , & |x| \leq a \\ \Psi_{a0} e^{-\beta|x|} , & |x| > a \end{cases} \implies \nabla_{\perp} \Psi_a = \begin{cases} \hat{x} \alpha \Psi_{a1} \cos(\alpha x) , & |x| \leq a \\ -\hat{x} \frac{|x|}{x} \beta \Psi_{a0} e^{-\beta|x|} , & |x| > a \end{cases} .$$

Therefore, Ey E_y y H_y are expressed as:

$$E_y = \frac{i\omega\mu_0}{k_0^2 n^2 - k_z^2} \begin{cases} -\alpha H_{a1} \cos(\alpha x) , & |x| \leq a \\ \frac{|x|}{x} \beta H_{a0} e^{-\beta|x|} , & |x| > a \end{cases} , \quad H_y = \frac{i\omega\varepsilon_0 n^2}{k_0^2 n^2 - k_z^2} \begin{cases} \alpha E_{a1} \cos(\alpha x) , & |x| \leq a \\ -\frac{|x|}{x} \beta E_{a0} e^{-\beta|x|} , & |x| > a \end{cases} .$$

By imposing continuity of the tangential components E_y , E_z , H_y y H_z :

$$\begin{aligned} E_{s1} \cos(\alpha a) &= E_{s0} e^{-\beta a} , & E_{a1} \sin(\alpha a) &= E_{a0} e^{-\beta a} , \\ H_{s1} \cos(\alpha a) &= H_{s0} e^{-\beta a} , & H_{a1} \sin(\alpha a) &= H_{a0} e^{-\beta a} , \\ n_1^2 E_{s1} \sin(\alpha a) / \alpha &= -n_0^2 E_{s0} e^{-\beta a} / \beta , & n_1^2 E_{a1} \cos(\alpha a) / \alpha &= n_0^2 E_{a0} e^{-\beta a} / \beta , \\ H_{s1} \sin(\alpha a) / \alpha &= -H_{s0} e^{-\beta a} / \beta , & H_{a1} \cos(\alpha a) / \alpha &= H_{a0} e^{-\beta a} / \beta . \end{aligned}$$

Seeking non-trivial solutions yields:

$$\left[\frac{\cos(\alpha a)}{\beta a} + \frac{\sin(\alpha a)}{\alpha a} \right] \left[n_0^2 \frac{\cos(\alpha a)}{\beta a} + n_1^2 \frac{\sin(\alpha a)}{\alpha a} \right] \left[\frac{\sin(\alpha a)}{\beta a} - \frac{\cos(\alpha a)}{\alpha a} \right] \left[n_0^2 \frac{\sin(\alpha a)}{\beta a} - n_1^2 \frac{\cos(\alpha a)}{\alpha a} \right] = 0$$

Two types of conditions will be distinguished:

- Transverse Electric (TE) Modes:

$$\frac{\cos(\alpha a)}{\beta a} + \frac{\sin(\alpha a)}{\alpha a} = 0 , \quad (2.13)$$

$$\frac{\sin(\alpha a)}{\beta a} - \frac{\cos(\alpha a)}{\alpha a} = 0 . \quad (2.14)$$

- Transverse Magnetic (TM) Modes:

$$n_0^2 \frac{\cos(\alpha a)}{\beta a} + n_1^2 \frac{\sin(\alpha a)}{\alpha a} = 0 , \quad (2.15)$$

$$n_0^2 \frac{\sin(\alpha a)}{\beta a} - n_1^2 \frac{\cos(\alpha a)}{\alpha a} = 0 . \quad (2.16)$$

Assuming α, β y k_z are known, the amplitudes must satisfy the relations:

$$\begin{aligned} E_{s1} \left[n_0^2 \frac{\cos(\alpha a)}{\beta a} + n_1^2 \frac{\sin(\alpha a)}{\alpha a} \right] &= 0 , & E_{a1} \left[n_0^2 \frac{\sin(\alpha a)}{\beta a} - n_1^2 \frac{\cos(\alpha a)}{\alpha a} \right] &= 0 , \\ H_{s1} \left[\frac{\cos(\alpha a)}{\beta a} + \frac{\sin(\alpha a)}{\alpha a} \right] &= 0 , & H_{a1} \left[\frac{\sin(\alpha a)}{\beta a} - \frac{\cos(\alpha a)}{\alpha a} \right] &= 0 . \end{aligned}$$

The equations above impose that $E_1 = 0$ when any TE mode condition is satisfied. Similarly, the equations below impose $H_1 = 0$ in the case of TM modes, which enforces transversality in the oscillation of the respective fields.

A corollary for TE modes is that $E_z = E_x = 0$, meaning the electric field polarization will be exclusively in the $\hat{y}$ direction. Likewise, only a beam polarized in $\hat{x}$ could excite a TM mode. Therefore, this must be taken into account in an experiment: an arbitrary initial condition will propagate as a linear combination of the TE and TM modes supported by the waveguide.

2.2.1 Graphical Solutions and Comparison Between TE and TM Modes

Using the two TE mode equations along with the constraint $(\alpha a)^2 + (\beta a)^2 = k_0^2 a^2 (n_1^2 - n_0^2) \equiv V^2$ it is possible to obtain graphical solutions for the propagation constants k_z from the intersections $(\alpha a, \beta a)$, as plotted in Figure 2.2. The number of modes increases with the difference between refractive indices.

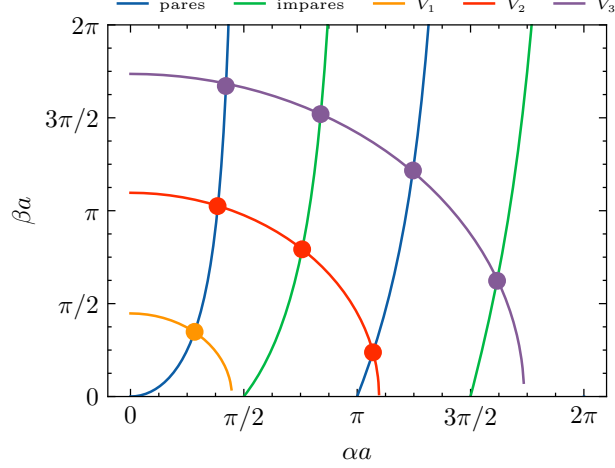


Figure 2.2: Graphical solutions for TE modes. A higher contrast $\Delta n = n_1 - n_0$, results in a greater number of guided modes supported by the waveguide.

After finding k_z , we have everything needed to substitute into the expressions obtained for the components of the electromagnetic field. Figure 2.3 plots the spatial form of four modes, which correspond to the four purple points in Figure 2.2.

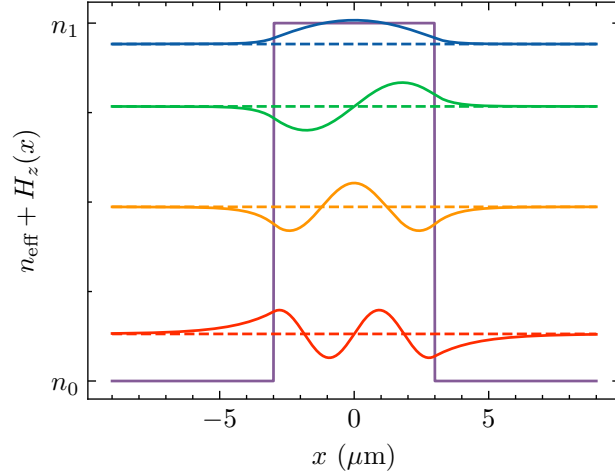


Figure 2.3: Spatial form of the longitudinal component of the magnetic field.

The cutoff condition in waveguides is equivalent to the energy no longer decaying in the region $|x| > a$, meaning that $\beta a \rightarrow 0$ must hold. Symmetric or even modes satisfy this condition when $\sin(V) = 0$, which implies $V = m\pi$, where m is an integer. In particular, considering $m = 0$ and sweeping $n_1 \rightarrow n_0$, there are always at least two modes—one TE and one TM—as long as $n_1 > n_0$. Antisymmetric or odd modes, on the other hand, must satisfy $\cos(V) = 0$, which implies $V = (m\pi \pm \pi/2)$. From this condition, it follows that the first pair of excited modes ($m = 0$) exists whenever $\lambda \leq \lambda_c \equiv 4a \sqrt{n_1^2 - n_0^2}$. Written compactly for the

m -th mode:

$$(\lambda_c)_m = \frac{4a}{m-1} \sqrt{n_1^2 - n_0^2},$$

which for $a = 4 \mu\text{m}$ and $\Delta n = 5 \times 10^{-4}$ implies a cutoff wavelength for the second mode of $\lambda_c \approx 615 \text{ nm}$.

2.3 Analytical solutions for circular optical fiber

In the previous section, the simplest system in which dielectric waveguides can be discussed was studied. The next step in complexity involves circular waveguides. For this purpose, it will be considered that the refractive index varies radially according to:

$$n(\rho) = \begin{cases} n_1, & \text{if } \rho \leq a \\ n_0, & \text{if } \rho > a \end{cases},$$

where the tuple (ρ, ϕ, z) defines the cylindrical coordinates most appropriate for this problem.

When considering the longitudinal components $\Psi = \{E_z, H_z\}$ of the electric and magnetic fields and using the method of separation of variables with $\Psi = R(\rho)\Phi(\phi)e^{ik_z z}$, equations (2.6) and (2.7) take the form:

$$\begin{aligned} \left[\frac{\partial^2}{\partial \rho^2} + \frac{\partial}{\rho \partial \rho} + \frac{\partial^2}{\rho^2 \partial \phi^2} + (k_0^2 n^2 - k_z^2) \right] R(\rho)\Phi(\phi) &= 0 \\ \rho^2 \frac{d^2 R}{R d\rho^2} + \rho \frac{dR}{R d\rho} + \rho^2 (k_0^2 n^2 - k_z^2) + \underbrace{\frac{d^2 \Phi}{\Phi d\phi^2}}_{-\ell^2} &= 0 \\ \therefore \Phi(\phi) &= A e^{i\ell\phi} \end{aligned}$$

After imposing periodicity conditions $\Phi(\phi) = \Phi(\phi + 2\pi)$, it necessarily follows that ℓ is an integer. Consequently, the equation for $R(\rho)$ is of the integer Bessel type. Therefore, seeking solutions such that $k_0^2 n_0^2 < \beta_z^2 < k_0^2 n_1^2$ and defining again $\alpha^2 \equiv k_0^2 n_1^2 - k_z^2$ y $\beta^2 \equiv k_z^2 - k_0^2 n_0^2$ we have:

$$\begin{aligned} \frac{d^2 R}{d\rho^2} + \frac{1}{\rho} \frac{dR}{d\rho} + \left(k_0^2 n^2 - k_z^2 - \frac{\ell^2}{\rho^2} \right) R &= 0 \\ \therefore R(\rho) &= \begin{cases} C_1 J_\ell(\alpha\rho) + D_1 Y_\ell(\alpha\rho), & \text{if } \rho \leq a \\ C_2 K_\ell(\beta\rho) + D_2 I_\ell(\beta\rho), & \text{if } \rho > a \end{cases}. \end{aligned}$$

It is necessary to impose $D_1 = D_2 = 0$ or the solution to be finite at $\rho = 0$ and as $\rho \rightarrow +\infty$. That is, the radial part of the solution is:

$$R(\rho) = \begin{cases} C_1 J_\ell(\alpha\rho), & \text{if } \rho \leq a \\ C_2 K_\ell(\beta\rho), & \text{if } \rho > a \end{cases}.$$

In this case, to impose the continuity conditions on $\mathbf{E}_{\parallel} = E_{\phi}\hat{\phi} + E_z\hat{z}$ and $\mathbf{H}_{\parallel} = H_{\phi}\hat{\phi} + H_z\hat{z}$, it becomes necessary to relate the remaining field components to E_z and H_z , for which equations (2.12) are required.

Since

$$\nabla_{\perp}\Psi = \begin{cases} \Psi_0^1 \left[\hat{\rho}\alpha J'_{\ell}(\alpha\rho) + i\hat{\phi}\ell J_{\ell}(\alpha\rho)/\rho \right] e^{i\ell\phi} e^{ik_z z}, & \text{if } \rho \leq a \\ \Psi_0^0 \left[\hat{\rho}\beta K'_{\ell}(\beta\rho) + i\hat{\phi}\ell K_{\ell}(\alpha\rho)/\rho \right] e^{i\ell\phi} e^{ik_z z}, & \text{if } \rho > a \end{cases}$$

Separating by components and substituting:

$$\begin{aligned} H_z &= e^{i\ell\phi} e^{ik_z z} \begin{cases} H_0^1 J_{\ell}(\alpha\rho), & \text{if } \rho \leq a \\ H_0^0 K_{\ell}(\beta\rho), & \text{if } \rho > a \end{cases} \\ H_r &= \frac{ie^{i\ell\phi} e^{ik_z z}}{k_0^2 n^2 - k_z^2} \begin{cases} k_z \alpha H_0^1 J'_{\ell}(\alpha\rho) - i\omega\varepsilon_0 n^2 \ell E_0^1 J_{\ell}(\alpha\rho)/\rho, & \text{if } \rho \leq a \\ k_z \beta H_0^0 K'_{\ell}(\beta\rho) - i\omega\varepsilon_0 n^2 \ell E_0^0 K_{\ell}(\beta\rho)/\rho, & \text{if } \rho > a \end{cases} \\ H_{\phi} &= \frac{ie^{i\ell\phi} e^{ik_z z}}{k_0^2 n^2 - k_z^2} \begin{cases} ik_z \ell H_0^1 J_{\ell}(\alpha\rho)/\rho + \omega\varepsilon_0 n^2 \alpha E_0^1 J'_{\ell}(\alpha\rho), & \text{if } \rho \leq a \\ ik_z \ell H_0^0 K_{\ell}(\beta\rho)/\rho + \omega\varepsilon_0 n^2 \beta E_0^0 K'_{\ell}(\beta\rho), & \text{if } \rho > a \end{cases} \\ E_z &= e^{i\ell\phi} e^{ik_z z} \begin{cases} E_0^1 J_{\ell}(\alpha\rho), & \text{if } \rho \leq a \\ E_0^0 K_{\ell}(\beta\rho), & \text{if } \rho > a \end{cases} \\ E_r &= \frac{ie^{i\ell\phi} e^{ik_z z}}{k_0^2 n^2 - k_z^2} \begin{cases} k_z \alpha E_0^1 J'_{\ell}(\alpha\rho) + i\omega\mu_0 \ell H_0^1 J_{\ell}(\alpha\rho)/\rho, & \text{if } \rho \leq a \\ k_z \beta E_0^0 K'_{\ell}(\beta\rho) + i\omega\mu_0 \ell H_0^0 K_{\ell}(\beta\rho)/\rho, & \text{if } \rho > a \end{cases} \\ E_{\phi} &= \frac{ie^{i\ell\phi} e^{ik_z z}}{k_0^2 n^2 - k_z^2} \begin{cases} ik_z \ell E_0^1 J_{\ell}(\alpha\rho)/\rho - \omega\mu_0 \alpha H_0^1 J'_{\ell}(\alpha\rho), & \text{if } \rho \leq a \\ ik_z \ell E_0^0 K_{\ell}(\beta\rho)/\rho - \omega\mu_0 \beta H_0^0 K'_{\ell}(\beta\rho), & \text{if } \rho > a \end{cases} \end{aligned}$$

Then, imposing continuity in z and ϕ :

$$H_0^1 J_{\ell}(\alpha a) = H_0^0 K_{\ell}(\beta a) \quad (2.17)$$

$$E_0^1 J_{\ell}(\alpha a) = E_0^0 K_{\ell}(\beta a) \quad (2.18)$$

$$-\omega\varepsilon_0 n_1^2 \alpha \beta^2 a E_0^1 J'_{\ell}(\alpha a) - ik_z \ell \beta^2 H_0^1 J_{\ell}(\alpha a) = \omega\varepsilon_0 n_0^2 \alpha^2 \beta a E_0^0 K'_{\ell}(\beta a) + ik_z \ell \alpha^2 H_0^0 K_{\ell}(\beta a) \quad (2.19)$$

$$-ik_z \ell \beta^2 E_0^1 J_{\ell}(\alpha a) + \omega\mu_0 \alpha \beta^2 a H_0^1 J'_{\ell}(\alpha a) = ik_z \ell \alpha^2 E_0^0 K_{\ell}(\beta a) - \omega\mu_0 \alpha^2 \beta a H_0^0 K'_{\ell}(\beta a) \quad (2.20)$$

Seeking non-trivial solutions:

$$\begin{vmatrix} K_{\ell}(\beta a) & -J_{\ell}(\alpha a) & 0 & 0 \\ 0 & 0 & K_{\ell}(\beta a) & -J_{\ell}(\alpha a) \\ ik_z \ell \alpha^2 K_{\ell}(\beta a) & ik_z \ell \beta^2 J_{\ell}(\alpha a) & \omega\varepsilon_0 n_0^2 \alpha^2 \beta a K'_{\ell}(\beta a) & \omega\varepsilon_0 n_1^2 \alpha \beta^2 a J'_{\ell}(\alpha a) \\ \omega\mu_0 \alpha^2 \beta a K'_{\ell}(\beta a) & \omega\mu_0 \alpha \beta^2 a J'_{\ell}(\alpha a) & -ik_z \ell \alpha^2 K_{\ell}(\beta a) & -ik_z \ell \beta^2 J_{\ell}(\alpha a) \end{vmatrix} = 0$$

Finally, the transcendental equation satisfied by α , β and k_z is:

$$\left(\frac{J'_\ell(\alpha a)}{\alpha a J_\ell(\alpha a)} + \frac{K'_\ell(\beta a)}{\beta a K_\ell(\beta a)} \right) \left(n_1^2 \frac{J'_\ell(\alpha a)}{\alpha a J_\ell(\alpha a)} + n_0^2 \frac{K'_\ell(\beta a)}{\beta a K_\ell(\beta a)} \right) = \ell^2 \left[\left(\frac{1}{\alpha a} \right)^2 + \left(\frac{1}{\beta a} \right)^2 \right] \left(\frac{k_z}{k_0} \right)^2. \quad (2.21)$$

Since, in principle, the values of k_z are already determined by the equation above, it is possible to obtain two relations between H_0^1 and E_0^1 :

$$\frac{E_0^1}{H_0^1} = -\frac{ik_z \ell}{\omega \epsilon_0} \left[\left(\frac{1}{\alpha a} \right)^2 + \left(\frac{1}{\beta a} \right)^2 \right] \left[n_1^2 \frac{J'_\ell(\alpha a)}{\alpha a J_\ell(\alpha a)} + n_0^2 \frac{K'_\ell(\beta a)}{\beta a K_\ell(\beta a)} \right]^{-1}, \quad (2.22)$$

$$\frac{H_0^1}{E_0^1} = \frac{ik_z \ell}{\omega \mu_0} \left[\left(\frac{1}{\alpha a} \right)^2 + \left(\frac{1}{\beta a} \right)^2 \right] \left[\frac{J'_\ell(\alpha a)}{\alpha a J_\ell(\alpha a)} + \frac{K'_\ell(\beta a)}{\beta a K_\ell(\beta a)} \right]^{-1}. \quad (2.23)$$

Taking the square root of the quotient of the equations: (2.22) and (2.23) we have:

$$\frac{E_0^1}{H_0^1} = i \sqrt{\frac{\mu_0}{\epsilon_0}} \frac{\sqrt{\frac{J'_\ell(\alpha a)}{\alpha a J_\ell(\alpha a)} + \frac{K'_\ell(\beta a)}{\beta a K_\ell(\beta a)}}}{\sqrt{n_1^2 \frac{J'_\ell(\alpha a)}{\alpha a J_\ell(\alpha a)} + n_0^2 \frac{K'_\ell(\beta a)}{\beta a K_\ell(\beta a)}}}. \quad (2.24)$$

2.3.1 TE and TM modes

The simplest case to study is by imposing $\ell = 0$. Equation (2.21) implies:

$$\begin{aligned} \frac{J'_0(\alpha a)}{\alpha a J_0(\alpha a)} + \frac{K'_0(\beta a)}{\beta a K_0(\beta a)} &= 0, \quad (\text{TE modes}) \\ n_1^2 \frac{J'_0(\alpha a)}{\alpha a J_0(\alpha a)} + n_0^2 \frac{K'_0(\beta a)}{\beta a K_0(\beta a)} &= 0, \quad (\text{TM modes}) \end{aligned}$$

from equation (2.24), it is straightforward to note that the conditions for transverse modes imply that the longitudinal components become zero in the respective cases: $H_0^1 = 0$ for TE and $E_0^1 = 0$ for TM.

The cutoff conditions occur when $\beta a \rightarrow 0$. Using the asymptotic expressions for Bessel functions with small arguments and their recurrence relations, we have:

$$\frac{\alpha a J_0(\alpha a)}{J_1(\alpha a)} = -\frac{\beta a K_0(\beta a)}{K_1(\beta a)} \approx (\beta a)^2 \ln \left(\frac{\beta a e^\gamma}{2} \right) \rightarrow 0,$$

thus it becomes necessary that $J_0(\alpha a)|_{\alpha a \rightarrow \nu} = 0$. Denoting $x_{0,m} = 2.405, 5.520, 8.654, \dots$ as the m -th zero of the function $J_0(x)$, the cutoff wavelength is given by:

$$(\lambda_c)_{0,m} = 2\pi a \frac{\sqrt{n_1^2 - n_0^2}}{x_{0,m}}, \quad (\text{TE and TM modes}), \quad (2.25)$$

which translates to a cutoff wavelength of: $\lambda_c \approx 616$ nm for the same parameters as in the

section 2.2, $a = 4 \mu\text{m}$ y $\Delta n = 5 \times 10^{-4}$. Contrary to the case of slab waveguides, in optical fibers it is necessary to be below a non-zero maximum cutoff threshold $(\lambda_c)_{0,1}$ for TE or TM modes to exist. As will be seen later, this is because purely transverse modes correspond to excited modes: the fundamental mode is hybrid.

2.3.2 HE and EH modes

Interpreting (2.21) as a quadratic equation in $J'_\ell(\alpha a)/\alpha a J_\ell(\alpha a)$:

$$\frac{J'_\ell(\alpha a)}{\alpha a J_\ell(\alpha a)} = -\left(\frac{n_1^2 + n_0^2}{2n_1^2}\right) \frac{K'_\ell(\beta a)}{\beta a K_\ell(\beta a)} \pm \sqrt{\left(\frac{n_1^2 - n_0^2}{2n_1^2}\right)^2 \left(\frac{K'_\ell(\beta a)}{\beta a K_\ell(\beta a)}\right)^2 + \left(\frac{k_z \ell}{k_0 n_1}\right)^2 \left[\left(\frac{1}{\alpha a}\right)^2 + \left(\frac{1}{\beta a}\right)^2\right]^2}$$

If the recurrence relations of the Bessel functions $J_\ell(r)$, are used, it is possible to obtain two types of solutions that are usually called HE and EH due to the longitudinal field with greater relative value:

$$\frac{J_{\ell-1}(\alpha a)}{\alpha a J_\ell(\alpha a)} = -\left(\frac{n_1^2 + n_0^2}{2n_1^2}\right) \frac{K'_\ell(\beta a)}{\beta a K_\ell(\beta a)} + \frac{\ell}{(\alpha a)^2} - \sqrt{\Delta}, \quad (\text{HE modes}) \quad (2.26)$$

$$\frac{J_{\ell+1}(\alpha a)}{\alpha a J_\ell(\alpha a)} = \left(\frac{n_1^2 + n_0^2}{2n_1^2}\right) \frac{K'_\ell(\beta a)}{\beta a K_\ell(\beta a)} + \frac{\ell}{(\alpha a)^2} - \sqrt{\Delta}, \quad (\text{EH modes}) \quad (2.27)$$

$$\Delta = \left(\frac{n_1^2 - n_0^2}{2n_1^2}\right)^2 \left(\frac{K'_\ell(\beta a)}{\beta a K_\ell(\beta a)}\right)^2 + \left(\frac{k_z \ell}{k_0 n_1}\right)^2 \left[\left(\frac{1}{\alpha a}\right)^2 + \left(\frac{1}{\beta a}\right)^2\right]^2.$$

For $\ell = 0$, the relations obtained for the TE_m (HE_{0m}) and TM_m (EH_{0m}). For $\ell \neq 0$ it becomes necessary to consider the asymptotic expressions of the Bessel functions when $\beta a \rightarrow 0$ and $\alpha a \rightarrow V$:

$$\frac{K'_\ell(x)}{x K_\ell(x)} = -\frac{K_{\ell-1}(x)}{x K_\ell(x)} - \frac{\ell}{x^2} \approx \begin{cases} \ln(xe^\gamma/2) - \frac{1}{x^2} & \text{si } \ell = 1 \\ -\frac{1}{2(\ell-1)} - \frac{\ell}{x^2} & \text{si } \ell > 1 \end{cases}.$$

The case $\ell = 1$ is of special interest. The condition for HE modes is developed as:

$$\frac{\alpha a J_1(\alpha a)}{J_0(\alpha a)} = \left\{ \left(\frac{n_1^2 + n_0^2}{2n_1^2}\right) \left[\ln\left(\frac{2}{e^\gamma \beta a}\right) + \frac{1}{(\beta a)^2} \right] + \frac{1}{(\alpha a)^2} - \sqrt{\Delta} \right\}^{-1} \rightarrow 0.$$

It is sufficient to consider the function $J_1(\alpha a)|_{\alpha a \rightarrow V} = 0$. On the other hand, the EH condition eliminates the zero $x = 0$ from the ratio $\frac{\alpha a J_1(\alpha a)}{J_2(\alpha a)}$ which causes it to be "out of phase" relative

to the HE condition. Using reasoning similar to the $\ell = 0$, case, the cutoff conditions are:

$$(\lambda_c)_{1,m} = 2\pi a \frac{\sqrt{n_1^2 - n_0^2}}{x_{1,m}}, \quad (\text{HE modes}_{1m})$$

$$(\lambda_c)_{1,m} = 2\pi a \frac{\sqrt{n_1^2 - n_0^2}}{x_{1,m+1}}, \quad (\text{EH modes}_{1m})$$

with $x_{1,m} = 0, 3.832, 7.016, 10.173, \dots$. In particular, the HE_{11} mode always exists. The next largest finite cutoff wavelengths are $\lambda_c \approx 252$ nm for the HE_{12} and EH_{11} modes.

The case $\ell \geq 2$ has a different development due to the asymptotic forms involved and will not be relevant for this thesis. However, they are included in Figure 2.4 for completeness.

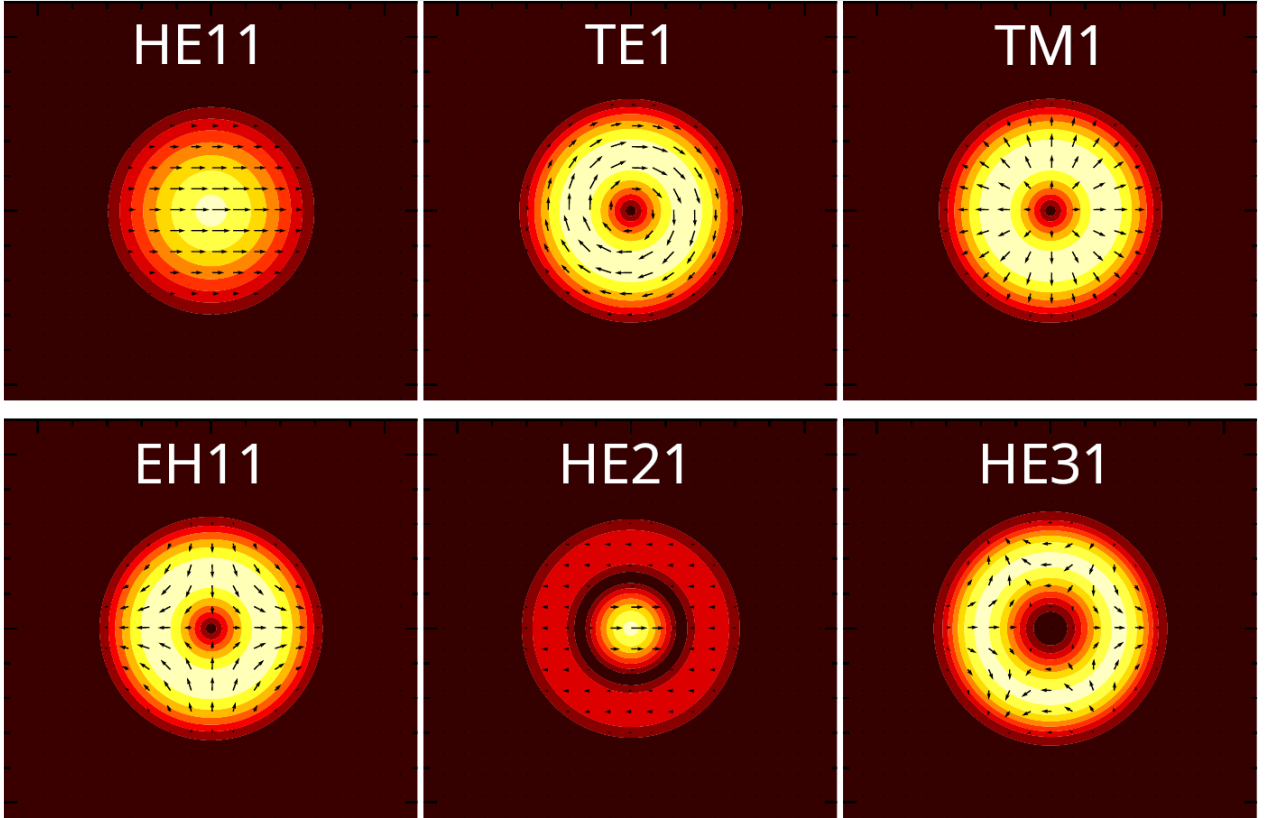


Figure 2.4: First guided normal modes for a circular waveguide or optical fiber.

2.4 Normal modes in waveguides

If the waveguide structure does not vary in the $\hat{z}$ direction, the solution for the electric field can be written, by separation of variables, as a plane wave of the form $\mathbf{E}(\mathbf{r}) = \mathbf{E}_v(x, y)e^{i\beta_v z}$. In turn, it is convenient to separate the Laplacian operator as $\nabla^2 = \nabla_{\perp}^2 + \frac{\partial^2}{\partial z^2}$.

Defining the operator D_{ij} such that $D_{ij}E_j \equiv \partial_i \left(\frac{\partial_j n^2}{n^2} E_j \right)$, the transverse part of the Helmholtz

equations (2.6) can be written as:

$$\begin{pmatrix} \nabla_{\perp}^2 + k_0^2 n^2 + D_{xx} & D_{xy} \\ D_{yx} & \nabla_{\perp}^2 + k_0^2 n^2 + D_{yy} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix} = \beta_v^2 \begin{pmatrix} E_x \\ E_y \end{pmatrix}. \quad (2.28)$$

Equation (2.28) constitutes an eigenvalue problem for β_v^2 , whose eigenfunctions $\mathbf{E}_v^{\perp}(x, y)$ describe the transverse profiles of the guided modes. In general, this is a coupled vectorial second-order system, whose resolution requires numerical techniques. However, in the *weak guidance* regime, where the refractive index varies slightly relative to a homogeneous background n_0 , the system can be decoupled and treated as a scalar problem for a dominant component of the electric field².

Under this approximation, quasi-transverse modes are typically considered, where the electric field is primarily polarized in a fixed direction (for example, E_y), and the Helmholtz equation reduces to:

$$\left[\nabla_{\perp}^2 + k_0^2 n^2(x, y) \right] \phi_v(x, y) = \beta_v^2 \phi_v(x, y), \quad (2.29)$$

where $\phi_v(x, y)$ represents the transverse field profile. This approximation is valid for many experimental geometries, especially when the light is linearly polarized and the index variations do not induce coupling between orthogonal polarizations.

The modes obtained from this equation allow for the calculation of propagation constants β_v , spatial field profiles $\phi_v(x, y)$, and other relevant quantities such as confinement factors or group velocities. Furthermore, these modes serve as a basis for subsequent numerical methods, such as Eigenmode Expansion (EME) or Coupled-Mode Theory (CMT), used to describe the dynamics in photonic lattices.

2.5 Coupled-Mode Theory

2.5.1 Derivation from a Variational Principle

From equations (2.8) and (2.9), it is possible to express the propagation constant k_z in terms of the Poynting vector $\mathbf{S} = \frac{1}{2} \text{Re}\{\mathbf{E} \times \mathbf{H}^*\}$, following a derivation analogous to that presented in [32].

$$k_z = \frac{\frac{1}{4i} \iint \left[(-\nabla_{\perp} \times \mathbf{E} + i\omega\mu_0\mathbf{H}) \cdot \mathbf{H}^* + (\nabla_{\perp} \times \mathbf{H} + i\omega\epsilon_0 n^2 \mathbf{E}) \cdot \mathbf{E}^* \right] dx dy}{\frac{1}{4} \iint (\mathbf{E} \times \mathbf{H}^* + \mathbf{E}^* \times \mathbf{H}) \cdot \hat{\mathbf{z}} dx dy}. \quad (2.30)$$

²see Appendix B.

When proposing a solution in the form of a superposition of modes from the individual waveguides, such as:

$$\mathbf{E} = \sum_i a_i \mathbf{e}_i, \quad \mathbf{H} = \sum_i a_i \mathbf{h}_i,$$

where a_i represents the amplitude of the i -th mode, the coefficients remain undetermined but subject to minimizing the expression for k_z . Substituting these expansions into (2.30), a simplified expression for the propagation constant of the system k_z is obtained:

$$(\mathbf{E} \times \mathbf{H}^* + \mathbf{E}^* \times \mathbf{H}) \cdot \hat{\mathbf{z}} = \sum_{ij} a_i^* (\mathbf{e}_j \times \mathbf{h}_i^* + \mathbf{e}_i^* \times \mathbf{h}_j) \cdot \hat{\mathbf{z}} a_j \equiv \sum_{ij} a_i^* p_{ij} a_j.$$

The integrand of the numerator is expanded as:

$$\begin{aligned} \text{integrand of the numerator} &= i \sum_{ij} \left[a_i^* k_z^j (\hat{\mathbf{z}} \times \mathbf{e}_j) \cdot \mathbf{h}_i^* a_j + a_j (\omega \epsilon_0 (n^2 - n_j^2) \mathbf{e}_j - k_z^j \hat{\mathbf{z}} \times \mathbf{h}_j) \cdot \mathbf{e}_i^* a_i^* \right], \\ &= i \sum_{ij} a_i^* \left[k_z^j p_{ij} + \omega \epsilon_0 (n^2 - n_j^2) \mathbf{e}_j \cdot \mathbf{e}_i^* \right] a_j, \end{aligned}$$

where k_z^j is the propagation constant of the j -th isolated mode³.

Defining the following fundamental quantities:

$$P_{ij} \equiv \frac{1}{4} \iint (\mathbf{e}_j \times \mathbf{h}_i^* + \mathbf{e}_i^* \times \mathbf{h}_j) \cdot \hat{\mathbf{z}}, dx dy, \quad (2.31)$$

$$= \frac{1}{4\omega\mu_0} \iint \left[(k_z^i + k_z^j) (\mathbf{e}_i^* \cdot \mathbf{e}_j - e_{z,i}^* e_{z,j}) + i (\mathbf{e}_i^* \cdot \nabla_{\perp} e_{z,j} + \mathbf{e}_j \cdot \nabla_{\perp} e_{z,i}^*) \right] dx dy, \quad (2.32)$$

$$H_{ij} \equiv P_{ij} k_z^j + \frac{\omega \epsilon_0}{4} \iint (n^2 - n_j^2) \mathbf{e}_j \cdot \mathbf{e}_i^* dx dy, \quad (2.33)$$

the expression (2.30) for the propagation constant k_z can be written compactly as a Rayleigh-Ritz quotient:

$$k_z = \frac{\sum_{ij} a_i^* H_{ij} a_j}{\sum_{ij} a_i^* P_{ij} a_j}. \quad (2.34)$$

Differentiating with respect to a_k^* to optimize the value of k_z yields:

$$\frac{\partial k_z}{\partial a_k^*} = \frac{\sum_{ij} H_{ij} a_j \delta_{ik}}{\sum_{ij} a_i^* P_{ij} a_j} - \frac{(\sum_{ij} a_i^* H_{ij} a_j)(\sum_{ij} \delta_{ik} P_{ij} a_j)}{(\sum_{ij} a_i^* P_{ij} a_j)^2} = \frac{\sum_j (H_{jk} - k_z P_{kj}) a_j}{\sum_{ij} a_i^* P_{ij} a_j} \stackrel{!}{=} 0. \quad (2.35)$$

³in general, $k_z \neq k_z^j$.

Recovering $k_z \rightarrow -i\frac{\partial}{\partial z}$, the equations for N non-orthogonal coupled modes are obtained:

$$-i \sum_j P_{kj} \frac{da_j}{dz} = \sum_j H_{kj} a_j, \text{ with } k = 1, 2, \dots, N. \quad (2.36)$$

Clearly, from the expression (2.31) for P_{ij} , Hermiticity is satisfied. At first glance, it may seem that H_{ij} is not Hermitian, but let us see that it indeed is. To do this, it suffices to consider the difference $H_{ij} - H_{ji}^*$ and note that it vanishes:

$$\begin{aligned} H_{ij} - H_{ji}^* &= P_{ij} k_z^j - P_{ji}^* k_z^i + \frac{\omega \epsilon_0}{4} \iint (n_i^2 - n_j^2) \mathbf{e}_i^* \cdot \mathbf{e}_j dx dy \\ &= \frac{1}{4} \iint (k_z^j - k_z^i) (\mathbf{e}_j \times \mathbf{h}_i^* + \mathbf{e}_i^* \times \mathbf{h}_j) \cdot \hat{\mathbf{z}} + \omega \epsilon_0 (n_i^2 - n_j^2) \mathbf{e}_j \cdot \mathbf{e}_i^* dx dy \\ &= \frac{1}{4} \iint (i \nabla_\perp \times \mathbf{e}_j + \omega \mu_0 \mathbf{h}_j) \cdot \mathbf{h}_i^* - (i \nabla_\perp \times \mathbf{h}_j - \omega \epsilon_0 n_j^2 \mathbf{e}_j) \cdot \mathbf{e}_i^* dx dy \\ &\quad - \frac{1}{4} \iint (i \nabla_\perp \times \mathbf{e}_i^* + \omega \mu_0 \mathbf{h}_i^*) \cdot \mathbf{h}_j + (i \nabla_\perp \times \mathbf{h}_i^* + \omega \epsilon_0 n_i^2 \mathbf{e}_i^*) \cdot \mathbf{e}_j dx dy \\ &\quad + \frac{\omega \epsilon_0}{4} \iint (n_i^2 - n_j^2) \mathbf{e}_j \cdot \mathbf{e}_i^* dx dy \\ &= \frac{i}{4} \iint (\nabla_\perp \times \mathbf{e}_j) \cdot \mathbf{h}_i^* - (\nabla_\perp \times \mathbf{h}_j) \cdot \mathbf{e}_i^* + (\nabla_\perp \times \mathbf{e}_i^*) \cdot \mathbf{h}_j - (\nabla_\perp \times \mathbf{h}_i^*) \cdot \mathbf{e}_j dx dy \\ &= \frac{i}{4} \iint \nabla_\perp \cdot (\mathbf{e}_j \times \mathbf{h}_i^* + \mathbf{e}_i^* \times \mathbf{h}_j) dx dy = \frac{i}{4} \oint_C (\mathbf{e}_j \times \mathbf{h}_i^* + \mathbf{e}_i^* \times \mathbf{h}_j) \cdot \hat{\mathbf{n}} ds \\ &= 0, \end{aligned}$$

where it has been used that the fields must decay to zero at infinity. Equations (2.36) can be derived from a Principle of Least Action after defining the discrete field Lagrangian L of the Schrödinger type:

$$L = - \sum_{ij} a_i^* \left(i P_{ij} \frac{d}{dz} + H_{ij} \right) a_j. \quad (2.37)$$

The generalized momenta of this Lagrangian are $\Pi_k = \frac{\partial L}{\partial \dot{a}_k} = -i \sum_j a_j^* P_{jk}$, so the associated Hamiltonian H is:

$$H = \sum_j \Pi_j \dot{a}_j - L = \sum_{ij} -i a_i^* P_{ij} \dot{a}_j + a_i^* \left(i P_{ij} \frac{da_j}{dz} + H_{ij} a_j \right) = \sum_{ij} a_i^* H_{ij} a_j. \quad (2.38)$$

Since the Lagrangian does not depend explicitly on z , the Hamiltonian H is a conserved

quantity. Varying the Lagrangian yields:

$$\begin{aligned}
\delta L &= \sum_{ij} \frac{\partial L}{\partial a_j} \delta a_j + \frac{\partial L}{\partial a_i^*} \delta a_i^* + \frac{\partial L}{\partial \dot{a}_j} \delta \dot{a}_j + \frac{\partial L}{\partial \dot{a}_i^*} \delta \dot{a}_i^* \\
&= \sum_{ij} \left(\frac{\partial L}{\partial a_j} - \frac{d}{dz} \frac{\partial L}{\partial \dot{a}_j} \right) \delta a_j + \left(\frac{\partial L}{\partial a_i^*} - \frac{d}{dz} \frac{\partial L}{\partial \dot{a}_i^*} \right) \delta a_i^* + \frac{d}{dz} \left(\frac{\partial L}{\partial \dot{a}_j} \delta a_j + \frac{\partial L}{\partial \dot{a}_i^*} \delta a_i^* \right) \\
&\Rightarrow \frac{d}{dz} \sum_{ij} \left(\frac{\partial L}{\partial \dot{a}_j} \delta a_j + \frac{\partial L}{\partial \dot{a}_i^*} \delta a_i^* \right) = 0 .
\end{aligned}$$

The pair of transformations associated with the unitary group $U(1)$, $a_j \rightarrow a'_j = e^{i\phi} a_j$, $a_i^* \rightarrow a'^*_i = e^{-i\phi} a_i^*$ leave the Lagrangian invariant. If an infinitesimal variation $\phi \ll 1$ is taken, then $\delta a_j = i\phi a_j$ and $\delta a_i^* = -i\phi a_i^*$, is taken, then P , which is identified with the total power of the system and is directly related to the Poynting vector, is:

$$P = \sum_{ij} a_i^* P_{ij} a_j = \iint \mathbf{S} \cdot \hat{\mathbf{z}} dx dy . \quad (2.39)$$

Thus, there are two conserved quantities in the dynamics, H y P , which will be useful for verifying the validity of numerical solutions to equation (2.36).

2.5.2 TE Dimer in Slab-Type Waveguides

Using the equations for the transverse components (??) in the case of fundamental TE modes (2.14), we have:

$$\begin{aligned}
\mathbf{e}_{j\perp} &= \hat{\mathbf{y}} \frac{i\omega\mu_0 H_{a1}}{k_0^2 n_j^2 - (k_z^j)^2} \begin{cases} -\alpha \cos(\alpha(x - x_j)), & |x - x_j| \leq a \\ \beta \sin(\alpha a) e^{-\beta(|x - x_j| - a)}, & |x - x_j| > a \end{cases} , \\
\mathbf{h}_{j\perp} &= \hat{\mathbf{x}} \frac{ik_z^j H_{a1}}{k_0^2 n_j^2 - (k_z^j)^2} \begin{cases} \alpha \cos(\alpha(x - x_j)), & |x - x_j| \leq a \\ -\beta \sin(\alpha a) e^{-\beta(|x - x_j| - a)}, & |x - x_j| > a \end{cases} ,
\end{aligned}$$

with $x_j = jd$, $d \geq 2a$ and $L = \int dy$. The off-diagonal elements have the form:

$$\begin{aligned}
(\mathbf{e}_{j\perp} \times \mathbf{h}_{i\perp}^*) \cdot \hat{\mathbf{z}} &= \frac{k_z^i \omega \mu_0 H_{a1}^2 \beta \sin(\alpha a)}{(k_0^2 n_j^2 - k_z^2)(k_0^2 n_i^2 - k_z^2)} \begin{cases} \alpha \cos(\alpha(x - x_i)) e^{-\beta(|x - x_j| - a)}, & |x - x_i| \leq a \\ i \leftrightarrow j, & |x - x_j| \leq a \\ \beta \sin(\alpha a) e^{-\beta(|x - x_i| + |x - x_j| - 2a)}, & \text{otherwise} \end{cases} , \\
P_{ij} &= \frac{L k_z^i k_0 \mu_0 c H_{a1}^2 \sin^2(\alpha a) e^{-\beta d} e^{\beta a}}{2\beta} \left[\frac{e^{-\beta a} + \beta(d - 2a) e^{\beta a}}{\beta^2} - \frac{2 \cosh(\beta a)}{\alpha^2 + \beta^2} \right] , \\
\tilde{H}_{ij} \equiv H_{ij} - P_{ij} k_z^j &= \frac{L k_0 \mu_0 c H_{a1}^2}{2\beta} \sin^2(\alpha a) e^{2\beta a} e^{-\beta d} .
\end{aligned}$$

The diagonal elements have the form:

$$\begin{aligned} (\mathbf{e}_{j\perp} \times \mathbf{h}_{j\perp}^*) \cdot \hat{\mathbf{z}} &= \frac{k_z^j \omega \mu_0 H_{a1}^2}{(k_0^2 n_j^2 - k_z^2)^2} \begin{cases} \alpha^2 \cos^2(\alpha(x - x_j)), & |x - x_j| \leq a \\ \beta^2 \sin^2(\alpha a) e^{2\beta a} e^{-2\beta|x-x_j|}, & |x - x_j| > a \end{cases}, \\ P_{jj} &= \frac{L k_z^j k_0 \mu_0 c H_{a1}^2 a}{2\alpha^2} \left[1 + \frac{\sin^2(\alpha a)}{a\beta^3} k_0^2 (n_1^2 - n_0^2) \right], \\ \tilde{H}_{jj} \equiv H_{jj} - P_{jj} k_z^j &= \frac{L k_0^3 \mu_0 c H_{a1}^2 \sin^2(\alpha a) e^{2\beta a} e^{-2\beta d}}{4\beta^3} (n_1^2 - n_0^2) \sinh(2\beta a). \end{aligned}$$

The dynamics is governed by the coupled system of equations (2.36), which, after pre-multiplying by the inverse of the matrix P_{ij} , takes the compact form:

$$-i \frac{d}{dz} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} \tilde{k} & C \\ C & \tilde{k} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix},$$

with, for example, $\tilde{k} \equiv k_z^{\text{TE}} + \delta k$, $k_z^{\text{TE}} \sim 1.3 \times 10^5 \text{ cm}^{-1}$, $C \equiv \frac{P_{ii}\tilde{H}_{ij} - P_{ij}\tilde{H}_{ii}}{P_{ii}^2 - P_{ij}^2} \sim 1 \text{ cm}^{-1}$ and $\delta k \equiv \frac{P_{ii}\tilde{H}_{ii} - P_{ij}\tilde{H}_{ij}}{P_{ii}^2 - P_{ij}^2} \sim -0.02 \text{ cm}^{-1}$ for $\lambda = 730 \text{ nm}$, $n_1 = n_0 + 4 \times 10^{-4}$ and a distance between waveguides of $d = 18 \text{ }\mu\text{m}$. The matrix $\hat{P}^{-1}\hat{H}$ can be written in terms of the Pauli matrices as $\tilde{k}I + C\sigma_x$, which allows using their properties to solve the system dynamics:

$$\begin{pmatrix} a_1(z) \\ a_2(z) \end{pmatrix} = e^{i[\tilde{k}I + C\sigma_x]z} \begin{pmatrix} a_1(0) \\ a_2(0) \end{pmatrix} = e^{i\tilde{k}z} \begin{pmatrix} \cos(Cz) & i \sin(Cz) \\ i \sin(Cz) & \cos(Cz) \end{pmatrix} \begin{pmatrix} a_1(0) \\ a_2(0) \end{pmatrix}.$$

One way to experimentally determine the coupling value C is by individually exciting one mode. With this, the ratio between the powers $|a_1(z)|^2$ and $|a_2(z)|^2$ allows solving for the coupling given a known propagation distance $z = L < \frac{\pi}{2C}$:

$$C = \frac{1}{L} \tan^{-1} \left(\left| \frac{a_2(L)}{a_1(L)} \right|^2 \right). \quad (2.40)$$

2.5.3 Bands and Topology: Photonic Su-Schrieffer-Heeger Network

Although it is possible to reformulate Maxwell's equations as an eigenvalue problem in analogy with quantum mechanics to establish an analogue to Bloch's theorem [33], the Hermitian matrix $\hat{C} \equiv \hat{P}^{-1}\hat{H}$ from equation (2.36) provides sufficient justification to invoke it: It is possible to express the solutions of the system in a quasimomentum basis whose periodicity matches that of the matrix $\hat{C}$. To illustrate this, the celebrated Su-Schrieffer-Heeger (SSH) model will be used, which is one of the simplest systems exhibiting *topological edge states* [31, 34, 35]. Originally developed to explain the formation of localized electronic states in polyacetylene, it involves considering only two types of coupling: one strong (double bond) and one weak (single bond). In terms of Solid State Physics, it corresponds to a

one-dimensional lattice with a two-site basis, whose modal amplitudes are denoted a_n and b_n . Figure 2.5 illustrates the model to be studied, which corresponds to the following Hamiltonian H .

$$H = \sum_n (tb_n^* a_n + t' a_{n+1}^* b_n + c.c.) .$$

Under periodic boundary conditions (Born–von Karman), it is possible to diagonalize the

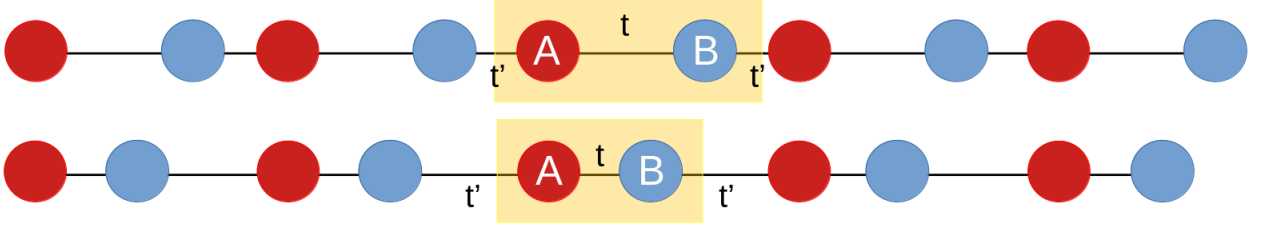


Figure 2.5: Schematic of the SSH model. The unit cell is enclosed in yellow, where sites A and B and the intracell (intercell) coupling t (t') are distinguished. Two different configurations are shown.

Hamiltonian in the Bloch basis with $a_n = \frac{1}{\sqrt{N}} \sum_k a_k e^{iknd}$, $b_n = \frac{1}{\sqrt{N}} \sum_k b_k e^{iknd}$ as follows:

$$\begin{aligned} H &= \frac{1}{N} \sum_n \sum_{kk'} \left[(t + t' e^{ikd}) a_k b_k^* e^{i(k-k')nd} + (t + t' e^{-ikd}) a_k^* b_{k'} e^{-i(k-k')nd} \right] \\ &= \sum_k \begin{pmatrix} a_k^* & b_k^* \end{pmatrix} \underbrace{\begin{pmatrix} 0 & t + t' e^{-ikd} \\ t + t' e^{ikd} & 0 \end{pmatrix}}_{\hat{H}(k)} \begin{pmatrix} a_k \\ b_k \end{pmatrix}, \end{aligned}$$

where it has been used that

$$\frac{1}{N} \sum_n e^{i(k-k')nd} = \begin{cases} \frac{N}{N} = 1, & \text{if } k = k' \\ \frac{1}{N} \frac{1 - e^{i(k-k')Nd}}{1 - e^{i(k-k')d}} e^{i(k-k')d} = 0, & \text{if } k \neq k' \end{cases} = \delta_{kk'},$$

since $k, k' \in \left\{ \frac{2\pi l}{Nd}, l = 0, \pm 1, \dots, \pm \frac{N}{2} \right\}$ belong to the 1D reciprocal lattice. The spectrum of $\hat{H}(k)$ is $\lambda_k = \pm \sqrt{t^2 + t'^2 + 2tt' \cos(kd)}$, which would seem to indicate that the system is symmetric between t y t' . However, the picture is not complete without the eigenvectors of $\hat{H}(k)$,

$$\mathbf{v}_{\pm} = \frac{1}{\sqrt{2}} \begin{pmatrix} \pm e^{-i\phi(k)} \\ 1 \end{pmatrix}, \text{ with } \phi(k) = \tan^{-1} \left(\frac{t' \sin(kd)}{t + t' \cos(kd)} \right). \text{ If the vector } \mathbf{d}(k) \equiv \begin{pmatrix} t + t' \cos(kd) \\ t' \sin(kd) \\ 0 \end{pmatrix} \text{ is defined,}$$

it is possible to write the Hamiltonian $\hat{H}(k)$ in terms of the Pauli matrices as $\hat{H}(k) = \mathbf{d} \cdot \boldsymbol{\sigma} = d_x \sigma_x + d_y \sigma_y$. This vector $\mathbf{d}(k)$ is a vectorial representation of the 2×2 bulk Hamiltonian in momentum space. The phase $\phi(k)$ is related to the angle formed by the projection of the vector $\mathbf{d}(k)$ onto the xy plane, $\phi(k) = \tan^{-1} \left(\frac{d_y}{d_x} \right)$. In particular, when $\delta \equiv t/t' < 1$, the vector $\mathbf{d}(k)$ encloses the origin once, while for $\delta > 1$ it does not enclose the origin, as seen in Figure 2.6. Formally, the Zak phases, $\mathcal{Z}_{\pm}$, associated with the bulk eigenstates (a region sufficiently

far from the edges) change depending on the value of δ . By definition [36],

$$\mathcal{Z}_{\pm} = i \oint \mathbf{v}_{\pm}^{\dagger} \frac{d\mathbf{v}_{\pm}}{dk} dk = -\frac{1}{2} \int_{-\frac{\pi}{d}}^{\frac{\pi}{d}} \frac{d\phi(k)}{dk} dk = -\frac{d}{2} \int_{-\frac{\pi}{d}}^{\frac{\pi}{d}} \frac{1 + \delta \cos(kd)}{\delta^2 + 1 + 2\delta \cos(kd)} dk = -\frac{\pi}{2} \left(1 + \frac{1 - \delta^2}{|1 - \delta^2|} \right).$$

That is,

$$\mathcal{Z}_{\pm} = \begin{cases} -\pi, & \text{if } |\delta| < 1 \\ 0, & \text{if } |\delta| > 1 \end{cases}.$$

The change in the Zak phase as the parameter δ varies is known as a *topological transition* since it is not possible to smoothly transition from one regime to the other. This has implications for the system with open boundary conditions (Dirichlet) due to the *bulk-boundary correspondence* [37]. Specifically, as illustrated in Figure 2.7, the topology of the Hamiltonian $\hat{H}(k)$ induces the appearance of zero-energy edge states when open boundary conditions are imposed.

The Hamiltonian of the SSH model exhibits various symmetries: time-reversal, inversion, particle-hole, translational, and chiral. The latter is particularly relevant, as the edge states of the SSH model are *topologically protected* against any disorder that respects chiral symmetry [38]. The chiral operator in question is the Pauli matrix σ_z , since

$$\{\hat{H}(k), \sigma_z\} = \{d_x \sigma_x + d_y \sigma_y, \sigma_z\} = d_x \{\cancel{\sigma_x}, \sigma_z\} + d_y \{\cancel{\sigma_y}, \sigma_z\} \stackrel{0}{=} 0.$$

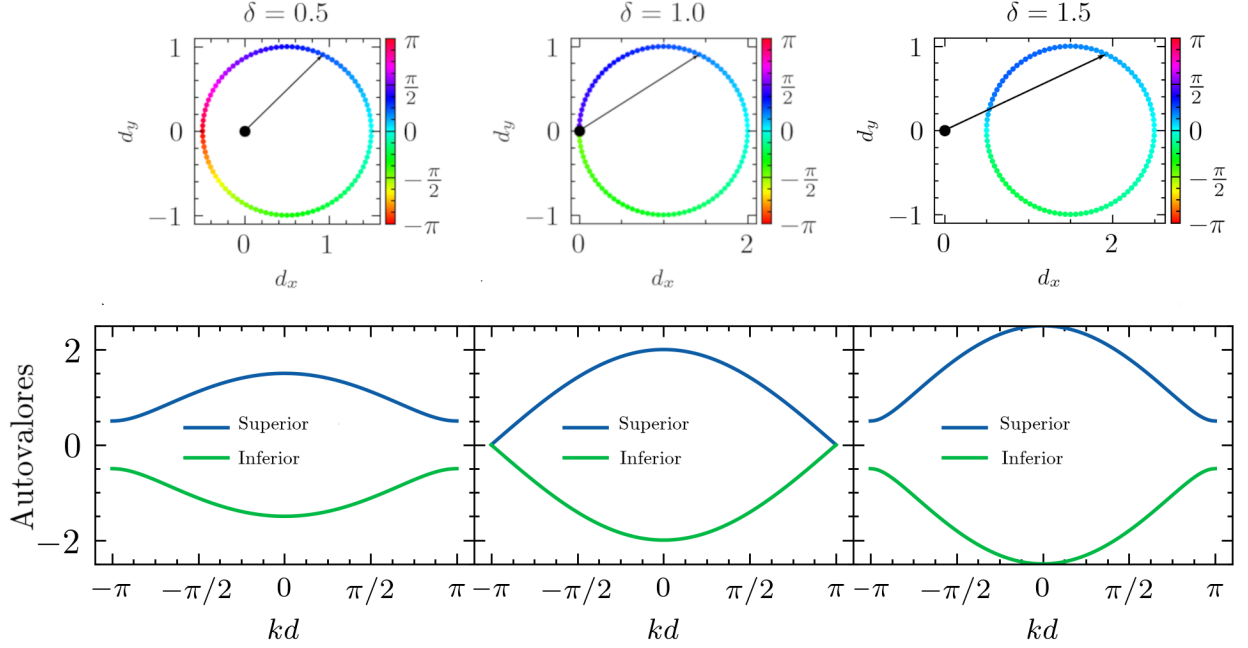


Figure 2.6: Topology of the SSH lattice. Top: Vectors $\mathbf{d}(\mathbf{k})$ and their phase. For $\delta < 1$ ($\delta > 1$), the origin is (not) enclosed. Bottom: Bands as a function of quasimomentum k in the first Brillouin zone for $\delta < 1$, $\delta = 1$ and $\delta > 1$. Here $t' = 1$ is fixed and δ' is varied for a number $N = 100$ of unit cells.

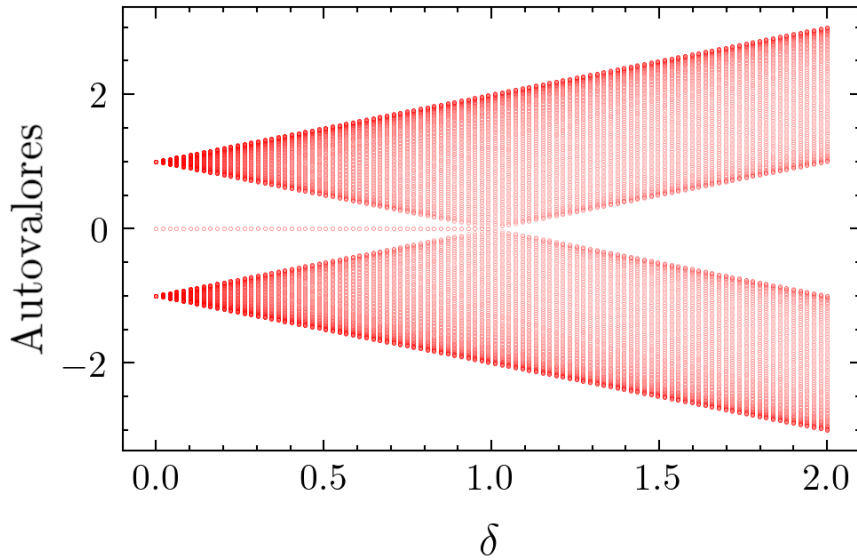


Figure 2.7: Spectrum of the finite SSH lattice. The emergence of zero-energy edge states for $\delta < 1$ is observed. Open boundary conditions as a function of the parameter $\delta = t/t'$, with fixed $t' = 1$ and a number $N = 100$ of unit cells.

3. Métodos Numéricos

En este capítulo se presentan los métodos numéricos fundamentales utilizados en el desarrollo de esta tesis. Dado que muchos sistemas fotónicos carecen de soluciones analíticas exactas, los enfoques numéricos resultan esenciales para modelar su comportamiento electromagnético. Estos métodos permiten aproximar soluciones con precisión controlada, incluso en geometrías donde no existe alguna simetría a explotar.

El punto de partida lo constituyen las ecuaciones (2.6) y (2.7), que describen la propagación de campos electromagnéticos en medios dieléctricos. Cuando se aplica la aproximación de guiado débil los términos del lado derecho de ambas ecuaciones pueden despreciarse. Esta simplificación elimina los efectos cruzados entre componentes de campo, reduciendo el sistema a ecuaciones escalares de tipo Helmholtz:

$$\left[\nabla^2 + k_0^2 n^2(\mathbf{r}) \right] \Psi(\mathbf{r}) = 0 , \quad (3.1)$$

donde la ecuación (3.1) no solo sirve como marco teórico, sino también como base para los algoritmos desarrollados en los siguientes capítulos. Su versatilidad permite adaptar diferentes esquemas numéricos según las simetrías del problema, como se detallará en las secciones posteriores.

3.1 Expansión en Modos Normales

Este método numérico, desarrollado e implementado durante la investigación del estudiante en su Magíster, es útil cuando los sistemas fotónicos en estudio son invariantes en la dirección de propagación z . Esto es, $n(\mathbf{r}) \equiv n(x, y)$. Las soluciones de la ecuación (3.1) se pueden expandir en ondas planas con perfiles transversales: $\Psi(\mathbf{r}) = \Psi(x, y) \exp(i\beta z)$. Con esto, cada modo transversal ν debe cumplir la siguiente ecuación a resolver numéricamente:

$$\left[\nabla_{\perp}^2 + k_0^2 n^2(x, y) \right] \Psi_{\nu}(x, y) = \beta_{\nu}^2 \Psi_{\nu}(x, y) , \quad \text{con } \nabla_{\perp}^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} . \quad (3.2)$$

Y el campo total propagado es una combinación lineal de los modos Ψ_{ν} :

$$\Psi(\mathbf{r}) = \sum_{\nu} a_{\nu} \Psi_{\nu}(x, y) e^{i\beta_{\nu} z} , \quad \text{con } a_{\nu} \propto \Psi_{\nu}(x, y) \cdot \Psi(x, y, z=0) . \quad (3.3)$$

En vez de integrar directamente la ecuación de valores propios (3.2), la estrategia será discretizar el espacio y aproximar al operador laplaciano transversal $\nabla_{\perp}^2$ como una matriz, pues

$$\frac{\partial^2 \Psi(x, y)}{\partial x^2} \sim \frac{\Psi[i+1, j] - 2\Psi[i, j] + \Psi[i-1, j]}{\Delta x^2} .$$

$\nabla_{\perp}^2 \sim \hat{D}_{xx} \otimes I_y + I_x \otimes \hat{D}_{yy}$ La matriz es nula en las posiciones no diagonales más allá de un espacio (*sparse-like*), por lo que es posible optimizar el proceso de cómputo al utilizar la librería de Python `scipy.sparse.linalg`, especialmente diseñada para el álgebra lineal de matrices de escasos elementos. El anexo Appendix C contiene una implementación de este algoritmo en Python bajo [licencia GNU GPL v3](#).

En la Figura 3.1 se valida el método numérico mediante comparación con las soluciones analíticas obtenidas en la subsección 2.2.1. La complejidad del problema es del orden $O(N^2)$ debido a la construcción de las matrices de $N \times N$, lo que se refleja en la dependencia del tiempo de ejecución en función del paso $\Delta x \equiv \frac{L}{N-1}$.

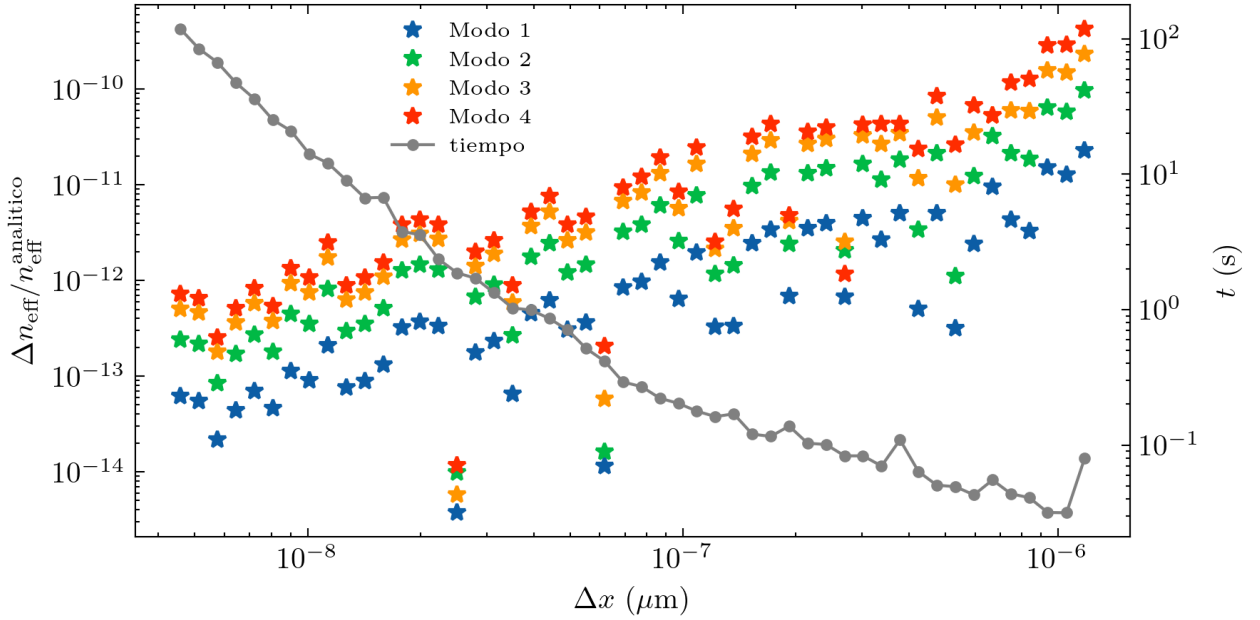


Figure 3.1: Error relativo y tiempo de ejecución de las soluciones numéricas a la ecuación (3.2) con respecto a las soluciones analíticas, en función del paso Δx .

3.2 Método de Propagación de Haces

Otra forma de abordar la integración numérica de la ecuación (3.1) consiste en separar el campo en su envolvente lenta y una fase rápidamente oscilante, considerando que en el rango visible el orden de magnitud es, $k_0 \sim 10^5 \text{ cm}^{-1}$: $\Psi(\mathbf{r}) = \phi(x, y, z) \exp(ik_0 n_0 z)$. Luego de reemplazar en la ecuación (3.1) se obtiene la ecuación óptica de Schrödinger [39]:

$$-2ik_0 n_0 \frac{\partial}{\partial z} \phi(x, y, z) = \left[\nabla_{\perp}^2 + k_0^2 (n^2(\mathbf{r}) - n_0^2) \right] \phi(x, y, z), \quad (3.4)$$

donde se ha utilizado la aproximación paraxial $\left| \frac{\partial^2 \phi}{\partial z^2} \right| \ll 2k_0 n_0 \left| \frac{\partial \phi}{\partial z} \right|$. Los algoritmos que resuelven la ecuación (3.4) son conocidos como Métodos de Propagación de Haces o *Beam Propagation Methods* (BPM) escalares, utilizados ampliamente en esta área de investigación [40–44]. Fuera de la aproximación de guiaje débil, los efectos de polarización cruzada son más relevantes y

se hace necesaria una descripción vectorial del problema.

3.2.1 Implementación mediante transformada de Fourier (FFTBPM)

La ecuación (3.4) se puede escribir en términos de operadores lineales diferenciales como [44]:

$$\frac{\partial \phi}{\partial z} = (\hat{A} + \hat{B})\phi, \text{ con } \hat{A} \equiv i \frac{\nabla_{\perp}^2}{2k_0 n_0} \text{ y } \hat{B} \equiv i \frac{k_0}{2n_0} [n^2(\mathbf{r}) - n_0^2]. \quad (3.5)$$

La solución formal a la ecuación (3.5) es $\phi(\mathbf{r}) = \exp[(\hat{A} + \hat{B})(z - z_0)]\phi(x, y, z_0)$. Es conveniente trabajar con el operador $\hat{A}$ en el espacio de Fourier y con el operador $\hat{B}$ en el espacio real. Esto se debe a que $\hat{A}$ contiene derivadas espaciales —en particular, el operador laplaciano transversal $\nabla_{\perp}^2$ — las cuales se convierten en multiplicaciones simples al aplicar la transformada de Fourier. De esta forma, la operación $\hat{A}\phi$ se vuelve computacionalmente eficiente al transformarla en el dominio de los vectores de onda $\mathbf{k}_{\perp}$, donde actúa como un operador diagonal. Por otro lado, $\hat{B}$ depende del perfil espacial del índice de refracción $n(\mathbf{r})$, el cual está definido localmente en el espacio real. Evaluar este término en Fourier implicaría una convolución, lo cual es innecesariamente costoso y menos intuitivo. Por este motivo, se mantiene en el espacio real.

Dado que $\hat{A}$ y $\hat{B}$ no conmutan en general, se puede expandir el operador exponencial a orden $O(\Delta z^3)$ como $\exp[(\hat{A} + \hat{B})\Delta z] \approx \exp\left(\frac{\hat{A}\Delta z}{2}\right)\exp(\hat{B}\Delta z)\exp\left(\frac{\hat{A}\Delta z}{2}\right) + O(\Delta z^3)$.

El algoritmo implementado en el Anexo Appendix D es el siguiente:

1. Se comienza con un perfil $\phi(x, y, z_0)$
2. Se actúa en el espacio de Fourier transformando el perfil y multiplicado por la fase asociada a $\hat{A}$: $\exp\left(\frac{ik^2\Delta z}{4k_0n_0}\right)\mathcal{F}(\phi(z_0))$, donde $k = \sqrt{k_x^2 + k_y^2}$ son las frecuencias de Fourier.
3. Se aplica transformada de Fourier inversa y se multiplica por la fase asociada a $\hat{B}$:

$$\exp\left[\frac{i\Delta z k_0^2(n^2 - n_0^2)}{2k_0n_0}\right]\mathcal{F}^{-1}\left(\exp\left(\frac{ik^2\Delta z}{4k_0n_0}\right)\mathcal{F}(\phi(z_0))\right)$$
4. Se regresa al espacio de Fourier y se multiplica por la fase asociada a $\hat{A}$:

$$\exp\left(\frac{ik^2\Delta z}{4k_0n_0}\right)\mathcal{F}\left\{\exp\left[\frac{ik_0^2(n^2 - n_0^2)}{2k_0n_0}\Delta z\right]\mathcal{F}^{-1}\left[\exp\left(\frac{ik^2\Delta z}{4k_0n_0}\right)\mathcal{F}(\phi(z_0))\right]\right\}$$
5. Se vuelve al espacio real, habiendo avanzado un paso Δz :

$$\phi(z_0 + \Delta z) = \mathcal{F}^{-1}\left\{\exp\left(\frac{ik^2\Delta z}{4k_0n_0}\right)\mathcal{F}\left\{\exp\left[\frac{ik_0^2(n^2 - n_0^2)}{2k_0n_0}\Delta z\right]\mathcal{F}^{-1}\left[\exp\left(\frac{ik^2\Delta z}{4k_0n_0}\right)\mathcal{F}(\phi(z_0))\right]\right\}\right\}$$
6. Se itera hasta llegar a la distancia de propagación z deseada.

En la Figura 3.2 se comparan los resultados experimentales con las simulaciones numéricas obtenidas mediante BPM. Los perfiles de intensidad normalizados muestran un acuerdo cualitativo notable, particularmente en la distribución espacial de los lóbulos principales.

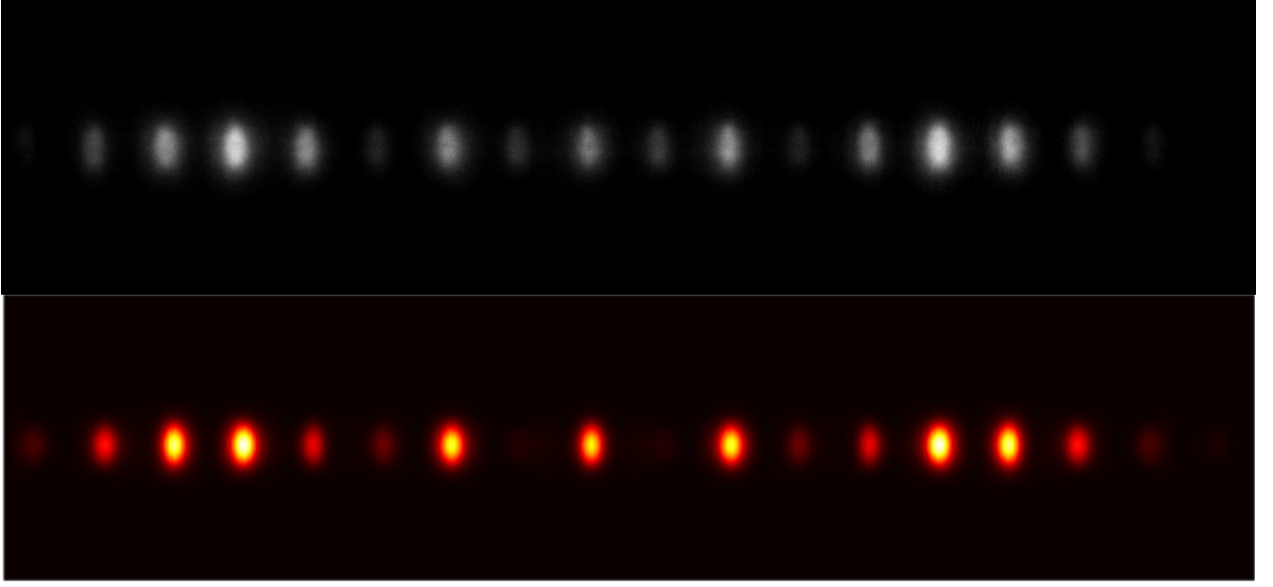


Figure 3.2: Excitación del sitio central de una red unidimensional. **Arriba:** Medición experimental a $\lambda = 730$ nm. **Abajo:** Simulación numérica realizada con BPM.

Este nivel de concordancia cualitativa valida la aproximación de guiado débil empleada en la ecuación (3.1), así como la aproximación paraxial, y refuerza la confiabilidad del método numérico para el diseño de dispositivos fotónicos con geometrías análogas.

3.3 Comentarios sobre Teoría de Modos Acoplados

La simulación numérica de las ecuaciones (2.36) representa la formulación más compacta posible, dado que la matriz de acoplamientos $\hat{C} \equiv \hat{P}^{-1}\hat{H}$ codifica todas las propiedades de la red que se desean estudiar de manera semi-empírica. En este sentido, la diagonalización de $\hat{C}$ constituye una herramienta numérica valiosa para analizar el comportamiento de redes fotónicas ante variaciones de parámetros como la dimerización (definida como la razón entre dos constantes de acoplamiento) o el desintonizado (la diferencia entre las constantes de propagación de modos distintos). Además, este enfoque permite explorar incluso sistemas “no físicos”, gracias a los grados de libertad disponibles en los elementos de $\hat{C}$: por ejemplo, casos en los que el acoplamiento a segundos vecinos sea mayor que a primeros vecinos, o sistemas con acoplamientos complejos.

En la aproximación de guiado débil se pueden despreciar las componentes longitudinales del campo eléctrico:

$$\nabla \cdot (n^2 \mathbf{E}) \stackrel{\nabla n^2 \sim 0}{\sim} n^2 \nabla \cdot \mathbf{E} = 0 ,$$

donde el orden de magnitud de la componente longitudinal del campo es $|E_z| \sim |\nabla_{\perp} \cdot \mathbf{E}_{\perp}|/\beta$. Así, los modos estudiados son cuasi-transversales, y la matriz $\hat{P}$ se reduce a:

$$P_{ij} \sim \frac{k_z^i + k_z^j}{4\omega\mu_0} \iint (\mathbf{e}_i^{\perp})^* \cdot \mathbf{e}_j^{\perp} dx dy . \quad (3.6)$$

La ecuación (3.6), que cuantifica el solapamiento entre modos, junto con la ecuación (2.33) que describe los elementos de matriz del operador Hamiltoniano, constituyen herramientas fundamentales para el análisis comparativo entre el método de Expansión en Modos Normales (EME) y la Teoría de Modos Acoplados (CMT). Esta comparación se realiza a través de un estudio sistemático del espectro de autovalores de la matriz de acoplamientos $\hat{C}$, el cual contiene información esencial sobre las constantes de propagación efectivas del sistema.

La Figura 3.3 presenta un análisis cuantitativo de esta comparación, mostrando la evolución de los autovalores para un acoplador fotónico dieléctrico en función de la distancia de separación entre guías. Se observa que ambos métodos convergen asintóticamente para distancias mayores a $d > 20 \mu\text{m}$, donde la aproximación de campo débil de CMT resulta válida. Sin embargo, en el régimen de acoplamiento fuerte ($d < 13 \mu\text{m}$), las discrepancias comienzan a hacerse notar.

Más allá del análisis espectral, la validez de los autoestados calculados puede verificarse indirectamente mediante la ecuación (3.3), que relaciona la evolución del campo con la proyección sobre los modos propios del sistema y comparando cualitativamente los perfiles dinámicos obtenidos con los resultados experimentales y numéricos mostrados en la Figura 3.2. Este análisis (ver Figura 3.4) permite establecer los límites de aplicabilidad de cada método numérico: mientras que EME proporciona resultados precisos en todo el rango de parámetros a costa de mayor complejidad computacional, CMT ofrece una alternativa eficiente para sistemas débilmente acoplados donde la aproximación de modos localizados sigue siendo válida: distancias de separación $d \gtrsim 15 \mu\text{m}$.

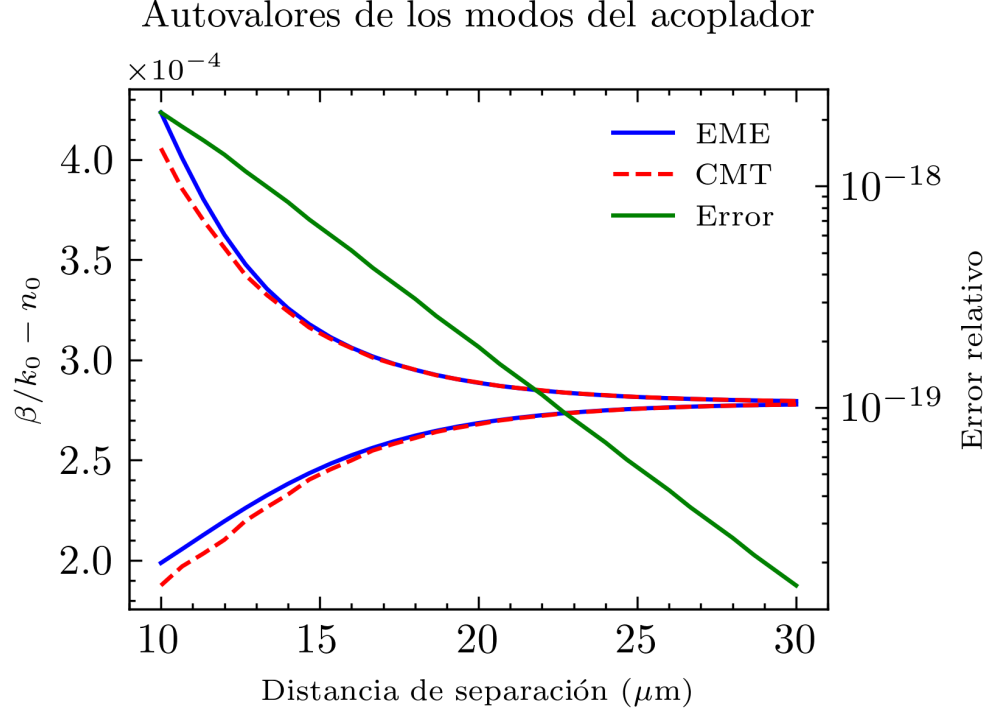


Figure 3.3: Comparación de los autovalores obtenidos con el método de Expansión en Modos Normales (EME) y con la Teoría de Modos Acoplados (CMT). Se observa mayor divergencia cuando la distancia entre guías es pequeña. Parámetros utilizados: $n_0 = 1.48$, $\Delta n = 4 \times 10^{-4}$, $\lambda = 730 \text{ nm}$ y $a = 3 \mu\text{m}$.

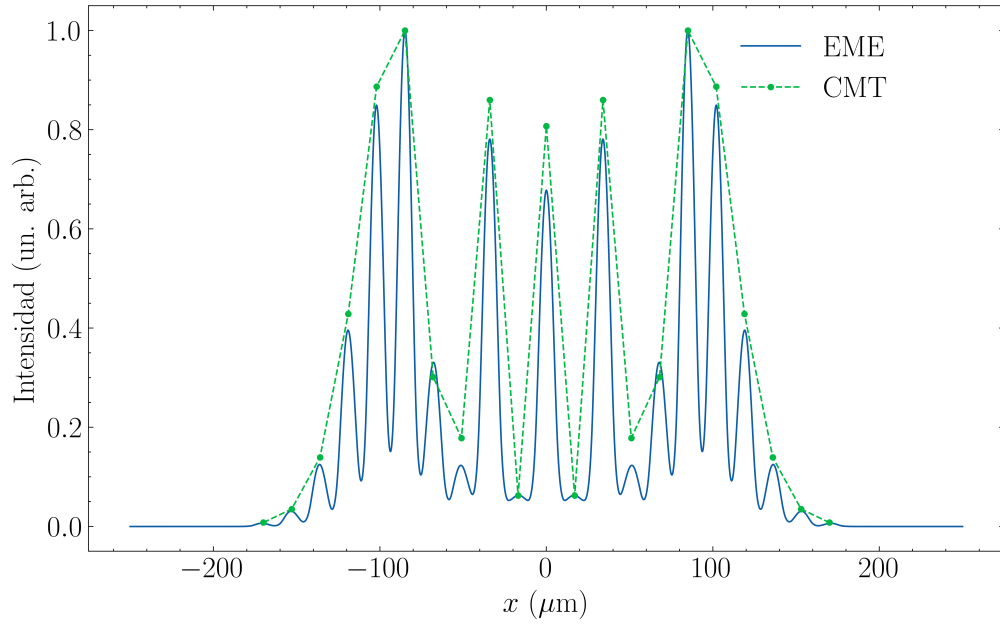


Figure 3.4: Perfil dinámico de un arreglo unidimensional de 21 guías a distancia $d = 17 \mu\text{m}$ simulado con CMT y con EME. Los resultados son similares a los de la Figura 3.2.

4. Métodos Experimentales

En este capítulo se detallan los procedimientos experimentales desarrollados para la fabricación y caracterización de redes fotónicas basadas en guías de onda. La metodología comprende tres aspectos fundamentales: (1) escritura directa de guías mediante láser de femtosegundos, (2) sistemas de excitación óptica con láser supercontinuo, y (3) técnicas avanzadas de modulación espacial de luz para el control de condiciones iniciales.

4.1 Escritura directa de guías de onda

La fabricación de guías de onda se realizó mediante la técnica de escritura directa con láser de femtosegundos: pulsos ultracortos emitidos desde un láser de fibra Yb-dopado ATSEVA ANTAUS de 1030 nm, a una tasa de repetición de 500 kHz, son fuertemente enfocados en una muestra de borosilicato, produciendo cambios leves y permanentes en el índice de refracción local $\Delta n \sim 10^{-4} - 10^{-3}$. Este método, disponible e implementado por miembros anteriores del Grupo de Redes Fotónicas, permite crear estructuras tridimensionales mediante movimientos controlados por computador de la plataforma XYZ Aerotech Nanopositioner [45].

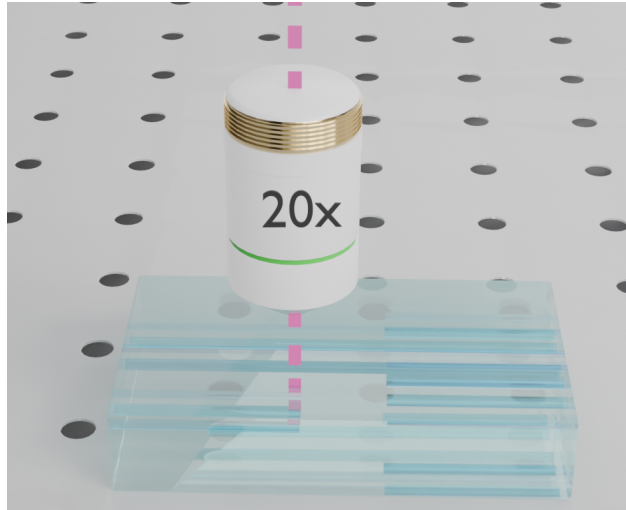


Figure 4.1: Esquema de la técnica de escritura de guías de onda. El haz láser es focalizado mediante un objetivo microscopio (20x) mientras la muestra se desplaza en tres ejes mediante una plataforma controlada por computadora.

Los parámetros de escritura que funcionan para excitar modos en el espectro visible son:

- Ancho del pulso: 215 fs.
- Tasa de repetición: 500 kHz.
- Velocidad de escritura: 0.1 - 20.0 mm/s.

- Potencia de llegada medida: 70.0 - 150.0 mW.

4.2 Montaje de excitación láser supercontinuo

Para la caracterización óptica se implementó un sistema de excitación basado en un láser supercontinuo (SC) de banda ancha (470 nm - 2200 nm). Específicamente, el modelo es un YSL SC-5. La configuración permite seleccionar longitudes de onda específicas mediante un modulador acusto-óptico AOTF-PRO (430 nm - 1450 nm), con una resolución espectral de ± 5 nm y una potencia de salida de 1 mW por longitud de onda.

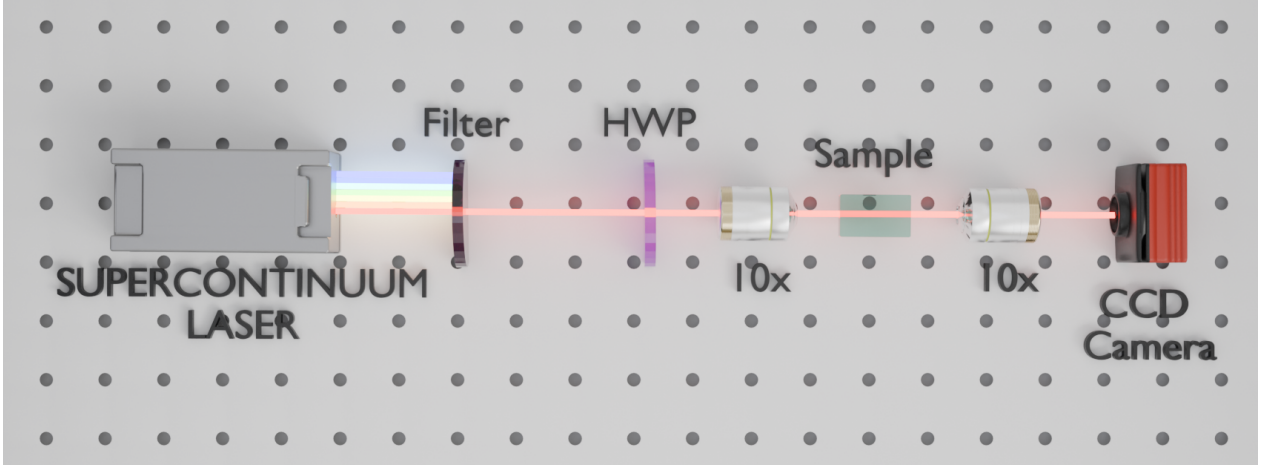


Figure 4.2: Montaje de excitación por láser supercontinuo. Filtro: Modulador acústico e iris. HWP: Retardador de media onda.

4.3 Montaje de modulación espacial de luz

Para usar condiciones iniciales distintas a una gaussiana se hace necesario incorporar métodos de modulación espacial de luz. En esta tesis se implementó una técnica conocida como grabado de fase mediante holograma [46].

4.3.1 Etapa premodulación

El modulador espacial de luz utilizado es un HOLOEYE PLUTO-VIS SLM - Reflective LCOS (resolución 1920×1080 px, tamaño de píxel 8 μm), cuya respuesta óptica ocurre con polarización paralela al plano de la mesa óptica. Se utiliza un retardador de media onda ($\lambda/2$) seguido de un polarizador Glan-Thompson 100.000:1 con el objetivo de que la polarización de la luz láser coincida con la de la respuesta del SLM. Posteriormente se magnifica y se colima el haz para que abarque todo el área de píxeles disponible utilizando un objetivo 20x y un lente de foco 100mm (telescopio).

4.3.2 Etapa de modulación

Una rejilla de difracción que maximiza la potencia del primer orden de difracción es utilizada. Para modular en amplitud se debe multiplicar la rejilla por la máscara de amplitud deseada, mientras que para modular en fase basta con sumar el nivel de gris correspondiente a la fase deseada. En la Figura 4.3 se bosqueja el algoritmo implementado en Python en el anexo Appendix E.

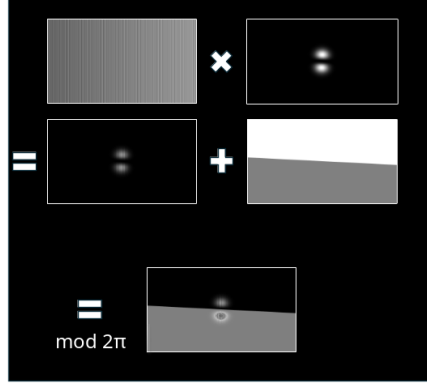


Figure 4.3: Algoritmo de modulación espacial de luz para máscaras de amplitud y fase arbitrarias. Los parámetros de la rejilla de difracción están sujetos a la longitud de onda usada (730 nm).

4.3.3 Etapa de acoplamiento

La imagen modulada pasa por un par de lentes L_2 de foco 1000 mm y L_3 de foco 50 mm para reducir el tamaño al orden de los micrómetros. La inclinación de la cara de entrada de la muestra debe coincidir con el plano de la imagen modulada, por lo que se generan dos pares de haces gaussianos, unos verticales y otros horizontales, de manera de que al trasladar el lente objetivo 4x, los máximos de interferencia se generen en el punto medio de los centros de los haces gaussianos.

4.3.4 Etapa de captura en cámara

Una vez calibrada la inclinación de la muestra, se fija su posición en el montaje experimental. La imagen de salida se magnifica mediante un objetivo 10x y se captura utilizando un perfilómetro de haz (Beam Profiler BC106N-VIS, Thorlabs) con resolución espacial de $6.45 \mu\text{m}/\text{px}$ [47].

Configuración interferométrica opcional

El montaje puede modificarse para incorporar un interferómetro Mach-Zehnder mediante:

- Un *beam splitter* entre la lente L_1 y el SLM, utilizando el haz previo a la modulación como referencia.
- Un segundo *beam splitter* antes de la cámara para recombinar los haces.
- Un filtro de densidad neutra (ND-Filter) para equilibrar las intensidades ($I_{\text{ref}} \approx I_{\text{muestra}}$).

Para la generación de modos dipolares, es fundamental ajustar la diferencia de fase entre los haces recombinados a un valor cercano a π . Esto se evidencia experimentalmente mediante la aparición de un mínimo de intensidad en el centro del patrón de interferencia, correspondiente a una interferencia destructiva en esa región.

4.3.5 Circuito óptico

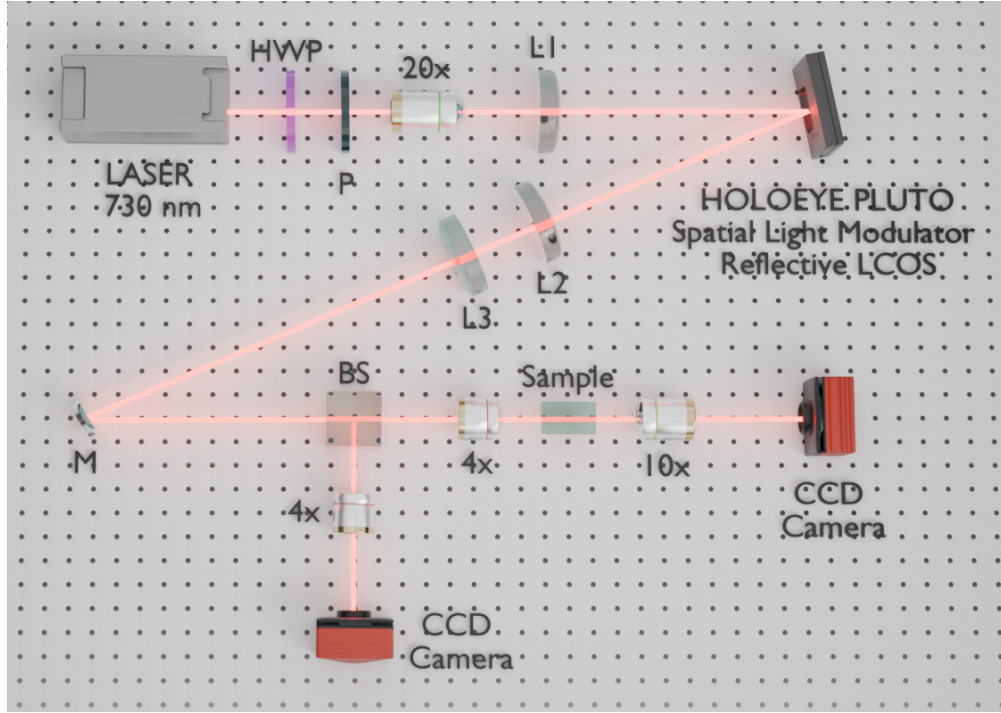


Figure 4.4: Diagrama completo del sistema experimental para modulación espacial de luz, mostrando los componentes ópticos y sus disposiciones relativas.

4.3.6 Generación de imágenes complejas

La capacidad de control preciso sobre amplitud y fase se verifica mediante la generación de vórtices ópticos, cuantificados por su Momentum Angular Orbital (OAM). El análisis interferométrico permite:

- Determinar la carga topológica ℓ contando las ramas espirales en el patrón de interferencia.
- Establecer el signo de ℓ mediante la dirección de rotación del frente de fase.

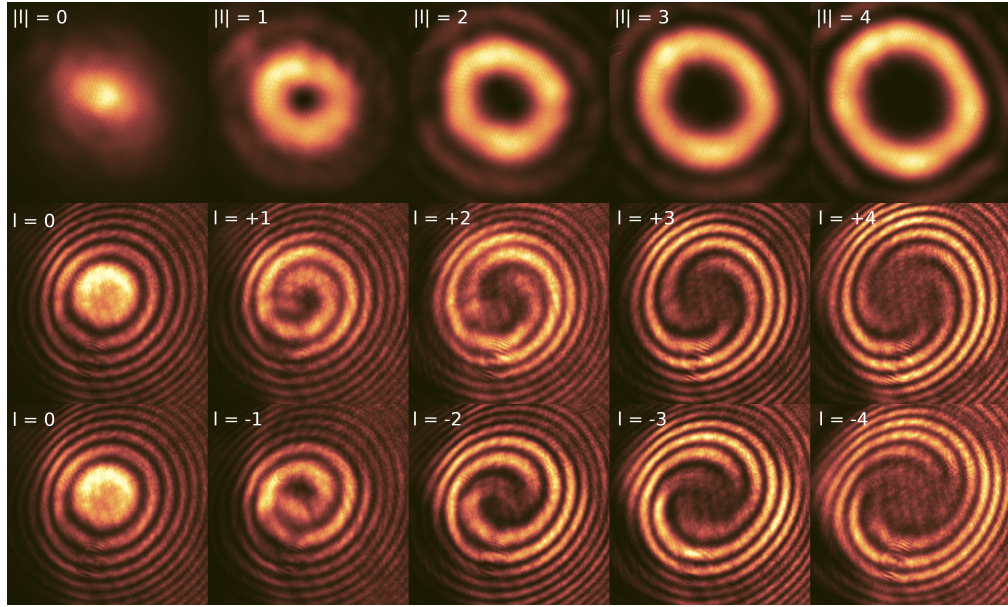


Figure 4.5: Caracterización de vórtices ópticos mediante interferometría Mach-Zehnder. **Arriba:** Intensidad para diferentes cargas OAM (ℓ). **Centro y Abajo:** Patrones de interferencia correspondientes, donde el número de ramas espirales indica $|\ell|$ y el sentido de giro da cuenta del signo de ℓ .

4.4 Análisis de imágenes

A partir de las imágenes capturadas es posible extraer información de la potencia que contiene cada sitio del sistema fotónico discreto en estudio. Para ello la imagen completa debe ser seccionada equispaciadamente en rectángulos que encierren las regiones donde existen guías de onda, iluminadas o no. La potencia de cada sitio será entonces la suma de la intensidad de cada píxel encerrado en su rectángulo respectivo.

El procesamiento de imágenes incluye:

- Corrección de fondo
- Normalización por intensidad máxima.
- Segmentación automática mediante umbral.

Los datos obtenidos permiten reconstruir la distribución de potencia en la red fotónica y analizar fenómenos de acoplamiento entre guías vecinas.

5. Acoplamiento de Modos p_y y Ángulo de Invisibilidad

En electrostática, las interacciones dipolares pueden describirse mediante los polinomios de Legendre de segundo orden, particularmente con $P_2(\cos(\theta)) = 3\cos^2(\theta) - 1$, donde θ representa el ángulo entre dos dipolos eléctricos. El valor angular $\theta_m \approx 0.62$ radianes se conoce como el *ángulo mágico*, ya que anula el término de interacción dipolar dominante [48].

Este capítulo se centra en el estudio experimental de los modos dipolares verticales, también denominados modos p_y , cuya existencia requiere superar la condición de corte: una longitud de onda suficientemente corta, combinada con un contraste de índice Δn y una dimensión transversal adecuada. Estos modos presentan una distribución transversal caracterizada por dos lóbulos simétricos respecto al eje horizontal y una diferencia de fase de π entre ellos.

5.1 Acopladores

Al estudiar el acoplamiento evanescente entre modos p_y en guías elípticas, se distinguen dos configuraciones límite: cuando las guías están dispuestas horizontalmente (alineadas a lo largo del eje x), el acoplamiento —denotado κ_π — es positivo; en cambio, para acopladores verticales (alineados en y), el acoplamiento κ_σ es negativo [30]. Este cambio de signo se interpreta como un efecto de interferencia entre los lóbulos del modo p_y , cuyo perfil lobular con simetría impar respecto al eje horizontal genera distintas condiciones de acoplamiento dependiendo de la orientación relativa de las guías.

Este comportamiento presenta una analogía¹ con los enlaces químicos σ y π que se forman en cadenas de carbono: en química cuántica, los enlaces σ corresponden a superposiciones frontales (axiales) de orbitales, mientras que los enlaces π surgen de superposiciones laterales, típicamente entre orbitales p con nodos en el eje de enlace. De forma análoga, llamamos aquí *acoplamiento* κ_σ al que se da por superposición axial entre modos con simetría adecuada, generando un perfil sin nodo transversal; mientras que el *acoplamiento* κ_π o $\kappa_{\theta=0}$ corresponde a una superposición lateral, en la que la distribución modal presenta un nodo central.

Para verificar este efecto, se fabricaron experimentalmente 20 dímeros con una separación

¹Esta analogía es puramente geométrica y modal. No implica la existencia de nubes electrónicas, potenciales interatómicos ni energía de enlace en sentido químico. Sin embargo, se mantiene como una herramienta conceptual útil para describir la orientación relativa de los modos y el efecto de la simetría en la intensidad y signo del acoplamiento. Su utilidad reside en la claridad con que comunica diferencias en la naturaleza del acoplamiento, especialmente en sistemas multiorbitales como los estudiados en esta tesis.

fija de $25\ \mu\text{m}$ y una longitud de propagación de 15 mm, variando el ángulo entre guías desde 0.00 rad hasta 1.57 rad. La excitación inicial fue generada mediante un modulador espacial de luz (SLM), programado para acoplar selectivamente el modo p_y en una de las guías.

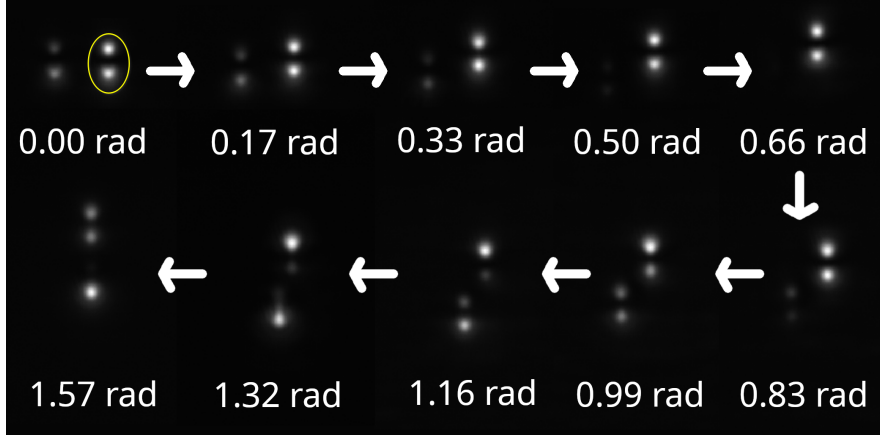


Figure 5.1: Barrido angular experimental para una distancia de propagación de 15 mm. Se observa el efecto del ángulo mágico entre 0.50 y 0.66 rad.

Las imágenes obtenidas fueron analizadas siguiendo el procedimiento descrito en la Sección 4.4. Posteriormente, mediante la ecuación (2.40) que describe el acoplamiento dinámico, se caracterizó cuantitativamente el acoplamiento efectivo como función del ángulo θ , medido desde la horizontal. Para reflejar correctamente la continuidad del acoplamiento con el ángulo, se ajustó el signo del acoplamiento en el ajuste teórico, como muestra la Figura 5.2.

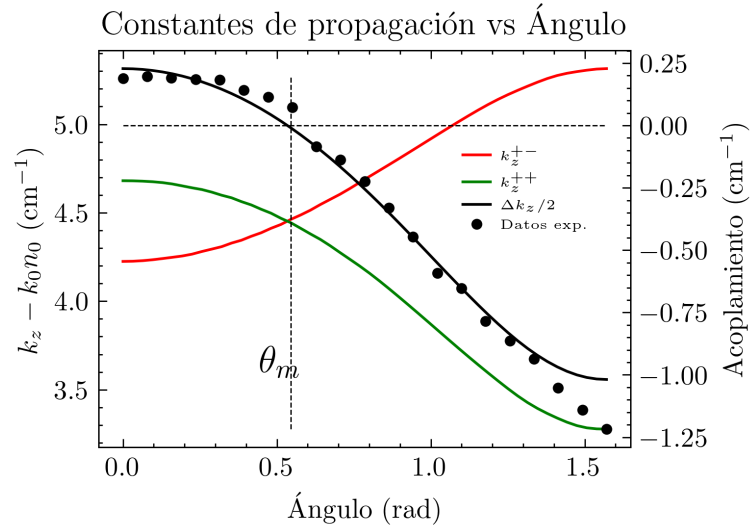


Figure 5.2: Constantes de propagación y acoplamientos en función del ángulo para modos p_y , calculados numéricamente mediante EME. Los resultados se comparan con los datos experimentales extraídos de la Figura 5.1.

5.2 Redes tipo panal de abeja

La red tipo panal de abeja, conocida por ser la estructura cristalina subyacente del grafeno, presenta propiedades relevantes para esta tesis, particularmente en sus bandas de Bloch: ambas son dispersivas y exhiben un punto de Dirac [49]. Una vez determinados los parámetros de fabricación en la sección anterior, se estudió el mismo efecto en una red tipo panal de abeja manteniendo constante la distancia entre sitios mientras se variaba el ángulo (Figura 5.3).

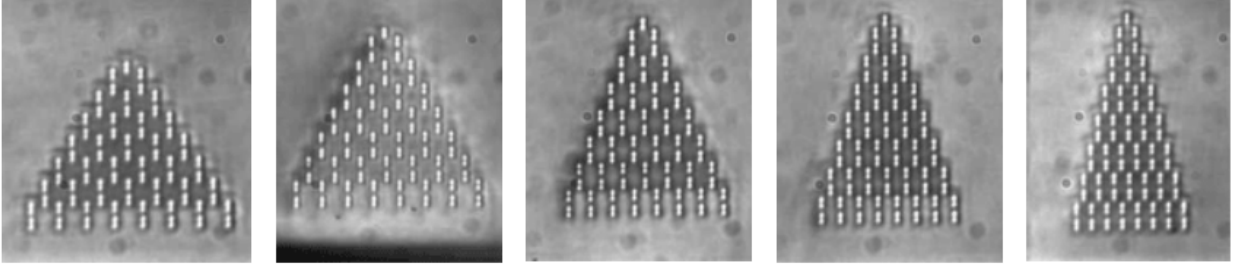


Figure 5.3: Imágenes de microscopio de redes fotónicas tipo panal de abeja para modos p_y bajo iluminación con luz blanca.

Un modelo de primeros vecinos consideraría únicamente el acoplamiento vertical κ_σ y el acoplamiento angular κ_θ . Sin embargo, la descripción precisa de los datos experimentales requiere incluir acoplamientos hasta terceros vecinos, así como correcciones por no ortogonalidad. La Figura 5.4 muestra un esquema del modelo completo. El Hamiltoniano del sistema se expresa como:

$$\hat{H}_\Sigma = \hat{H}_{NN} + \hat{H}_{NNN} + \hat{H}_{NNNN}, \quad (5.1)$$

donde $\hat{H}_{NN}$, $\hat{H}_{NNN}$ y $\hat{H}_{NNNN}$ contienen los acoplamientos a primeros vecinos (κ_σ y κ_θ), segundos vecinos (t_π y $t_{\theta 2}$), y terceros vecinos ($t_{\theta 1}$ y $t_{\theta 3}$), respectivamente (ver Figura 5.4).

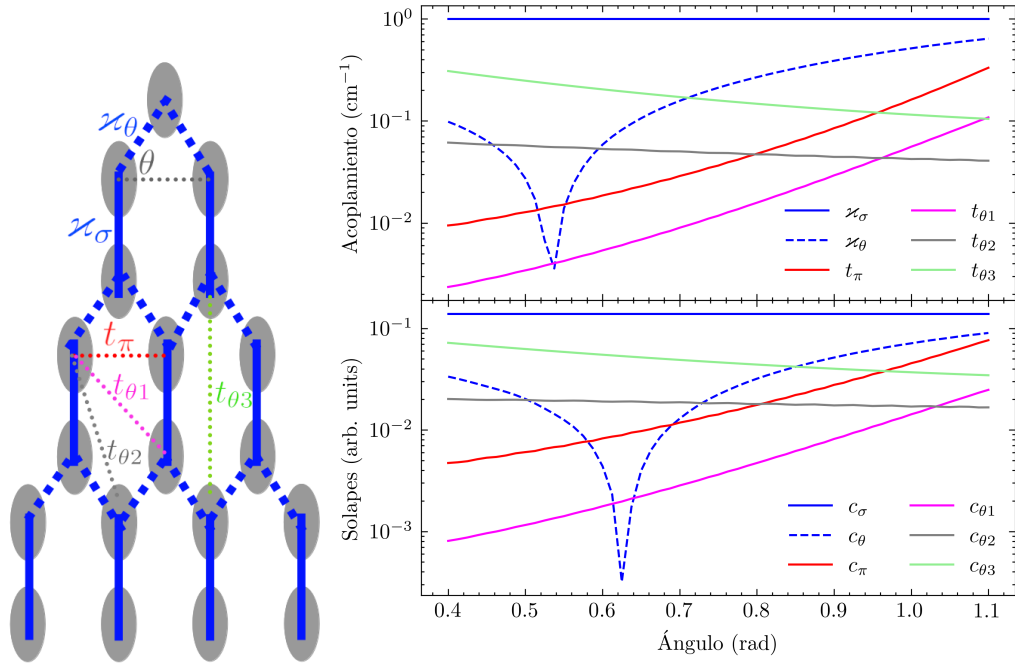


Figure 5.4: **Izquierda:** Esquema del modelo de panal de abeja para modos p_y con acoplamientos hasta terceros vecinos. **Derecha:** (Arriba) Valores absolutos de los acoplamientos en función del ángulo θ . (Abajo) Integrales de solapamiento normalizadas en función del ángulo θ .

Además, se define la matriz no ortogonal $\hat{c}$ a partir de la matriz $\hat{P}$ [ecuación (3.6)] como $c_{ij} = P_{ij}/I$, donde $I = P_{ii}$ es la intensidad del modo p_y , de modo que los elementos diagonales de $\hat{c}$ sean iguales a 1. De manera similar a los acoplamientos, en la Figura 5.4 se grafica la magnitud de los solapes c_{ij} en función del ángulo.

El Hamiltoniano no hermítico $\hat{H}'_\Sigma \equiv \hat{c}^{-1} \hat{H}_\Sigma$ describe la dinámica de la red e incorpora tanto efectos no ortogonales como acoplamientos de largo alcance. En la Figura 5.5 se muestra el espectro en función del ángulo θ . Se distinguen principalmente dos regiones: una para ángulos $0.4 < \theta < 0.7$, en la que los autovalores se agrupan en tres bandas cuasiplanas, y otra para ángulos $0.7 < \theta < 1.1$, en la que la brecha disminuye en magnitud y, por tanto, se incrementa el transporte en la red. Se calcula el grado de participación inverso (IPR, por sus siglas en inglés) de cada autoestado como $\text{IPR} = \sum_i I_i^2 / (\sum_i I_i)^2$. Esto permite la descripción adecuada de los datos experimentales mostrados en el panel derecho de la Figura 5.5.

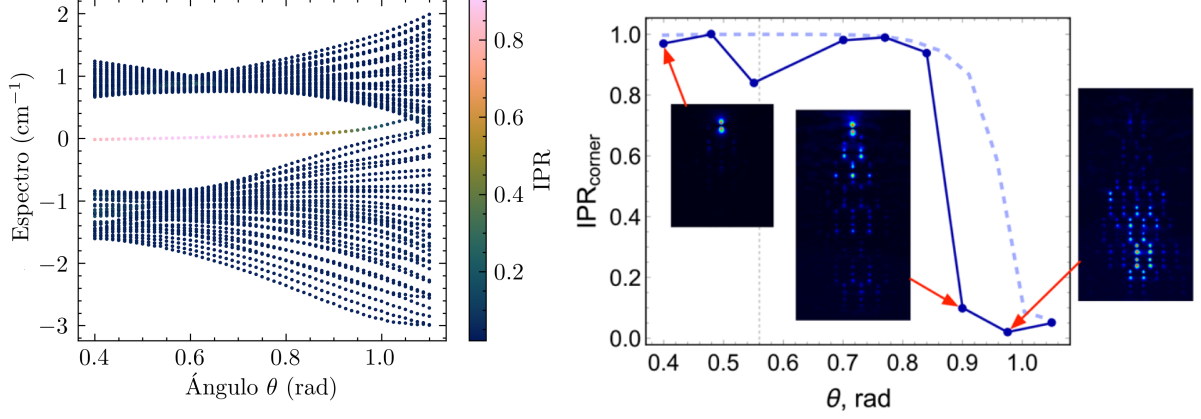


Figure 5.5: **Izquierda:** Espectro de la matriz $\hat{H}_\Sigma$ en función del ángulo. Cada punto se colorea según su grado de participación inverso (IPR). **Derecha:** Grado de participación inverso para la excitación del sitio de esquina en función del ángulo θ .

La fuerte localización observada al excitar el sitio de esquina se explica porque la proyección de la condición inicial sobre los autoestados corresponde mayoritariamente al autoestado de autovalor nulo. La existencia de este autoestado se debe a la topología que presenta el Hamiltoniano $\hat{H}_\Sigma$ en el espacio recíproco o de Bloch. Si bien es posible calcular la polarización de bulto mediante bucles de Wilson y determinar que su valor está discretizado como $\mathbf{P} = (0.5, 0.5)$, resulta más ilustrativo notar que la red puede descomponerse en cadenas cuasi-SSH (ver Figura 5.6). En ambos esquemas, la cuantización es consecuencia de la simetría de inversión presente en la red.

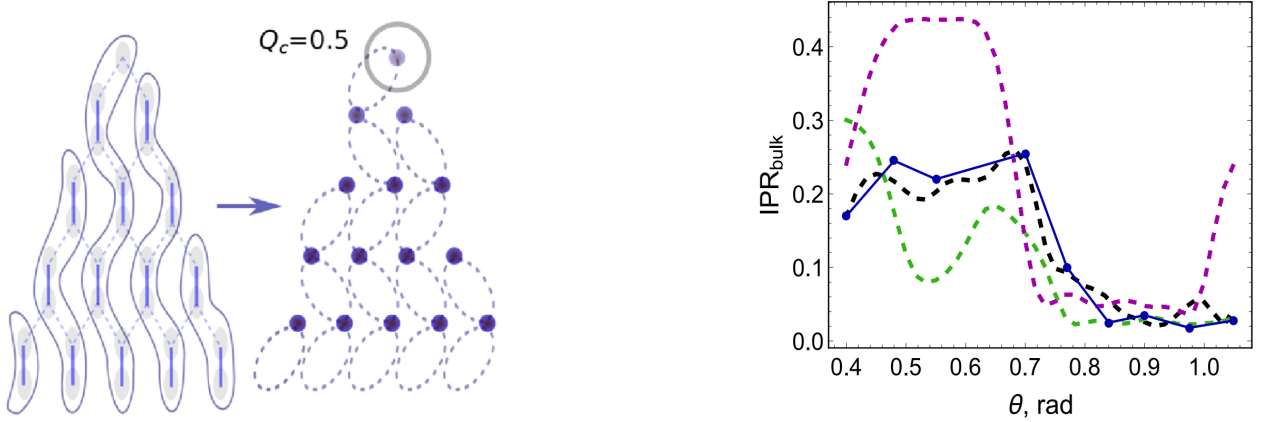


Figure 5.6: **Izquierda:** Descomposición de la red en el límite de bandas planas en cadenas SSH de dímeros aislados. **Derecha:** IPR para la excitación experimental del sitio central de la red (puntos unidos con líneas para guiar la vista) y simulados numéricamente para tres casos: sin acoplamientos de largo alcance ni correcciones no-ortogonales (curva morada discontinua), con acoplamientos de largo alcance pero sin correcciones no-ortogonales (curva verde discontinua), y con acoplamientos de largo alcance y correcciones no-ortogonales (curva negra discontinua).

Se estudió experimentalmente la dinámica asociada a bandas planas mediante la excitación localizada de un único sitio en el bulto de la red, con el objetivo de extraer el valor del *Inverse Participation Ratio* (IPR). Debido a que una excitación puntual tiene componentes en todas las bandas de Bloch, este tipo de preparación inicial amplifica las diferencias en la dispersión de cada banda. En este contexto, una propagación con muy poca difracción indica que todas las bandas involucradas son casi planas. Este enfoque permite observar de forma directa las propiedades del bulto de la red, ya que una excitación central evita efectos de borde y favorece la manifestación de la estructura modal global. En el límite ideal, un sistema compuesto únicamente por bandas perfectamente planas no permitiría transporte transversal, restringiendo la evolución dinámica a oscilaciones locales entre dímeros efectivos.

Los datos analizados revelan dos regímenes distintos: para redes con $0.4 < \theta < 0.8$ (cercanas al ángulo de invisibilidad) se observa un buen efecto de confinamiento (*caging*). Sin embargo, ángulos mayores ($\theta > 0.8$) favorecen una difracción significativa del campo, resultando en una deslocalización a la salida. Estos resultados experimentales (puntos en Figura 5.6) concuerdan con los cálculos teóricos (línea discontinua). Contrario a lo esperado, los acoplamientos de largo alcance y las correcciones por no ortogonalidad no degradan la dinámica de bandas planas cerca del ángulo de invisibilidad, sino que se complementan. Los valores máximos de IPR (0.20-0.25) para el estado dímero después de tres ciclos de confinamiento Aharonov-Bohm [50] indican una localización comparable con propuestas ópticas unidimensionales con todas sus bandas planas (ABF).

En conclusión, este estudio experimental demuestra que el *ángulo de invisibilidad* —definido como la condición angular para la supresión del acoplamiento dipolar entre modos P — constituye una herramienta eficaz para controlar el transporte en redes fotónicas discretas. Al explotar la simetría transversal de los modos como recurso activo, se logra inducir confinamiento óptico sin necesidad de desórdenes, no linealidades ni campos externos. Estos resultados fortalecen el marco conceptual de las redes multiorbitales como plataformas versátiles para estudiar dinámica de bandas planas y confinamiento controlado mediante geometría y simetría modal.

6. Acoplamiento Evanescente No Simétrico

Los modelos discretos en fotónica, como la teoría de modos acoplados (CMT), suelen asumir acoplamientos simétricos entre guías de onda, es decir, que la constante de acoplamiento entre dos sitios $C_{i \rightarrow j}$ es igual a $C_{j \rightarrow i}$. Esta suposición garantiza la hermiticidad del Hamiltoniano efectivo y suele estar justificada cuando las guías son idénticas y lo suficientemente separadas. Sin embargo, esta simetría puede romperse de forma inherente en sistemas reales, incluso en ausencia de pérdida o ganancia, cuando las guías presentan diferentes perfiles de índice de refracción, lo que experimentalmente se logra cambiando la potencia de escritura de las guías de ondas [51].

6.1 Origen de la dinámica no simétrica

El acoplamiento evanescente entre guías de ondas se produce por la superposición de las colas modales fuera del núcleo de confinamiento. Cuando dos guías tienen distintos contrastes de índice ($\Delta n_1 \neq \Delta n_2$), los modos guiados presentan colas evanescentes con diferente amplitud y extensión, lo que induce un acoplamiento direccionalmente desigual: $C_{1 \rightarrow 2} \neq C_{2 \rightarrow 1}$.

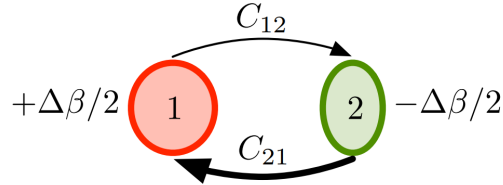


Figure 6.1: Esquema de acoplamiento no simétrico.

Este comportamiento *no simétrico* puede cuantificarse mediante la teoría de modos acoplados generalizada y no ortogonal, desarrollada en la sección 2.5, donde se incluye el efecto de solapamiento modal a través de la matriz $\hat{P}$, que codifica la normalización de potencias.

Considerando un dímero compuesto por dos guías no idénticas, las ecuaciones dinámicas para las amplitudes modales se escriben (véase la ecuación (2.36)) como:

$$-i \frac{d}{dz} \begin{pmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} H_{11} & H_{12} \\ H_{12} & H_{22} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \iff -i \frac{d}{dz} \mathbf{v} = \hat{H}' \mathbf{v}, \quad (6.1)$$

donde $\mathbf{v} = \hat{P}^{1/2} \mathbf{u}$ representa una base ortogonalizada, y $\hat{H}' = \hat{P}^{-1/2} \hat{H} \hat{P}^{-1/2}$ es el Hamiltoniano efectivo en dicha base. Como $\hat{H}$ y $\hat{P}$ son simétricas, $\hat{H}'$ también lo es, lo que garantiza autovalores reales. Al proponer soluciones estacionarias del tipo $\mathbf{v}(z) \sim e^{i\lambda z}$, se obtiene el problema generalizado de autovalores $\det(\hat{H} - \lambda \hat{P}) = 0$. Es importante destacar que esta

base ortogonalizada $\mathbf{v}$ no es accesible experimentalmente: en un montaje real, sólo es posible observar modos localizados en guías individuales, lo cual corresponde a la base física $\mathbf{u}$. Por lo tanto, el análisis experimental de la dinámica debe realizarse en esta base, donde la evolución está gobernada por la matriz de acoplamiento efectiva:

$$\hat{C} = \hat{P}^{-1} \hat{H} = \frac{1}{P_{11}P_{22} - P_{12}^2} \begin{pmatrix} P_{22}H_{11} - P_{12}H_{12} & P_{22}H_{12} - P_{12}H_{22} \\ P_{11}H_{12} - P_{12}H_{11} & P_{11}H_{22} - P_{12}H_{12} \end{pmatrix}.$$

Este sistema exhibe una dinámica no recíproca: la eficiencia de transferencia de energía depende del canal de entrada. Esta asimetría puede cuantificarse experimentalmente a través del *desbalance de intensidad*, definido para cada excitación $i = L, R$ como:

$$I^i(z) \equiv \frac{P_1^i(z) - P_2^i(z)}{P_1^i(z) + P_2^i(z)},$$

y su promedio absoluto como:

$$\bar{I}(z) \equiv \frac{|I^L(z) + I^R(z)|}{2}.$$

La cantidad $\bar{I}(z)$ resulta especialmente útil para detectar asimetrías globales en la dinámica: cualquier desviación de cero implica una evolución no simétrica, revelando la presencia de acoplamientos evanescentes no recíprocos.

6.2 Validación numérica

Utilizando el método de Expansión en Modos Normales (EME, Sección 3.1), se simuló la dinámica de dímeros asimétricos inyectando la misma condición inicial gaussiana en todos los casos. En la Figura 6.2 se muestra el promedio absoluto de desbalances de intensidad $\bar{I}(z)$ en función de la distancia para $n_2 - n_1 = 1.75 \times 10^{-5}$ (asimetría débil) y para $n_2 - n_1 = 5.00 \times 10^{-5}$ (asimetría fuerte).

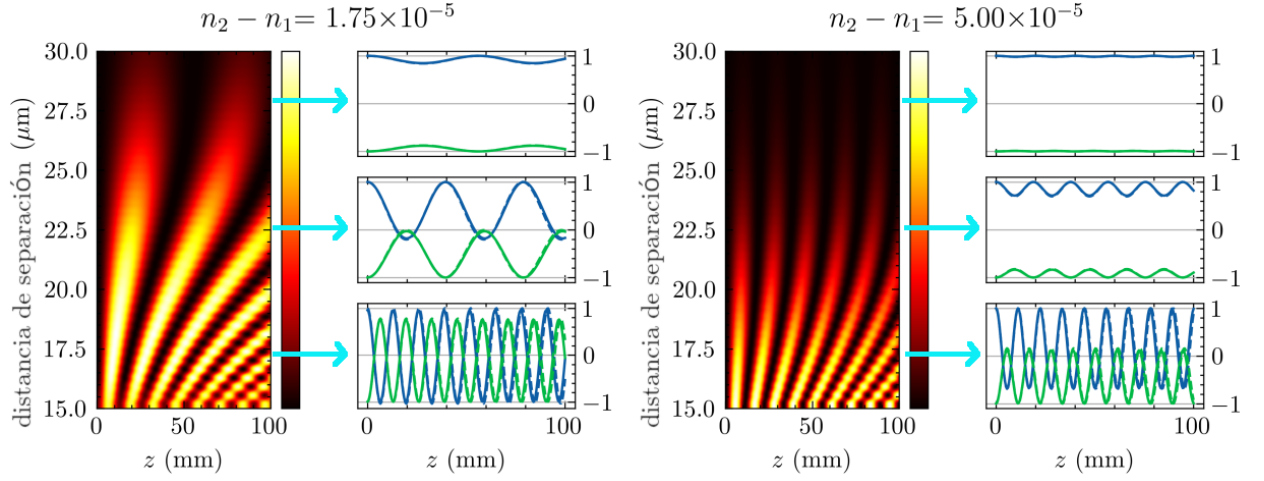


Figure 6.2: Promedio absoluto de desbalances de intensidad $\bar{I}(z)$ en función de la distancia de propagación z y de la distancia de separación entre guías, comparando dos regímenes: (1) asimetría débil ($\Delta n = 1.75 \times 10^{-5}$) y (2) asimetría fuerte ($\Delta n = 5.00 \times 10^{-5}$). Se incluyen los valores I^L (verde) e I^R (azul) correspondientes a tres casos específicos de estudio.

6.3 Comprobación experimental

Para comprobar este efecto, se fabricaron dímeros mediante la técnica de escritura por láser femtosegundo descrito en la sección 4.1, variando el canal de entrada i , la potencia de escritura P_g^i , la distancia entre guías d , y la longitud de propagación efectiva z . En particular, se exploró el caso $\Delta P = P_g^1 - P_g^2 = 5 \text{ mW}$, con separación $d = 18 \mu\text{m}$. La Figura 6.3 muestra que la excitación de la guía superior favorece la transferencia energética hacia la guía inferior, mientras que la excitación inversa alcanza una distribución energética 1:1 para $z \sim 11 \text{ mm}$.

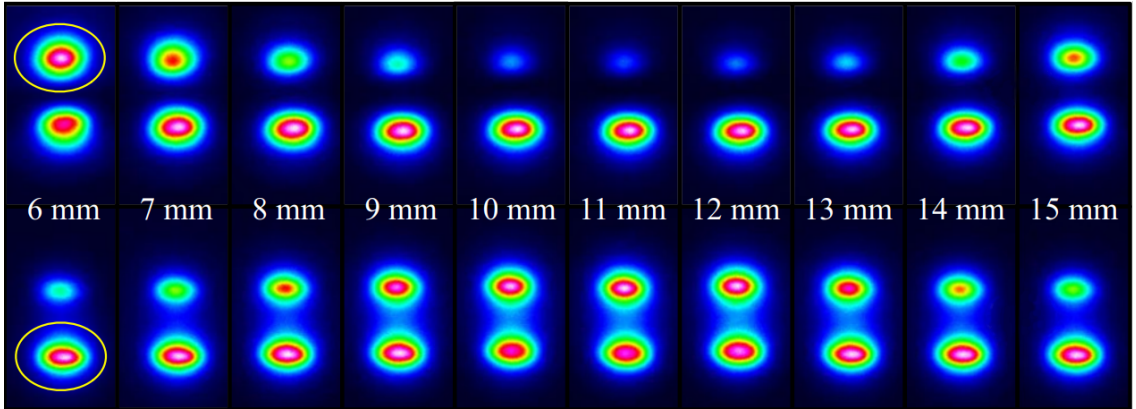


Figure 6.3: Perfiles de salida para distintas longitudes de propagación z_i , excitando la guía superior (izquierda) e inferior (derecha), con $d = 18 \mu\text{m}$.

Mediante guías tipo placa (*slab*) y modos TE (ver Sección 2.2), se obtienen expresiones analíticas para los acoplamientos:

$$\frac{C_{21}}{C_{12}} \approx \left(\frac{\Delta n_2}{\Delta n_1} \right)^2 e^{(\gamma_2 - \gamma_1)d} \approx \frac{P_2^1(z)}{P_1^2(z)},$$

donde γ_i representa el coeficiente de decaimiento evanescente del modo i . Esta expresión muestra que la asimetría aumenta con la separación entre guías, con la diferencia entre decaimientos de los modos $\gamma_2 - \gamma_1$ y con el cuadrado de la razón entre contrastes de índice $\Delta n_2/\Delta n_1$.

6.4 Implicancias y aplicaciones

Este estudio demuestra que el acoplamiento no simétrico es un fenómeno físico que, ausencia de pérdida o ganancia energética, es posible inducir de forma pasiva. Sus implicancias incluyen: *ruptura de reciprocidad*, sin requerir materiales magneto-ópticos ni modulación temporal; *asimetría modal*, dado que la desigualdad entre C_{ij} y C_{ji} afecta la formación de estados simétricos/antisimétricos y la interferencia; *dispositivos ópticos direccionales*, tales como divisores de haz asimétricos o aislantes ópticos; y su *extensión a redes* con acoplamientos direccionados. En conjunto, estos resultados permiten extender la teoría de modos acoplados y abren nuevas oportunidades en el diseño de redes ópticas con funcionalidad direccional.

7. Moléculas Fotónicas

La técnica de escritura de guías de onda mediante pulsos láser ultracortos, descrita en el Capítulo 4.1, permite la fabricación tridimensional de estructuras fotónicas con precisión micrométrica. No obstante, esta técnica presenta limitaciones inherentes a la forma elíptica del foco del láser, lo que impone restricciones sobre los modos transversales que pueden ser eficientemente excitados y, en particular, sobre los acoplamientos intermodales posibles entre modos de distinta simetría, como los S y P [41].

Para sortear estas restricciones y ampliar los grados de libertad en el diseño de redes fotónicas, se recurre aquí a un enfoque inspirado en la física molecular: al posicionar dos guías de onda lo suficientemente cerca, sus modos pueden acoplarse de forma coherente, generando nuevos modos colectivos extendidos. Este proceso es análogo a la formación de moléculas en física cuántica, donde dos orbitales atómicos se combinan linealmente para formar orbitales moleculares enlazantes y antienlazantes (ver Figura 7.1).

En este trabajo, llamaremos *molécula fotónica* a un sistema de dos guías de onda acopladas, cuyas funciones propias se combinan generando modos híbridos simétricos y antisimétricos, en analogía con los orbitales moleculares. Esta analogía se basa en la forma espacial de los modos y en su superposición controlada, pero no implica la existencia de fuerzas de enlace, electrones compartidos ni potenciales interatómicos. La relación con las moléculas cuánticas es, por tanto, formal y geométrica, y se restringe al paralelismo estructural entre los modos ópticos y los orbitales [29].

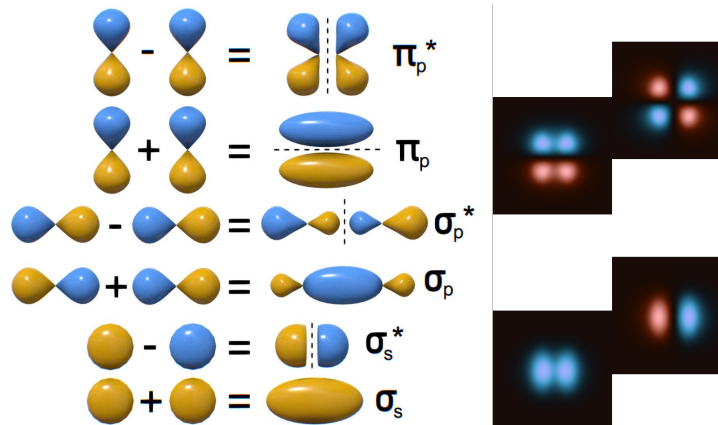


Figure 7.1: Analogía y diferencias entre moléculas cuánticas (izquierda) y fotónicas (derecha). Tanto los orbitales atómicos como los modos de guías individuales pueden hibridizarse, generando nuevos estados colectivos. En el caso fotónico, la hibridación ocurre entre modos ópticos guiados. En este caso, no hay análogo fotónico de los orbitales moleculares σ_p .

A partir de esta idea, en este capítulo se analiza experimentalmente una red fotónica construida a partir de dichas moléculas como bloques básicos. En particular, se estudia una red periódica tipo SSH [52], una extensión multiorbital del modelo de Su-Schrieffer-Heeger, que permite explorar el impacto del acoplamiento interorbital SP en la topología de bandas. El sistema presenta dos transiciones topológicas independientes, asociadas a la modulación de la dimerización geométrica y al desintonizado entre modos de diferente simetría. Estas transiciones se caracterizan mediante la observación de modos de borde, el cálculo de la polarización de bulto y el análisis espectral, permitiendo acceder a una rica fenomenología fotónica basada en la ingeniería modal.

7.1 Autoestados del acoplador fotónico para distancias de separación pequeñas

Como se mencionó en la Sección 3.3, la teoría de modos acoplados (CMT) describe adecuadamente los sistemas fotónicos en estudio cuando la distancia entre guías de ondas supera los $13\mu\text{m}$. Para separaciones menores, el sistema debe tratarse como una única macroguía. La expansión en modos normales (EME), presentada en la Sección 3.1, proporciona una herramienta numérica válida para ambos regímenes. Mediante simulaciones de pares de guías de onda a diferentes distancias, se caracterizó el comportamiento de sus autoestados. La Figura 7.2 muestra que el autoestado antisimétrico ($+-$) solo existe para distancias mayores a $13\mu\text{m}$, ya que por debajo de este umbral se convierte en un modo de radiación ($k_z^{+-} < k_0 n_0$). En cambio, el autoestado simétrico ($++$) persiste y puede describirse como una única entidad, lo que en esta tesis se denominará *modo S* de la molécula fotónica.

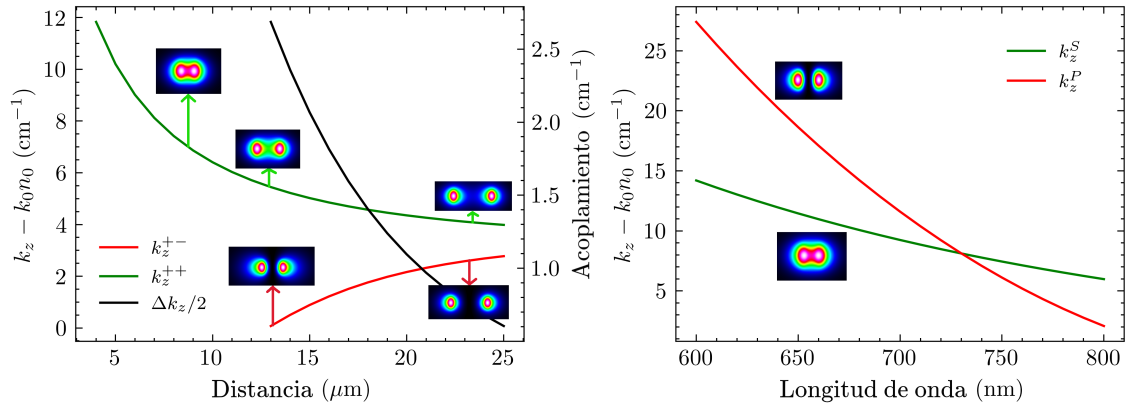


Figure 7.2: **Izquierda:** Constantes de propagación y acoplamientos en función de la distancia para modos fundamentales, calculados mediante EME. **Derecha:** Constantes de propagación de modos S y P en función de la longitud de onda.

Si bien las constantes de propagación de los modos S y P en una misma molécula presentan valores diferentes, estas pueden igualarse entre moléculas adyacentes mediante el ajuste del contraste de índice de refracción en las guías de onda [41, 52]. Por ejemplo, la Figura 7.2 muestra que a 730nm se igualan las constantes de propagación para $\Delta n_s = 5.200 \times 10^{-4}$ y

$$\Delta n_P = 10.423 \times 10^{-4}.$$

7.2 Moléculas Fotónicas en Red SP-SSH

Para la implementación experimental (Sección 4.1) de una red que presente acoplamiento SP [24, 41, 52], se utilizaron los modos P horizontales de la sección anterior, obtenidos mediante moléculas fotónicas. Un sintonizado preciso de las constantes de propagación de los modos S (k_z^S) y P (k_z^P) permitió considerar un grado de libertad análogo al del espín del electrón (*pseudospín*). El Hamiltoniano H de esta red [52], ilustrado en la Figura 7.3, es el siguiente:

$$\begin{aligned} H = \sum_n & \left[\frac{\delta\beta}{2} (p_{n,1}^* p_{n,1} - s_{n,1}^* s_{n,1} + p_{n,2}^* p_{n,2} - s_{n,2}^* s_{n,2}) + \kappa_{ss,2} s_{n,2}^* s_{n,1} - \kappa_{pp,2} p_{n,2}^* p_{n,1} \right. \\ & + \kappa_{ss,1} (s_{n-1,2}^* s_{n,1} + s_{n+1,2}^* s_{n,2}) - \kappa_{pp,1} (p_{n-1,2}^* p_{n,1} + p_{n+1,2}^* p_{n,2}) + \kappa_{sp,2} (s_{n,2}^* p_{n,1} - p_{n,2}^* s_{n,1}) \\ & \left. + \kappa_{sp,1} (s_{n-1,2}^* p_{n,1} - p_{n-1,2}^* s_{n,1} + s_{n+1,1}^* p_{n,2} - p_{n+1,1}^* s_{n,2}) \right] + \text{c.c.} \end{aligned}$$

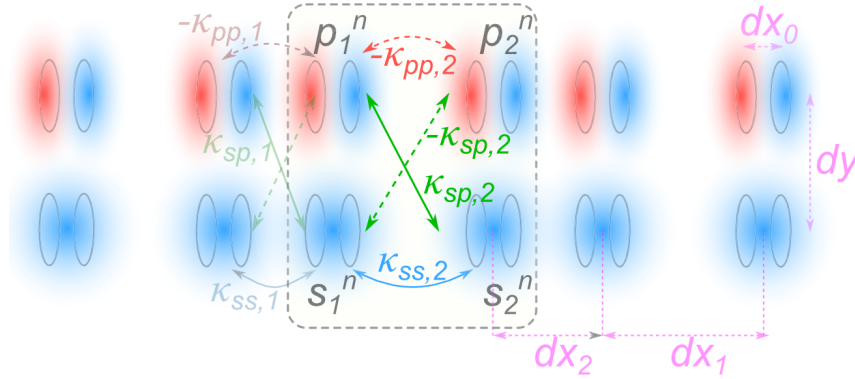


Figure 7.3: Esquema de la red SP-SSH y los acoplamientos considerados.

Si se elige la base $\mathbf{v}^n = (s_1^n, p_1^n, s_2^n, p_2^n)^T$, el Hamiltoniano de Bloch se puede escribir en bloques como:

$$\hat{H}(k) = \begin{pmatrix} \hat{\beta} & e^{-ika} \hat{\kappa}_{1-} + \hat{\kappa}_{2+} \\ e^{ika} \hat{\kappa}_{1+} + \hat{\kappa}_{2-} & \hat{\beta} \end{pmatrix},$$

donde $\hat{\beta} = \frac{k_z^S - k_z^P}{2} \hat{\sigma}_z$ y $\hat{\kappa}_{i\pm} = \begin{pmatrix} \kappa_{ss,i} & \pm \kappa_{sp,i} \\ \mp \kappa_{sp,i} & -\kappa_{pp,i} \end{pmatrix}$.

7.3 Topología en la Red SP-SSH

El modelo SP-SSH permite explorar nuevas fases topológicas más allá del modelo SSH convencional, ya que incorpora dos orbitales (modos S y P) cuyas combinaciones lineales pueden interpretarse como un pseudospín. Se hace necesario estudiar la topología del sistema puesto que el espectro del Hamiltoniano no basta para explicar completamente el comportamiento de este sistema, particularmente por la aparición de estados de borde a energía cero. Al

aplicar la transformación unitaria

$$\hat{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix},$$

se hace más evidente cómo interpretar el sistema. En particular, para $\Delta\beta = 0$ y $\kappa_{ss,1} = \kappa_{pp,1}$ se tiene

$$\hat{H} = \begin{pmatrix} \hat{H}_+ & 0 \\ 0 & \hat{H}_- \end{pmatrix},$$

con

$$\hat{H}_{\pm} = \begin{pmatrix} 0 & e^{-ika}(\kappa_{ss,1} \mp \kappa_{sp,1}) + (\kappa_{ss,2} \pm \kappa_{sp,2}) \\ e^{ika}(\kappa_{ss,1} \mp \kappa_{sp,1}) + (\kappa_{ss,2} \pm \kappa_{sp,2}) & 0 \end{pmatrix}.$$

Es decir, se tienen dos subsistemas tipo SSH, con acoplamiento intercelda (intracelda) $\kappa_{ss,1} \mp \kappa_{sp,1}$ ($\kappa_{ss,2} \pm \kappa_{sp,2}$). En el caso $\kappa_{ss,1} \neq \kappa_{pp,1}$ aún es posible capturar la topología mediante el determinante de $\hat{h} = e^{-ika}\hat{\kappa}_{1-} + \hat{\kappa}_{2+}$ (ver Figura 7.4) ya que en el plano complejo se puede contar el número de vueltas (*winding number*) en torno al origen al barrer todos los cuasimomentos en la primera zona de Brillouin. Estos bucles son una generalización de lo mostrado en la Figura 2.6 relativo al ángulo que forma la proyección en el plano xy del vector $\mathbf{d}(k)$.

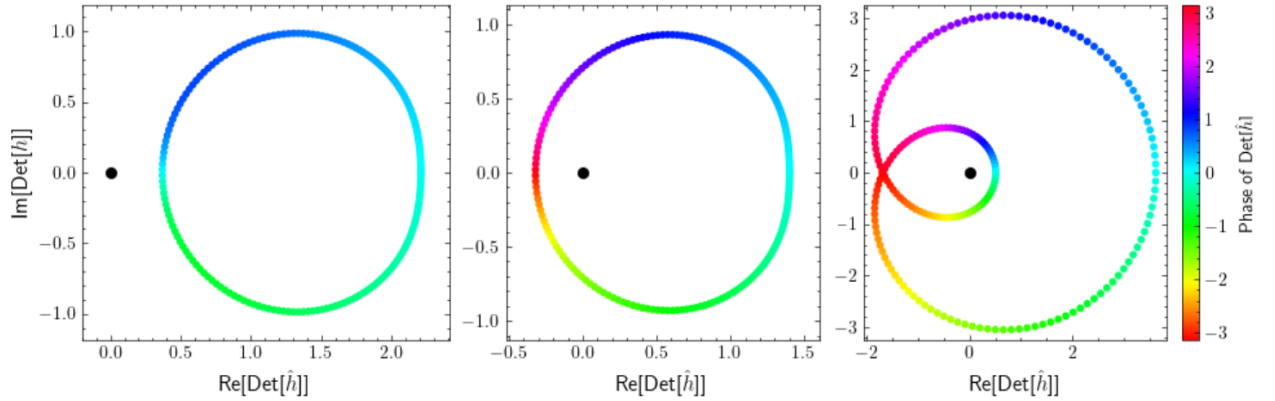


Figure 7.4: Topología de la Red SP-SSH para $\Delta\beta = 0$. Determinante de la matriz-bloque $\hat{h}$ para los cuasimomentos $-\frac{\pi}{a} \leq k < \frac{\pi}{a}$. Se aprecian exactamente tres fases topológicas distintas.

Para obtener todo el diagrama de fase (en particular, para $\Delta\beta \neq 0$), basta con variar $\Delta\beta$ y observar cuando hay cierre de brecha en las bandas, como se observa en la Figura 7.5. La red SP-SSH presenta dos mecanismos topológicos: uno debido a la hibridización SP y otro debido a la dimerización.

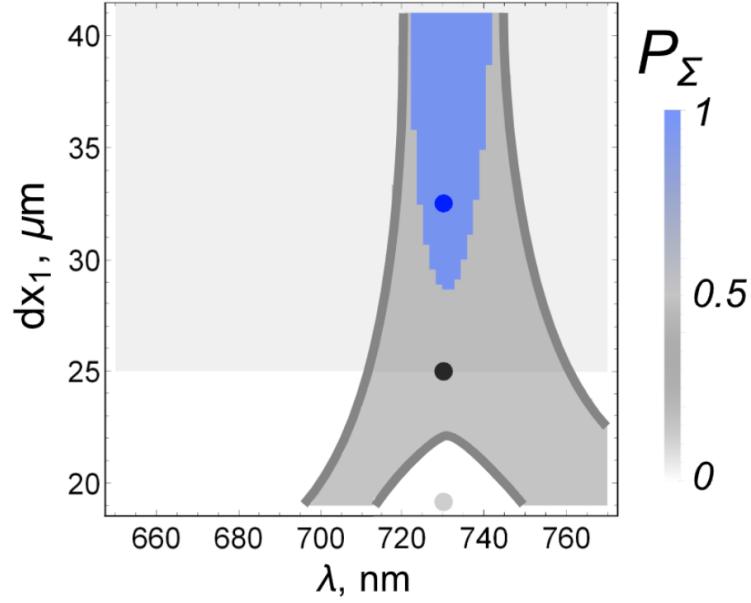


Figure 7.5: Diagrama de fases topológicas.

7.4 Implementación experimental

Para implementar experimentalmente el modelo SP-SSH, se fabricaron 25 redes fotónicas mediante la técnica de escritura con láser de femtosegundos descrita en el Capítulo 4.1. Los parámetros geométricos se ajustaron con separaciones entre moléculas: dx_1 variable, $dx_2 = 25 \mu\text{m}$, $dy = 20 \mu\text{m}$ y entre guías: $dx_0 = 7 \mu\text{m}$. Se optimizaron los contrastes de índice de refracción para lograr degeneración de modos: $k_z^S \sim k_z^P$ en $\lambda = 730 \text{ nm}$ [41]. Las mediciones se realizaron mediante barrido de longitud de onda (Capítulo 4.2) con $\lambda \in \{700, 780\} \text{ nm}$ y paso de 5 nm , con el objetivo de tener control del parámetro $\Delta\beta$, dependiente directamente de la longitud de onda utilizada. La excitación se aplicó a la guía S en un extremo de la red, midiendo la potencia relativa en el mismo extremo ($I_{\text{edge}}/I_{\text{total}}$). Aunque inicialmente no se distinguen claramente las fases topológicas con polarización de bulto¹ $P_\Sigma = 1/2$ o $P_\Sigma = 1$, el análisis numérico revela que durante la segunda transición topológica la razón $I_{\text{edge}}/I_{\text{total}}$ debería formar una meseta. La Figura 7.6 confirma este comportamiento, mientras que la Figura 7.7 muestra resultados experimentales representativos: Uno para una $\lambda = 680 \text{ nm}$, donde sólo se aprecia topología por dimerización y otro para $\lambda = 725 \text{ nm}$, donde ocurren dos transiciones topológicas al cambiar la dimerización.

¹invariante topológico que permite distinguir fases topológicas distintas.

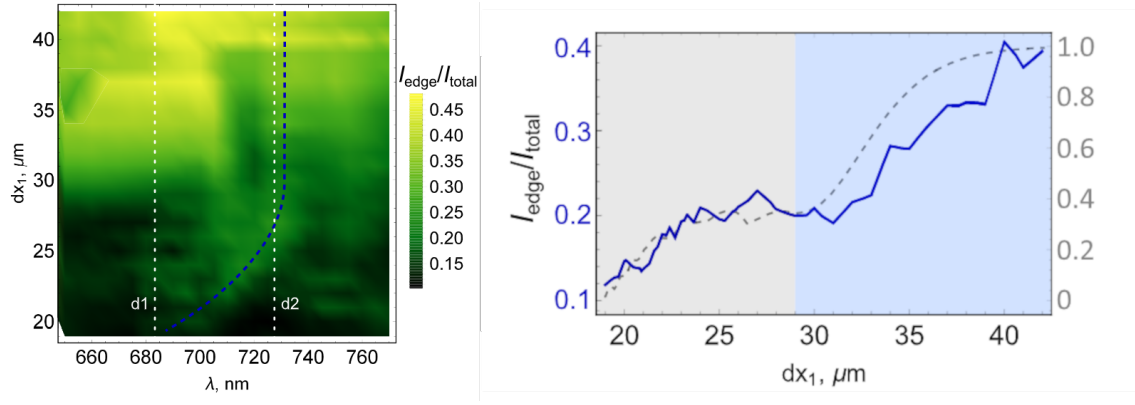


Figure 7.6: **Izquierda:** Resultados experimentales mostrando la meseta característica. **Derecha:** Comparación cuantitativa entre datos experimentales (línea continua) y simulaciones (línea discontinua).

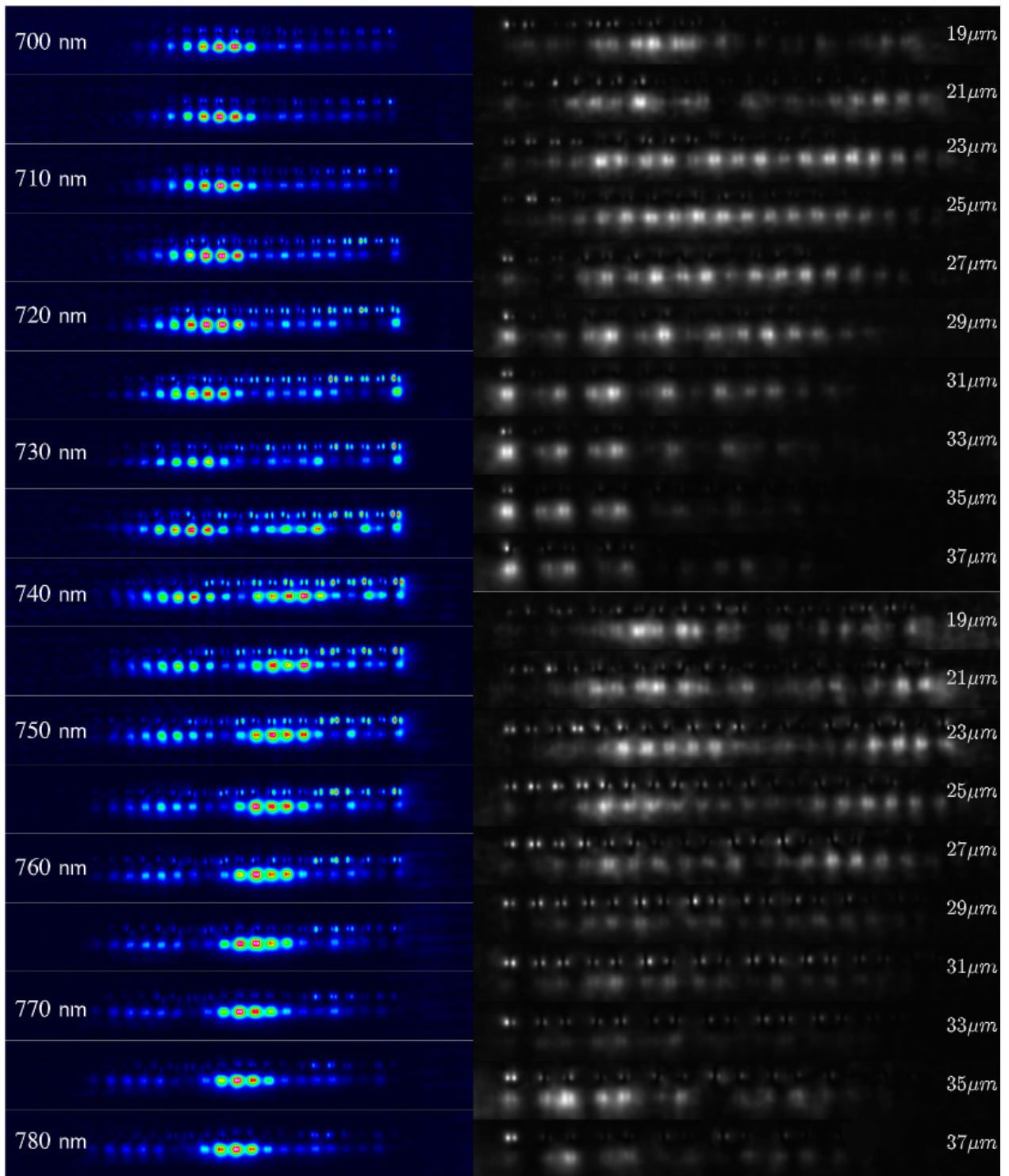


Figure 7.7: **Izquierda:** Red SP-SSH homogénea ($dx_1 = dx_2$) para diferentes longitudes de onda. **Derecha:** Arriba: Excitación en 680 nm para distintas dimerizaciones. Abajo: Misma excitación en 725 nm.

8. Conclusiones

Esta tesis ha investigado sistemáticamente las propiedades de redes fotónicas discretas con modos transversales no convencionales, centrándose en configuraciones que incluyen modos P , acoplamientos interorbitales SP , y acoplamientos evanescentes no simétricos. La investigación combinó herramientas analíticas, simulaciones numéricas y validación experimental para abordar diferentes manifestaciones de la estructura orbital de la luz guiada.

En el Capítulo 5, se caracterizó una red fotónica tipo panal de abeja compuesta por modos p_y , demostrando que el *ángulo de invisibilidad* induce un régimen de bandas cuasi-planas alrededor de $\theta \approx 0.56$ radianes. Mediante un análisis detallado que incorporó correcciones por no ortogonalidad y acoplamientos de largo alcance, se estableció que el confinamiento tipo Aharonov-Bohm persiste aún en presencia de estas perturbaciones. En particular, se comprobó que los efectos de no ortogonalidad son esenciales para describir correctamente la localización en los bordes de la red, lo cual fue cuantificado mediante el inverso del grado de participación (IPR) y corroborado experimentalmente.

El Capítulo 6 abordó el fenómeno de acoplamiento evanescente no simétrico en dímeros fotónicos con diferentes contrastes de índice de refracción. Se presentó una generalización de la teoría de modos acoplados que incluye términos de no ortogonalidad y permite modelar correctamente la dinámica direccional observada. Experimentalmente, se verificó un desbalance en la transferencia de energía dependiente del canal de entrada, en acuerdo con las predicciones teóricas basadas en matrices de acoplamiento efectivas no simétricas.

En el Capítulo 7, se estudió la implementación de redes SP-SSH construidas a partir de moléculas fotónicas. Se demostró que la dimerización estructural, combinada con la hibridación entre modos S y P , da lugar a dos transiciones de fase topológicas controlables de forma independiente. Se identificaron fases con distinta polarización de bulto cuantizada, y se diseñó un protocolo experimental para detectar modos de borde mediante la razón $I_{\text{borde}}/I_{\text{total}}$, en excelente acuerdo con simulaciones numéricas. Este modelo extiende la física del sistema SSH unidimensional al contexto multiorbital, revelando nuevos mecanismos de localización topológica mediados por grados de libertad internos.

Desde el punto de vista metodológico, esta tesis desarrolló un método numérico continuo, basado en la expansión en modos normales (EME), que permite extender la teoría de modos acoplados en el régimen de guías próximas o contrastes de índice de refracción arbitrarios. Este enfoque fue complementado con técnicas experimentales de fabricación y caracterización de redes multiorbitales que permitieron la calibración de constantes de propagación de modos

guiados distintos.

Los resultados obtenidos abren nuevas posibilidades en el diseño de dispositivos fotónicos que explotan mecanismos de localización y control basados en geometría, simetría transversal y topología. Asimismo, se establece un marco teórico-experimental adaptable a plataformas más complejas, por ejemplo, en sistemas con momentum angular orbital [42].

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Appendices

Appendix A Artículos publicados en orden cronológico

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Strain induced localization to delocalization transition on a Lieb photonic ribbon lattice

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Ribbon lattices are kind of transition systems in between one and two dimensions, and their study is crucial to understand the origin of different emerging properties. In this work, we study a Lieb ribbon lattice and the localization–delocalization transition occurring due to a reduction of lattice distances (compression) and the corresponding flat band deformation. We observe how above a critical compression ratio the energy spreads out and propagates freely across the lattice, therefore transforming the system from being a kind of insulator into a conductor. We implement an experiment on a photonic platform and show an excellent agreement with the predicted phenomenology. Our findings suggest and prove experimentally the use of compression or mechanical deformation of lattices to switch the transport properties of a given system.

The understanding of transport and localization properties of different materials is the most relevant aspect in solid-state physics, not only from a fundamental point of view but also in terms of concrete applications. When synthesizing materials some deformations could emerge during the process; such as, compression, strain, defects and/or dislocations. However, they can also be added deliberately to enhance or to induce certain transport properties. In particular, compression and strain of materials have driven much attention lately because their transport properties can be dramatically modified. For instance, graphene can go from a semimetallic to an insulating phase when an uniaxial compression is applied^{1,2}, phenomenon known as Lifshitz transition. In other two-dimensional (2D) materials similar transitions have been predicted. For example, black phosphorus switches from a semiconductor into a metal when it is subjected to an uniaxial strain³. This allows the control of the electronic transport properties on a nanodevice when an external electric field is applied⁴, which can be interpreted as a delocalization–localization transition. And, for germanium kagome lattices⁵ a transformation from a semimetallic into a semiconductor is observed when applying compression due to an increment of the orbital frustration that induces an electronic gap. Furthermore, the electronic structure and charge properties were studied in KCuSe and KCuTe⁶, finding that pressure effectively modifies the transport properties due to an enhancement of carrier mobility, which could have direct applications in optoelectronic technologies as, e.g., solar cells.

On the other hand, during the last decade, artificial lattices have arisen as feasible platforms to emulate and test most of the electronic properties predicted for solid-state-like materials^{7–14}. Some of these systems have shown the ability to carefully engineer compression and, thus, exploring interesting phenomena that are sometimes unrealizable in natural and synthesized materials. For example, the Lifshitz transition of graphene has been addressed using matter waves in optical lattices¹⁵, waveguides arrays¹⁶, arrays of microwave resonators¹⁷ and exciton-polariton lattices¹⁸. In the latter system, a predicted semi-Dirac scenario arises in graphene at a critical compression, which produces a highly anisotropic transport and particular localization features. Remarkable also, it has been experimentally shown in graphene photonic lattices that a smart design in term of compression or strain could induce a pseudomagnetic field, causing the rupture of Dirac cones and the appearance of Landau levels in the band structure^{19–21}, which constitutes a clear delocalization–localization transition.

Besides compression properties, very fundamental condensed-matter phenomena has been experimentally proved in photonic lattices; e.g., Anderson localization²², topological insulation²³, and Flat Band (FB) localization^{24,25}. FB lattices have become an ideal solution for observing transport and localization phenomena on a completely periodic and linear configuration^{26–28}, as well as for studies considering highly degenerated and interacting systems^{29,30}. Moreover, during the last years, several contributions in optics have demonstrated

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Photonic Molecule Approach to Multiorbital Topology

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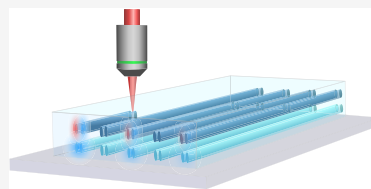
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ABSTRACT: The concepts of topology provide a powerful tool to tailor the propagation and localization of the waves. While electrons have only two available spin states, engineered degeneracies of photonic modes provide novel opportunities resembling spin degrees of freedom in condensed matter. Here, we tailor such degeneracies for the array of femtosecond laser written waveguides in the optical range exploiting the idea of photonic molecules: clusters of strongly coupled waveguides. In our experiments, we observe unconventional topological modes protected by the Z_3 invariant arising due to the interplay of interorbital coupling and geometric dimerization mechanism. We track multiple topological transitions in the system with the change in the lattice spacings and excitation wavelength. This strategy opens an avenue for designing novel types of photonic topological phases and states.

KEYWORDS: femtosecond laser writing, photonic orbitals, interorbital coupling, topological photonics, topological edge states



Topological photonics provides a versatile approach to shape the light flows by harnessing the topology of photonic bands.^{1–3} This enables disorder-robust topological states which can be achieved in plasmonic,⁴ photonic,^{5–7} or polaritonic^{8,9} systems operating from radiofrequency^{10,11} to the optical^{5,12} domain.

Experimental realization of optical topological structures is especially demanding because of relatively small feature size, narrow topological bandgaps, and losses. The major strategies to create topological structures for the optical domain include arrays of coupled ring resonators with engineered couplings^{6,13,14} or crystalline topological structures.^{7,12,15–17} While the former approach covers mostly the infrared range being hard to scale toward visible wavelengths, the latter strategy requires carefully designed lattice symmetry and does not provide much flexibility in reconfiguring the topological properties, creating pseudospin degrees of freedom, or tailoring effective spin–orbit interactions. Therefore, these approaches do not unlock the full functionalities available for photonic topological structures.

A recently suggested alternative is to optimize the responses of the *individual meta-atoms* achieving an accidental degeneracy of their modes.^{18–27} Such degenerate modes play a role analogous to the orbital or spin degrees of freedom in condensed matter or cold atom optical lattices.^{28–32} Since the number and the symmetry of the degenerate modes can be tailored on demand, this opens up a rich plethora of novel topological phases. While this appealing strategy can be implemented directly at microwave frequencies,³³ its realization in the visible range is severely hindered due to the fabrication limitations.

Here, we resolve this difficulty by experimentally realizing the concept of *photonic molecules*, a set of closely spaced waveguides which together give rise to photonic orbitals with different symmetries. Employing the femtosecond laser writing technique^{34–36}—an approach that has been consolidated as a key technique in physical lattice studies^{37–40}—to control the refractive index contrast, we create multiorbital photonic lattices such that fundamental *s* (symmetric or bonding) and horizontal excited *p* (antisymmetric or antibonding) modes become degenerate and interact with each other enabling rich physics.

The topological states in such lattice are protected by the Z_3 invariant and arise due to the combination of interorbital coupling and geometric dimerization mechanism. When those mechanisms add up constructively, the array hosts a pair of topological states at each end, which is a unique feature of optical lattices in the absence of long-range interactions. In our experiments, we track the topological transitions upon a change of the wavelength and lattice spacing observing multiple topological states.

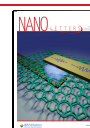
The simplest photonic molecule consists of a couple of closely spaced identical elliptical waveguides [see Figure 1a–top], which give rise to the pair of symmetric (*s*) and antisymmetric (*p*) horizontal modes with the typical intensity

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Observation of Invisibility Angle and Flat Band Physics in Dipolar Photonic Lattices

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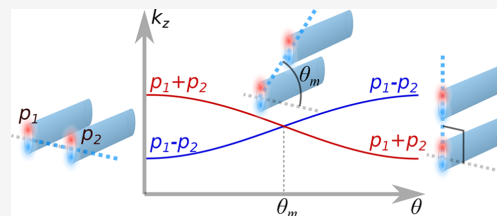
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ABSTRACT: Evanescently coupled waveguide arrays provide a tabletop platform to realize a variety of Hamiltonians, where physical waveguides correspond to the individual sites of a tight-binding lattice. Nontrivial spatial structure of the waveguide modes enriches this picture and uncovers further possibilities. Here, we demonstrate that the effective coupling between p -like modes of adjacent photonic waveguides changes its sign depending on their relative orientation vanishing for proper alignment at a so-called *invisibility angle*. Using femtosecond laser-written waveguides, we demonstrate this experimentally for p -mode dimers and graphene-like photonic lattices exhibiting quasi-flat bands at this angle. We observe diffraction-free propagation of corner and bulk states, providing robust experimental evidence of a two-dimensional Aharonov–Bohm-like caging in an optically switchable system.

KEYWORDS: flat bands, Aharonov–Bohm caging, photonic orbitals, laser-written waveguides



Photonic lattices have emerged as a versatile platform to explore localization and transport phenomena.^{1–3} As the paraxial equation governing light propagation in a single waveguide resembles Schrödinger equation in quantum mechanics, arrays of evanescently coupled waveguides provide a straightforward realization of paradigmatic tight-binding models⁴ with the possibility to accommodate such exotic features as negative couplings,^{5,6} artificial gauge fields,^{2,7} and non-Hermitian physics.^{8–10}

The development of this platform exhibits profound parallels with condensed matter physics and materials science¹¹ as illustrated in Figure 1. On a qualitative level, the formation of a material can be sketched as a coupling of atoms and formation of molecules; hybridization of molecular orbitals results in molecular aggregates and, finally, in the emergence of a 2D material. In the same spirit, the waveguides can be strongly coupled to each other to form photonic molecules.¹² The collective modes of those molecules with distinct symmetry can be readily identified as different orbitals, and their overlap determines the properties of a given photonic lattice. This parallel appears to be especially fruitful in light of the recent studies which demonstrated photonic molecule concept experimentally¹² and predicted a vast range of novel functionalities, including topological phases^{13,14} mediated by the coupling between different photonic orbitals.^{15–18}

Here, we add a crucial ingredient to this picture by introducing an *invisibility angle* in the photonic lattices. Specifically, we demonstrate theoretically and experimentally that a suitable arrangement of optical waveguides hosting p -type orbital modes results in a vanishing effective coupling between them. As a result, the lattice of such waveguides

features dispersionless bulk modes unlocking an all-bands-flat regime (ABF).^{19–21} A distinctive feature of this regime is photonic Aharonov–Bohm (AB) caging when an arbitrary initial state does not diffract into the bulk of the lattice, remaining confined in a finite spatial region, oscillating, and periodically recovering its initial profile.

The idea of an invisibility angle has some parallels with the magic angle concept in condensed matter literature,²² where it was introduced to denote unconventional electronic properties of twisted bilayer graphene, with flat bands and resulting superconducting behavior.^{23,24} The advancement of this concept resulted in the field of twistoronics²⁵ spanning such disciplines as condensed matter physics, polaritonics, and photonics.^{26,27}

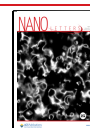
The origin of the flat bands can be traced to the compact localized states,^{28–31} which are perfectly confined lattice eigenmodes. Typically, flat bands in tight-binding lattices arise due to the destructive interference in the so-called connector sites^{32–35} and require fine-tuning of the lattice parameters.^{19,36} However, having an *all band flat* regime is much less trivial. While in optical one-dimensional systems the ABF regime could be achieved by several means including multiorbital mechanisms^{6,15,16,37–39} or effective magnetic

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Nonsymmetric evanescent coupling in photonics

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Asymmetrical interactions are ubiquitous in nature, and in physics their study becomes fundamental. Despite the prevalence of evanescent coupling in physics, little attention has been paid to wave function profiles, with symmetrical reciprocal interactions often assumed for practical reasons. This work challenges that assumption by analytically demonstrating the origin of nonsymmetrical coupling in a photonic platform, focusing on the behavior of evanescent tails from adjacent waveguides. We experimentally validate an asymmetrical dynamics by studying detuned photonic directional couplers, fabricated via femtosecond laser writing, and corroborate our findings through continuous numerical simulations.

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I. INTRODUCTION

A fundamental concept in lattice science is the study of localized wave functions bounded within well-defined spatial regions, such as waveguides or atoms. The tight-binding approximation [1] describes interactions between tightly bound orbitals at a given site that weakly interact with surrounding wave functions. This approach aims at examining the effects of weak coupling with neighboring sites. The approximation holds for well-separated sites where wave functions do not significantly overlap, as opposed to a strongly interacting regime [2,3]. In optics, analogous equations arise from coupled mode theory (CMT) [4], leading to key observations in transport and localization phenomena [5–9]. Here, the interwaveguide distance must exceed the waveguide widths to ensure minimal wave function distortion, allowing interactions between adjacent guided modes solely through evanescent fields. For more than two decades, research in photonic lattice phenomena [10–12] has predominantly assumed reciprocal coupling between waveguides. This assumption implies that the coupling constant between two waveguides is identical in both directions. Attempts to introduce nonreciprocity have mainly focused on simulating effective magnetic fields [9,13], which add a phase to the hopping terms but do not directly alter the wave function itself.

A nonreciprocal directional coupler (NRDC) was originally formulated long time ago in the context of resonant circuits [14]. Then, in the 1970s optical nonreciprocal devices started to be explored for integrated optics, and proposals for mode converters, isolators, and phase shifters were described [15]. Then, the validity of CMT was a main focus of research [16–19] looking for a better correspondence of the discrete model with exact numerical simulations. Some corrections to the simplest version of CMT were mandatory, with nonsymmetrical coupling constants being substantially different for nonidentical waveguides. At the end of the 1990s, magneto-optical waveguides were proposed for the construction of nonreciprocal Mach-Zehnder interferometers [20], and more

recently on-chip nonreciprocal optical resonators have been implemented in silicon [21]. Time modulation has been also suggested to implement nonreciprocity in compact integrated photonics, for example, using ringlike resonators or optomechanical cavities [22].

In this work, we demonstrate nonsymmetrical coupling interactions in photonic detuned dimers. We develop a standard formula for coupling coefficients considering steplike index waveguides, revealing the fundamental effect of evanescent profiles from neighboring waveguides. Our analysis shows that, opposite to the standard assumption, asymmetry is indeed an inherent property of photonic coupled systems, and it originates from the difference in refractive index profiles of two distant waveguides. Using the femtosecond (fs) laser writing technique [23], we fabricate photonic detuned couplers and experimentally demonstrate a nonsymmetrical dynamics. We corroborate all our observations through numerical simulations, without imposing any additional constraint but waveguide refractive index contrasts.

II. NONSYMMETRICAL PHOTONIC COUPLER MODEL

Let us start by studying a photonic coupler, such as the one sketched in Fig. 1(a), consisting of two waveguides separated a distance d . Coupled mode theory [24] generates different integrals which account for effective field interactions originated from the assumption that the total field is written as a linear combination of the modes at waveguides 1 and 2: $E_T = E_1 + E_2$. By assuming one-dimensional steplike index waveguides, we find different TE₀₀ (transverse electric) modes, such as the ones sketched in Fig. 1(b), for the indicated refractive index contrasts Δn_i . In this case, mode profiles at the waveguide region (core) are proportional to $\cos(\alpha_i x)$, while outside (cladding) they decay as $\exp(-\gamma_i |x|)$. Fields are better confined (larger α , γ) for a growing Δn , and the neighbor field will be larger at the stronger waveguide [see Fig. 1(b) inset]. Therefore, we naturally expect to observe an asymmetric coupling interaction $C_{ij} \neq C_{ji}$ for inhomogeneous

Observation of Non-Markovian Radiative Phenomena in Structured Photonic Lattices

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The spectral structure of a photonic reservoir shapes radiation phenomena for embedded quantum emitters. We implement an all-optical analog to study such an effect, particularly to observe the non-Markovian radiation dynamics of an emitter coupled to two-dimensional structured reservoirs. Its dynamics is simulated by light propagating through a photonic lattice, acting as a reservoir for an adjacent waveguide that mimics a coupled quantum emitter. We study radiation dynamics in square and Lieb lattices under different coupling regimes and observe how the flat band properties of the Lieb lattice significantly enhances light-matter coupling and non-Markovianity. Our platform opens a path for the experimental exploration of single-photon quantum optical phenomena in structured reservoirs to enhance light-matter interactions.

DOI: [10.1103/b3kw-vv3s](https://doi.org/10.1103/b3kw-vv3s)

The interaction between quantum emitters (QEs) and photonic modes plays a central role in quantum optics foundations and applications. Tailoring such interaction, by modifying the dispersion relation and dimensionality of the photonic medium [1–5], can lead to enhanced [6], suppressed [7], and directional interactions [8] and non-trivial radiation dynamics [9–12]. The band structure of the reservoir determines the density of states available to a coupled QE and, hence, its interaction strength. For example, placing a QE on resonance with the band edge of a photonics reservoir enhances light-matter interactions [13,14]. More recently, reservoirs with flat bands (FBs) that have, in general, a set of dispersive and nondispersive bands [15,16] with extended and compact states [17,18], respectively, have emerged as candidates to enhance such coupling [19–22].

Significant coupling and energy exchange between the QE and the reservoir manifest as non-Markovian radiative decay, a central example of the rich phenomenology lying beyond traditional quantum optics enabled by structured reservoirs [11,23–27]. While increasingly promising, embedding QEs in 2D materials with nontrivial band structures faces practical challenges that complicate experimental explorations of novel reservoir structures and collective effects of many indistinguishable emitters [28].

Photonic waveguide arrays (PWAs) offer a versatile platform to simulate QEs coupled to structured photonic reservoirs [10,29–32]. In such platforms, an array of evanescently coupled waveguides simulates a reservoir,

while a nearby individual waveguide plays the role of a QE, which decays into the reservoir (see Fig. 1). Light propagation through the whole PWA acts as an optical analog for the temporal evolution of the radiation phenomena [31]. In this sense, a PWA can reproduce quantum optical effects from cavity QED to waveguide QED in the weak and strong coupling regimes, in the same manner that coupled cavity arrays [33]. Although PWAs cannot simulate quantum systems in the nonlinear multiphoton regime, they can simulate collective effects of indistinguishable emitters [32] and the radiation dynamics of delayed-induced non-Markovian reservoirs [34]. Nonetheless, PWAs have an untapped potential to study the radiation phenomena of QEs coupled to arbitrary structured photonics reservoirs.

In this Letter, we experimentally study the decay dynamics of a single quantum emitter coupled to two-dimensional structured reservoirs, in an optical analog system. We

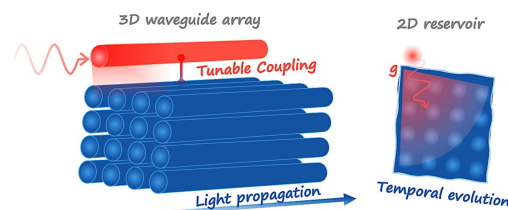


FIG. 1. Light propagating through a three-dimensional waveguide array simulates the temporal evolution of a quantum emitter coupled to a two-dimensional structured reservoir. Tunable couplings allow the experimental exploration of a wide range of coupling regimes.

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Appendix B Ortogonalidad de los modos normales

La ortogonalidad de los modos normales $\mathbf{E}_\nu^\perp$ puede demostrarse utilizando el convenio de Einstein en la ecuación (2.28) para simplificar la notación:

$$T_{ij}E_j^\nu = \beta_\nu^2 E_i^\nu, \quad (1)$$

donde el operador T_{ij} actúa como $T_{ij}E_j \equiv \delta_{ij}(\nabla_\perp^2 + k_0^2 n^2)E_j + \partial_i(E_j \partial_j \ln(n^2))$.

Para demostrar la ortogonalidad de los modos E_i , se puede demostrar equivalentemente que T_{ij} es hermítico:

$$\iint (T_{ij}E_j^\mu)^* E_i^\nu dA = \iint (E_i^\mu)^* (T_{ij}E_j^\nu) dA. \quad (2)$$

Esta propiedad se sigue de la identidad:

$$(T_{ji}E_i^\mu)^* E_j^\nu - (E_i^\mu)^* (T_{ij}E_j^\nu) = \nabla_\perp \cdot [E_i^\nu \nabla_\perp (E_i^\mu)^* - (E_i^\mu)^* \nabla_\perp E_i^\nu] + [E_j^\nu \partial_j (E_i^\mu)^* - (E_i^\mu)^* \partial_j E_j^\nu] \partial_i \ln(n^2).$$

Al integrar sobre el plano transversal el término de divergencia se anula para modos guiados (que decaen en el infinito), pero el segundo término desaparece sólo si $\nabla n^2 = \mathbf{0}$. Es decir, el operador T_{ij} sólo es Hermítico en la condición de guiage débil.

El sistema completo es hermítico, como se demuestra partiendo de la ecuación (2.5):

$$\begin{aligned} \iiint_V (\nabla \times \nabla \times \mathbf{E}^\nu) \cdot (\mathbf{E}^\mu)^* dV &= \iiint_V n^2 \frac{\omega_\nu^2}{c^2} \mathbf{E}^\nu \cdot (\mathbf{E}^\mu)^* dV \\ \iiint_V [\nabla \times \nabla \times (\mathbf{E}^\mu)^*] \cdot \mathbf{E}^\nu dV &= \iiint_V n^2 \frac{\omega_\mu^2}{c^2} (\mathbf{E}^\mu)^* \cdot \mathbf{E}^\nu dV \end{aligned}$$

Restando ambas ecuaciones y aplicando el teorema de divergencia:

$$\frac{\omega_\nu^2 - \omega_\mu^2}{c^2} \iiint_V n^2 \mathbf{E}^\nu \cdot (\mathbf{E}^\mu)^* dV = \oint_{\partial V} [\dots] \cdot \hat{\mathbf{n}} dA \xrightarrow{\partial V \rightarrow \infty} 0$$

Se obtienen así dos posibilidades: o los modos son degenerados ($\omega_\nu^2 = \omega_\mu^2$), o se tiene que $\iiint n^2 \mathbf{E}^\nu \cdot (\mathbf{E}^\mu)^* dV = 0$.

Appendix C Código en Python para cálculo de modos normales

1 # Scalar Eigenmode Solver

```

2
3 # Copyright (C) 2025  Diego Roman-Cortes
4
5 # This program is free software: you can redistribute it and/or modify
6 # it under the terms of the GNU General Public License as published by
7 # the Free Software Foundation, either version 3 of the License, or
8 # (at your option) any later version.
9
10 # This program is distributed in the hope that it will be useful,
11 # but WITHOUT ANY WARRANTY; without even the implied warranty of
12 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
13 # GNU General Public License for more details.
14
15 # You should have received a copy of the GNU General Public License
16 # along with this program.  If not, see <https://www.gnu.org/licenses/>.
17
18 # contact: diego.roman.c@ug.uchile.cl
19
20 import numpy as np
21 from scipy.sparse import diags, linalg, spdiags, kronsum
22
23 # Define the parameters
24 Nx = 240
25 Ny = 240
26 Lx = 80E-6
27 Ly = 80E-6
28
29 x, dx = np.linspace(-Lx/2, Lx/2, Nx, retstep=True)
30 y, dy = np.linspace(-Ly/2, Ly/2, Ny, retstep=True)
31
32 X, Y = np.meshgrid(x, y, indexing="xy")
33
34
35 n0 = 1.48 # Refractive index of the background
36 dn1 = 4.00E-3 # Amplitude of the refractive index modulation
37 wavelength = 730E-9 # Wavelength of the light
38 k0 = 2 * np.pi / wavelength # Wavenumber
39
40 n_eigen = 4
41
42 wx1 = 0.9E-6 # Width of the refractive index modulation
43 wy1 = 3.3E-6
44
45 a1 = 25E-6
46
47 ##create matrix
48 if dx == dy:
49     # Create the 2D Laplacian matrix for dx = dy using the 5-point stencil
50     # 0 1 0

```

```

51     # 1 -4 1
52     # 0 1 0
53
54
55     diag0 = -4 * np.ones(Nx * Ny)
56     diag1 = np.ones(Nx * Ny - 1)
57     diag1[np.arange(1, Nx * Ny) % Nx == 0] = 0 # Remove wrap-around
58     connections
59     diagNx = np.ones(Nx * Ny - Nx)
60
61     diags_data = [diag0, diag1, diag1, diagNx, diagNx]
62     diags_offsets = [0, -1, 1, -Nx, Nx]
63
64     T = spdiags(diags_data, diags_offsets, Nx * Ny, Nx * Ny) / (dx**2)
65 else:
66     # Just the Kronecker sum of the 1D Laplacian matrices
67     diagx = np.ones([Nx])
68     diagy = np.ones([Ny])
69
70     diagsx = np.array([diagx, -2 * diagx, diagx])
71     diagsy = np.array([diagy, -2 * diagy, diagy])
72     Dx = spdiags(diagsx, np.array([-1, 0, 1]), Nx, Nx) / (dx**2)
73     Dy = spdiags(diagsy, np.array([-1, 0, 1]), Ny, Ny) / (dy**2)
74
75     T = kronsum(Dx, Dy)
76
77 def dn_func(X, Y, wx, wy):
78     output = np.tanh(33.0*np.exp(-(X/wx)**2- ((Y)/wy)**2))
79     return output
80
81 # Create the refractive index modulation
82 dn_array = np.zeros(X.shape)
83 dn_array += dn1 * dn_func(X, Y, wx1, wy1)
84
85 # Create the sparse matrix for the Helmholtz problem
86 n = n0 + dn_array
87 n = diags(n.reshape(Nx*Ny), (0))
88
89 H = T + (k0*n)**2
90
91 # Solve the eigenvalue problem
92 eigenvalues, eigenvectors = linalg.eigsh(H, k=n_eigen, which='LA')
93
94 np.save(f"eigenvalues.npy", eigenvalues)
95 np.save(f"eigenvectors.npy", eigenvectors)

```

Appendix D Código en C de BPM

```
1 // Saves into a text file the output of a gaussian light beam propagating in a
   2D waveguide array
2
3 /*
4 Copyright (C) 2025 Diego Roman-Cortes
5
6 This program is free software: you can redistribute it and/or modify
7 it under the terms of the GNU General Public License as published by
8 the Free Software Foundation, either version 3 of the License, or
9 (at your option) any later version.
10
11 This program is distributed in the hope that it will be useful,
12 but WITHOUT ANY WARRANTY; without even the implied warranty of
13 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
14 GNU General Public License for more details.
15
16 You should have received a copy of the GNU General Public License
17 along with this program. If not, see <https://www.gnu.org/licenses/>.
18
19 contact: diego.roman.c@ug.uchile.cl
20 */
21
22 #include <stdio.h>
23 #include <stdlib.h>
24 #include <string.h>
25 #include <math.h>
26 #include <complex.h>
27 #include <fftw3.h>
28
29 int main(int argc, char* argv[]){
30     // number of points in grid
31     int Nx = 700;
32     int Ny = 700;
33     int Nz = 5000;
34
35     // parameters
36     double n0 = 1.48; // refraction index of borosilicate
37     double l0 = 730E-9; // wavelenght of light
38     double wx = 1.2E-6; // width of the waveguide
39     double wy = 3.0E-6; // height of the waveguide
40     double sigma = 8.0E-6; // width of LG-mode
41     double l = 0; // azimuthal parameter of LG-mode
42     double Lx = 350E-6; // width of the grid
43     double Ly = 350E-6; // height of the grid
44
45     double zmax = 50E-3; // propagation distance
46
47     // auxiliar variables
```

```

48     double dx = Lx/(Nx-1);
49     double dy = Ly/(Ny-1);
50     double dz = zmax/(Nz-1);
51     double k0 = 2*M_PI/10;
52     double beta = k0 * n0;
53     double xi, yj, r, phi;
54
55     //phi = atof(argv[1])*1E-9;
56     //printf("%f", phi);
57
58     double* dn = malloc(sizeof(double) * Nx * Ny);
59
60     // 1D array setup
61     double dn1 = 9.5E-4; // contrast of first waveguide
62
63     double d1x = 17E-6; // X separation of waveguides
64     double d1y = 18.5E-6; // Y separation of waveguides
65
66     // for animation
67     int frames = 50;
68     int rem, div;
69     char filename[10];
70
71     int i, j, k;
72
73     FILE *fp1, *fp2, *fp3;
74
75     //initialization of FFTW
76
77     fftw_init_threads();
78     fftw_complex *in = (fftw_complex*) fftw_malloc(sizeof(fftw_complex) * Nx *
79     Ny);
80     fftw_complex *aux = (fftw_complex*) fftw_malloc(sizeof(fftw_complex) * Nx
81     * Ny);
82     fftw_complex *out = (fftw_complex*) fftw_malloc(sizeof(fftw_complex) * Nx
83     * Ny);
84     fftw_plan_with_nthreads(8);
85
86     fftw_plan p_forward = fftw_plan_dft_2d(Nx, Ny, in, out, FFTW_BACKWARD,
87     FFTW_PATIENT);
88     fftw_plan p_inverse = fftw_plan_dft_2d(Nx, Ny, aux, in, FFTW_FORWARD,
89     FFTW_PATIENT);
90
91     fp1 = freopen("refractive2d.txt", "w", stdout);
92     // shape of refractive index contrast
93     for(i = 0; i < Nx; i++){
94         for(j = 0; j < Ny; j++){
95             xi = -0.5*Lx + i*dx;
96             yj = -0.5*Ly + j*dy;

```

```

92
93         for(int n=-9; n<10; n++){
94             dn[i+Nx*j] += dn1 * tanh(33.0 / (exp(((xi-n*d1x)/wx)*((xi-n*
95                 d1x)/wx) + ((yj)/wy)*((yj)/wy))));
96             }
97             printf("%e\n", dn[i+Nx*j]);
98         }
99         printf("\n");
100     }
101     fclose(fp1);
102
103     // initial field (gaussian)
104     for(i = 0; i < Nx; i++){
105         for(j = 0; j < Ny; j++){
106             xi = -0.5*Lx + i*dx;
107             yj = -0.5*Ly + j*dy;
108             r = sqrt((xi)*(xi) + (yj)*(yj));
109             in[i+Nx*j] += (cexp(-r*r/(sigma*sigma))); // hermite-gaussian mode
110         }
111     }
112     // save the input (gaussian) in a text file
113     fp2 = freopen("00.txt", "w", stdout);
114     for(i = 0; i < Nx; i++){
115         for(j = 0; j < Ny; j++){
116             xi = -0.5*Lx + i*dx;
117             yj = -0.5*Ly + j*dy;
118             printf("%e\n", cabs(in[i+Nx*j])*cabs(in[i+Nx*j]));
119         }
120         printf("\n");
121     }
122     fclose(fp2);
123
124     // frequency indices
125     int freqidx[Nx + Ny];
126
127     for(i=0; i < Nx/2; i++){
128         freqidx[i] = i;
129     }
130     for(j=0; j < Ny/2; j++){
131         freqidx[Nx+j] = j;
132     }
133     for(i=Nx/2; i < Nx; i++){
134         freqidx[i] = i-Nx;
135     }
136     for(j=Ny/2; j < Ny; j++){
137         freqidx[Nx+j] = j-Ny;
138     }
139

```



```

140     fftw_complex *phase = (fftw_complex*) fftw_malloc(sizeof(fftw_complex) *
141     Nx * Ny);
142     for(i = 0; i < Nx; i++){
143         for(j = 0; j < Ny; j++){
144             phase[i+j*Nx] = cexp(I*dz*( (2*M_PI) * (2*M_PI) * ( (freqidx[i]/Lx
145             ) * (freqidx[i]/Lx) + (freqidx[Nx+j]/Ly) * (freqidx[Nx+j]/Ly) )/(4*beta)));
146         }
147     }
148
149     // main loop
150
151     for(k=1; k <= Nz; k++){
152
153         fftw_execute(p_forward); // 'out' now points towards the DFT of 'in'
154
155         for(i = 0; i < Nx*Ny; i++){
156             aux[i] = out[i] * phase[i];
157         }
158
159         fftw_execute(p_inverse); // 'in' now points towards the inverse DFT of
160         'aux'
161
162         for(i = 0; i < Nx * Ny; i++){
163             in[i] /= (Nx * Ny); // normalization of FFT
164             in[i] *= cexp(-I * k0 * (((n0+dn[i])*(n0+dn[i]))- (n0*n0)) * dz
165             /(2*n0)); // potential operator in real space
166         }
167
168         fftw_execute(p_forward); // 'out' now points towards the DFT of 'in'
169
170         for(i = 0; i < Nx*Ny; i++){
171             aux[i] = out[i] * phase[i];
172         }
173
174         fftw_execute(p_inverse); // 'in' now points towards the inverse DFT of
175         'aux'
176
177         for(i = 0; i < Nx * Ny; i++){
178             in[i] /= (Nx * Ny); // normalization of FFT
179         }
180
181         // save to txt
182         rem = k % (Nz/frames);
183         if(rem == 0){
184             div = k / (Nz/frames);
185             sprintf(filename, "%02d.txt", div);
186
187             fp3 = freopen(filename, "w", stdout);
188
189             for(i = 0; i < Nx; i++){

```

```

184         for(j = 0; j < Ny; j++){
185             printf("%e\n", cabs(in[i+j*Nx])*cabs(in[i+j*Nx]));
186         }
187         printf("\n");
188     }
189     fclose(fp3);
190 }
191
192 }
193 fftw_cleanup_threads();
194 fftw_destroy_plan(p_forward);
195 fftw_destroy_plan(p_inverse);
196 fftw_free(in);
197 fftw_free(aux);
198 fftw_free(out);
199 fftw_free(phase);
200 free(dn);
201 return 0;
202 }

```

Appendix E Código en Python generador de hologramas

```
1 # Copyright (C) 2024 Diego Roman-Cortes
2 #
3 # This program is free software: you can redistribute it and/or modify
4 # it under the terms of the GNU General Public License as published by
5 # the Free Software Foundation, either version 3 of the License, or
6 # (at your option) any later version.
7 #
8 # This program is distributed in the hope that it will be useful,
9 # but WITHOUT ANY WARRANTY; without even the implied warranty of
10 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
11 # GNU General Public License for more details.
12 #
13 # You should have received a copy of the GNU General Public License
14 # along with this program. If not, see <https://www.gnu.org/licenses/>.
15 #
16 # e-mail: diego.roman.c@ug.uchile.cl
17
18 import numpy as np
19 import matplotlib.pyplot as plt
20 from PIL import Image, ImageChops
21 from scipy.special import eval_genlaguerre
22 from scipy import signal
23
24 my_dpi = 120 #120
25 plt.style.use('dark_background')
26
27 WIDTH = 1920
28 HEIGHT = 1080
29
30 sigma = HEIGHT/4
31
32 sigmax = HEIGHT/2.2/4*2.2
33 sigmay = HEIGHT/5.0*1.1
34
35 x = np.linspace(-WIDTH/2, WIDTH/2, num=WIDTH)
36 y = np.linspace(-HEIGHT/2, HEIGHT/2, num=HEIGHT)
37
38 Xn, Yn = np.meshgrid(x, y, indexing='xy')
39
40 angle = -0.05
41 X = Xn*np.cos(angle) - Yn*np.sin(angle)
42 Y = Yn*np.cos(angle) + Xn*np.sin(angle)
43
44 Z = np.zeros(X.shape, dtype=complex)
45 Z += (np.exp(-(X)/sigma/1.0)**2)*np.exp(-(Y)/sigma)**2) # any function of x
    and y
```

```

46 Z /= np.sqrt(np.sum(np.abs(Z)**2))
47
48 phase = (np.angle(Z)+np.pi) * 255.0 / (2*np.pi)
49
50 fig = plt.figure(figsize=(1920/my_dpi, 1080/my_dpi), dpi=my_dpi)
51
52 plt.imsave('fase.png', phase, cmap="gray", vmin=0, vmax=255)
53 plt.close("all")
54
55 fig = plt.figure(figsize=(1920/my_dpi, 1080/my_dpi), dpi=my_dpi)
56
57 blaze = (signal.sawtooth(Xn*2.0*np.pi/5.0) + 1)/2.0*255
58
59 plt.imsave('blaze.png', blaze, cmap="gray", vmin=0, vmax=255)
60 plt.close("all")
61
62 fig = plt.figure(figsize=(1920/my_dpi, 1080/my_dpi), dpi=my_dpi)
63
64 amplitude = (np.abs(Z)**2/np.max(np.abs(Z)**2)) * blaze/255
65 amplitude /= amplitude.max()
66 amplitude *= 255
67
68
69 plt.imsave('amplitude.png', amplitude, cmap="gray", vmin=0, vmax=255)
70 plt.close("all")
71
72
73 im1 = np.array(Image.open('amplitude.png').convert('L'), dtype="uint16")
74 im2 = np.array(Image.open('fase.png').convert('L'), dtype="uint16")
75
76 imf = (((amplitude + phase) % 255))
77 im = (imf).astype(np.uint8)
78
79 plt.imsave('vortex2.png', im, cmap="gray", vmin=0, vmax=255)

```